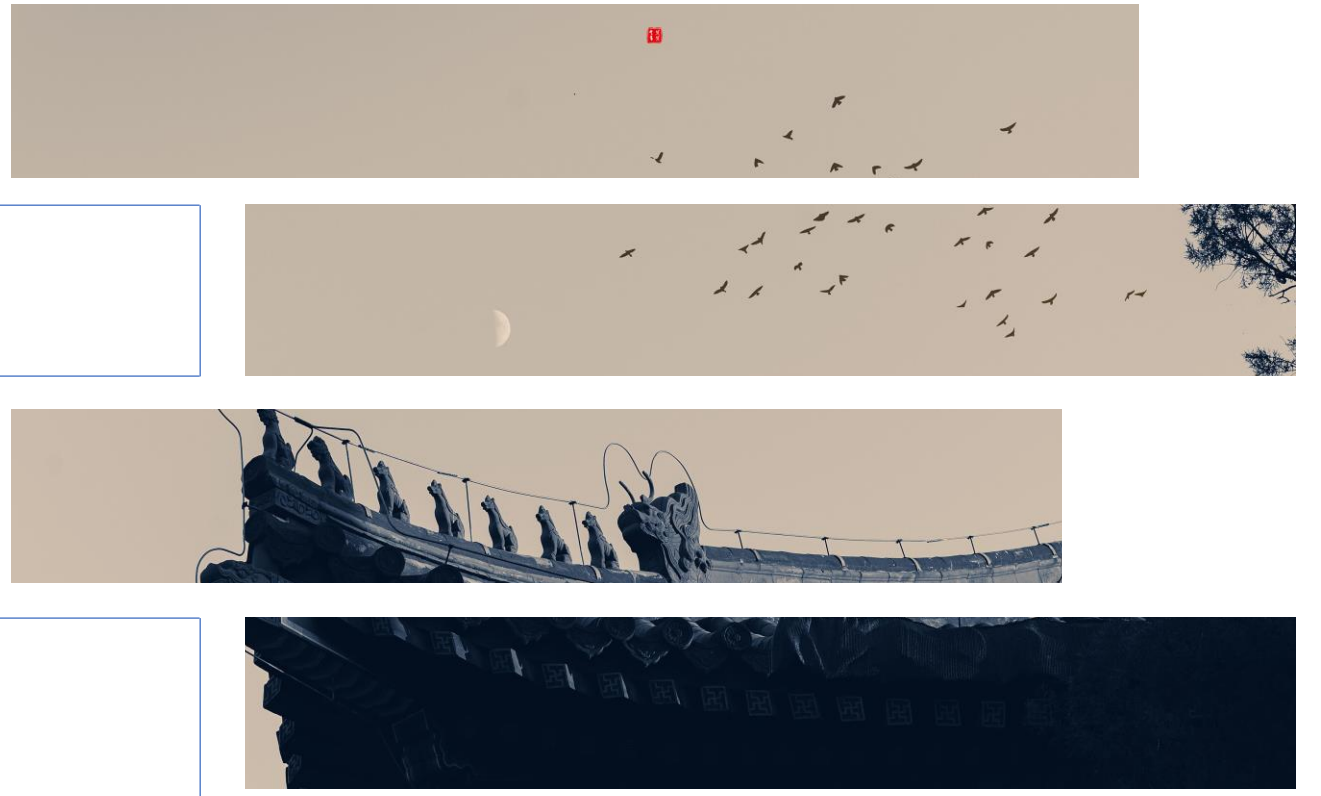
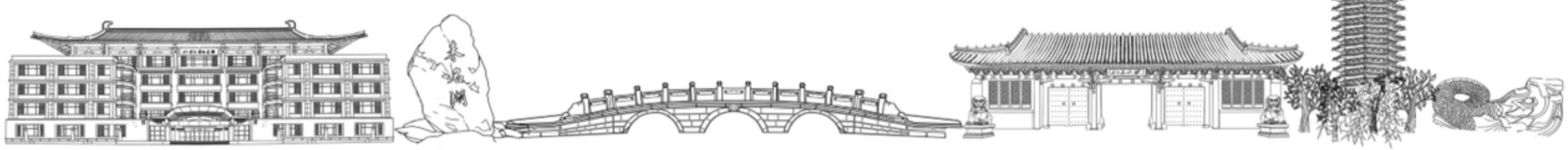
Cognitive Reasoning
Fall 2026



Lecture 04
Affordance, Functionality, and HOIs

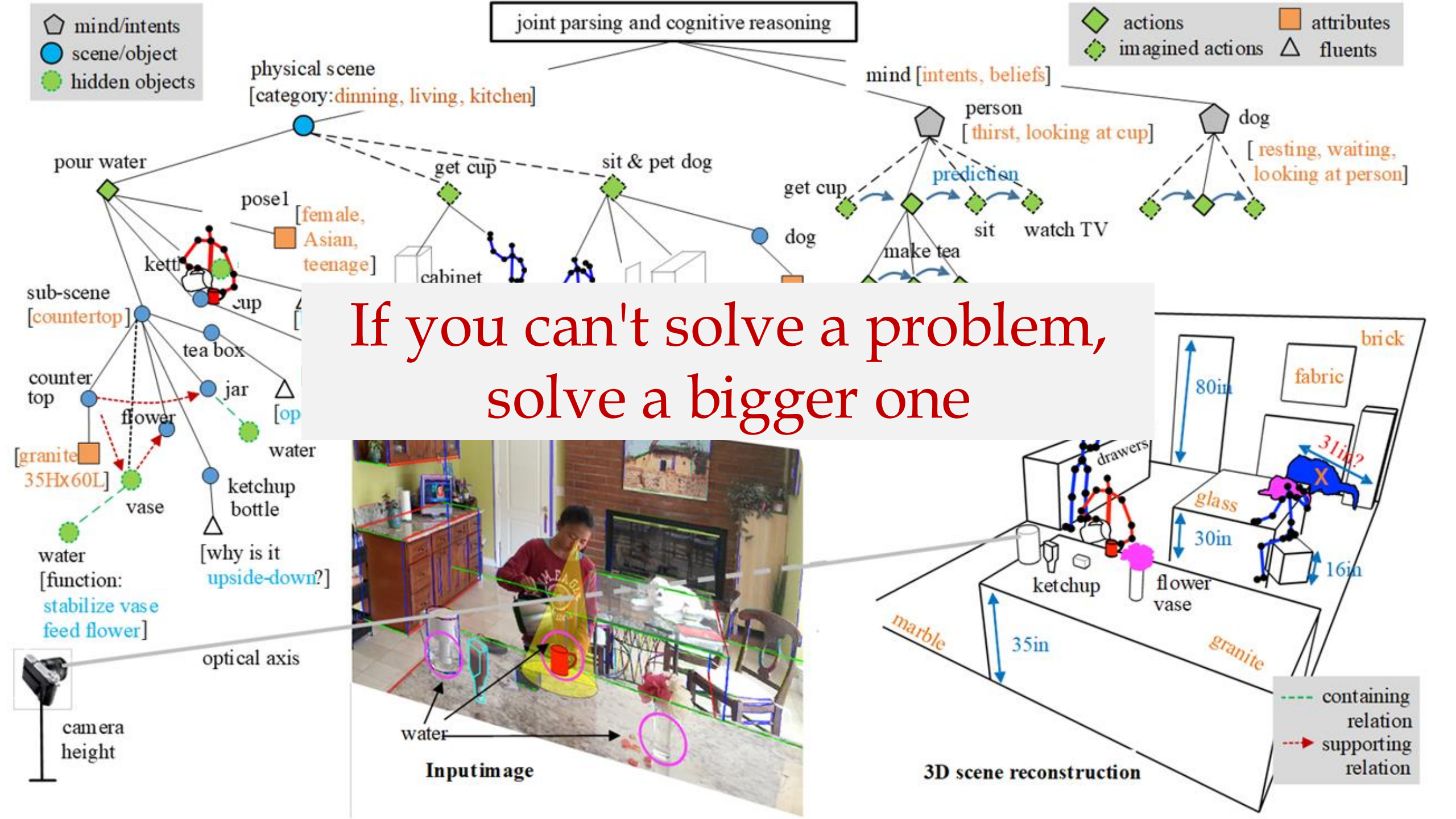
Instructor: Dr. Chi Zhang
School of Intelligence Science and Technology



Logistics

- Grouping
- Team leads report to https://www.wjx.top/vm/YM1N4un.aspx#
- Deadline: 10.07
- Proposal & Related work Check
- Deadline: 10.25




mind/intents
scene/object
hidden objects
joint parsing and cognitive reasoning
actions
imagined actions
attributes
fluents
physical scene
[category:dinning, living, kitchen]
mind [intents, beliefs]
person
[thirst, looking at cup]
dog
[resting, waiting, looking at person]
pour water
get cup
sit & pet dog
prediction
get cup
sit
watch TV
make tea
pose1
[female, Asian, teenage]
dog
cabinet
sub-scene
[countertop]
tea box
counter top
jar
flower
[granite 35Hx60L]
water
vase
ketchup bottle
water
[function: stabilize vase feed flower]
[why is it upside-down?]
optical axis
camera height
If you can't solve a problem,
solve a bigger one
brick
fabric
80in
31in?
X
glass
30in
16in
drawers
ketchup
flower vase
marble
35in
granite
water
Inputimage
3D scene reconstruction
containing relation
supporting relation

Input



Object Detection
Scene Parsing

Understand
Relations

Output:
Robot Execution

Affordance/Functionality
(Chapter 1)

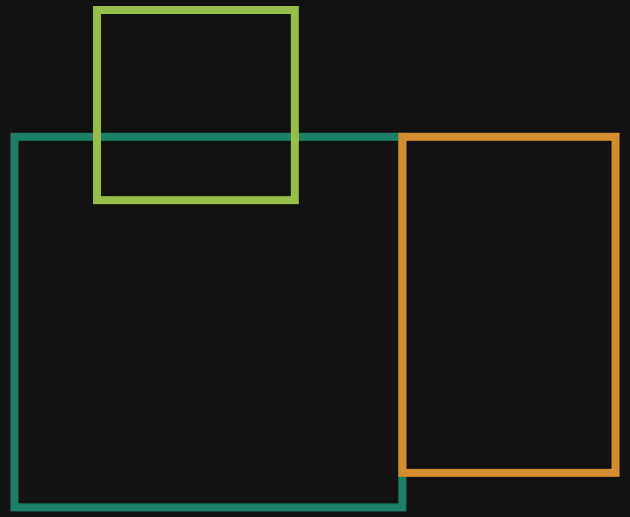
HOIs
(Chapter 2)

Humanoid
(Chapter 3)



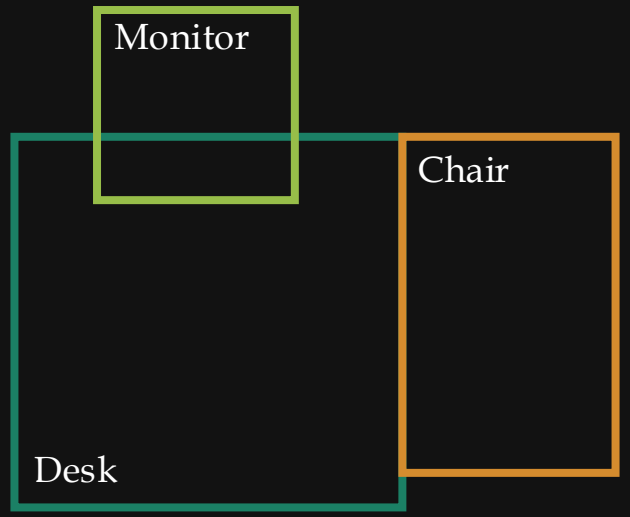
Chapter 2
Human-Object Interaction (HOI)

Here are some 2D bounding boxes in an image ...



Can you reconstruct the 3D scene?

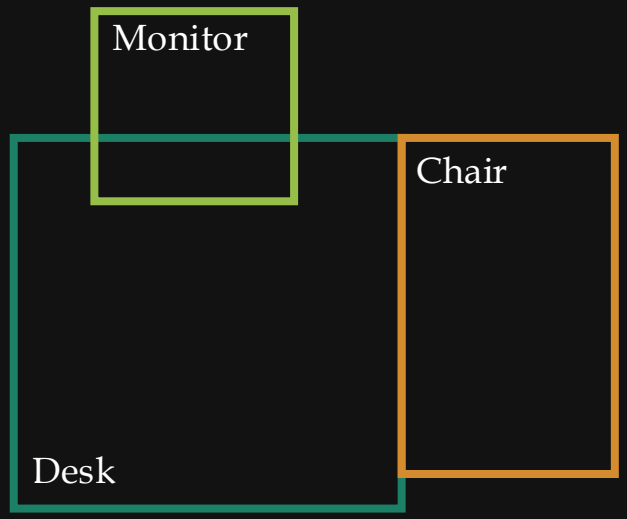
Now with labels ...

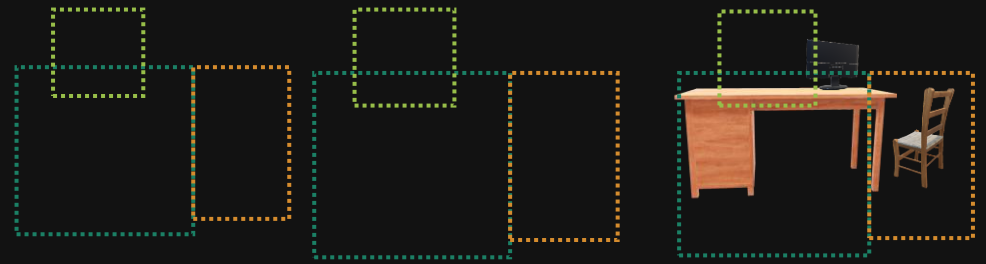


Can you reconstruct the 3D scene now?

Are you confident with your answer?

Now with labels ...

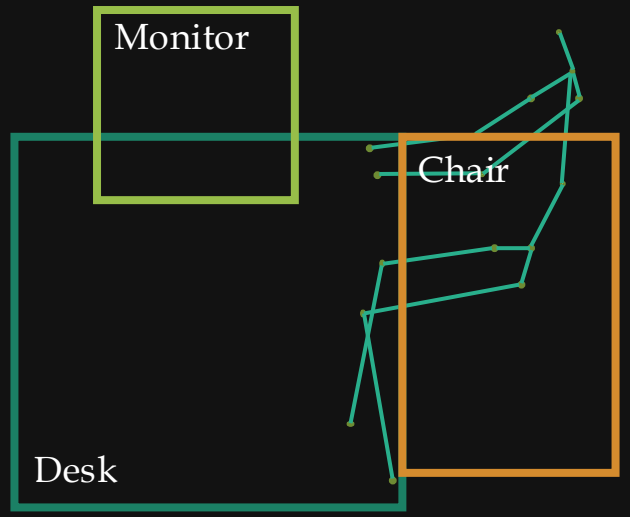




Can you reconstruct the 3D scene now?

Are you confident with your answer?

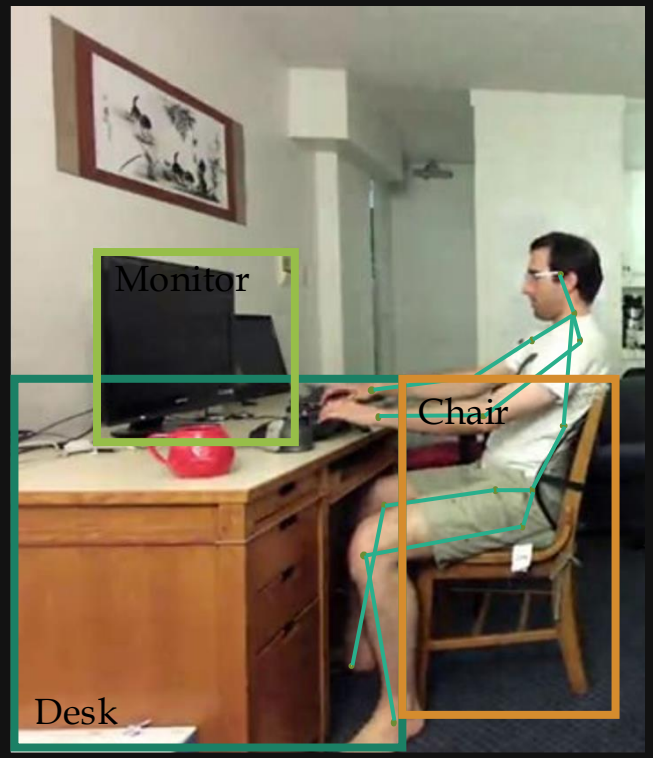
Now with a human ...



Can you reconstruct the 3D scene now?

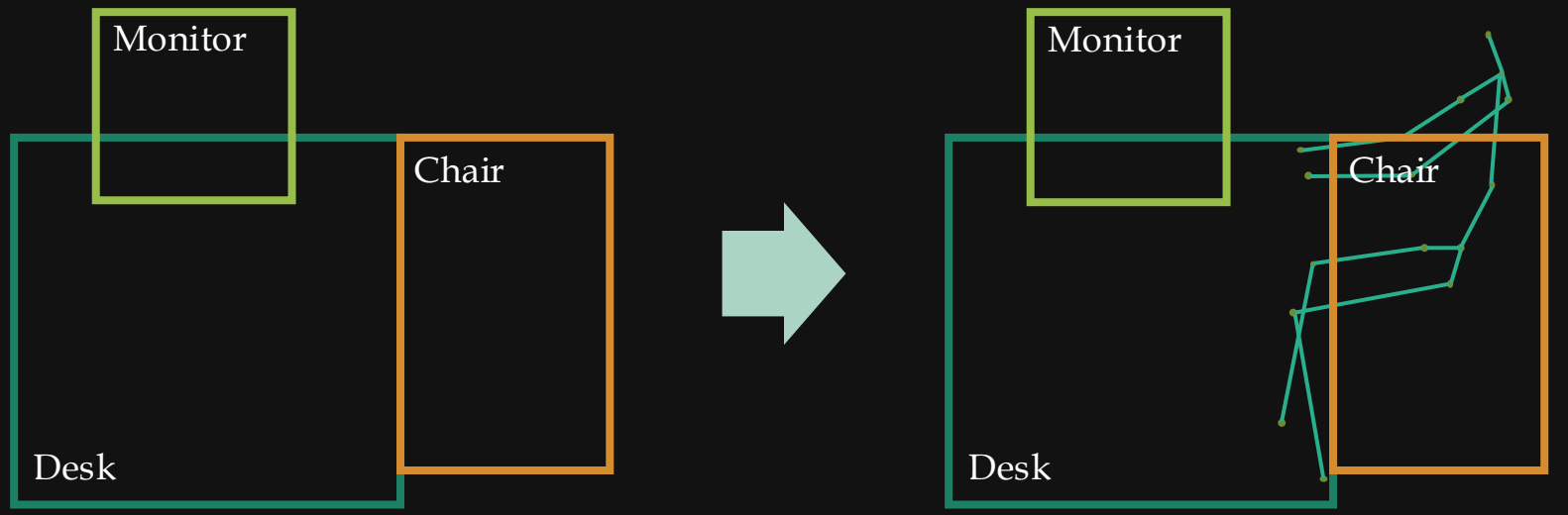
Are you more confident with your answer now?

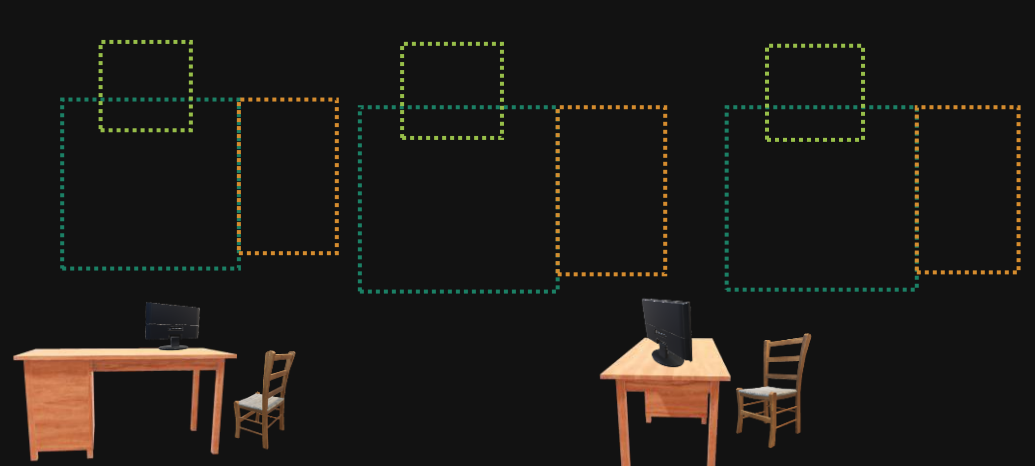
Now with the image ...

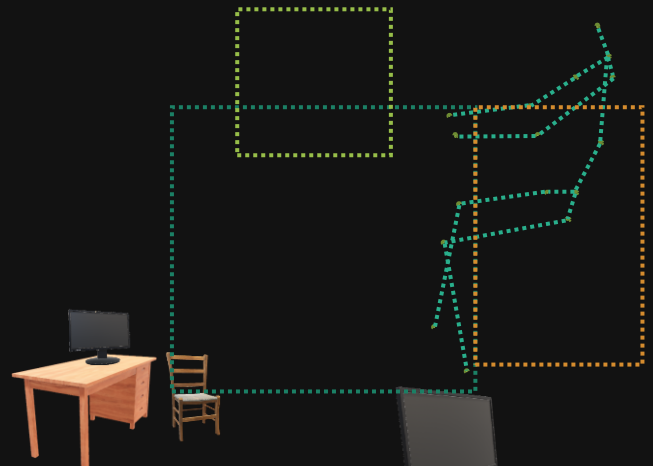


Did you guess right?

How does human pose improve object pose estimation?

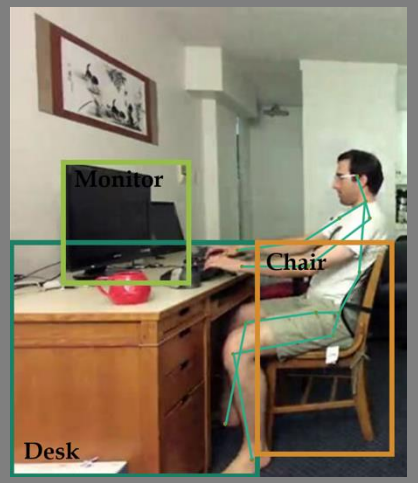






Joint Modeling of Human and Objects

HOI Detection



2D HOI

Section 1

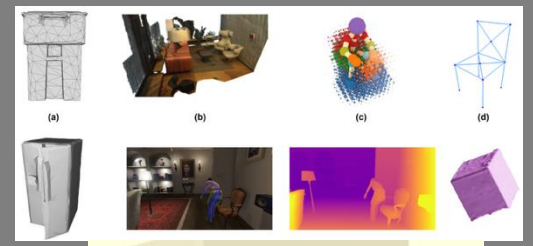
Reconstruct

Project

3D HOI

Section 2

3D Representation





3D Reconstruction

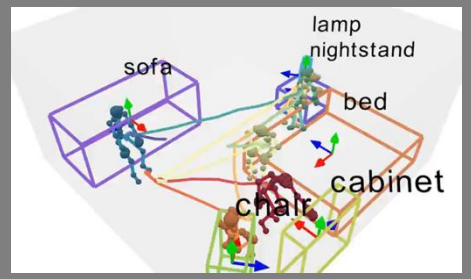
From Objects to Scenes

Joint Modeling of Human and Scenes

Human Motion Generation



Human-Centric Scene Generation

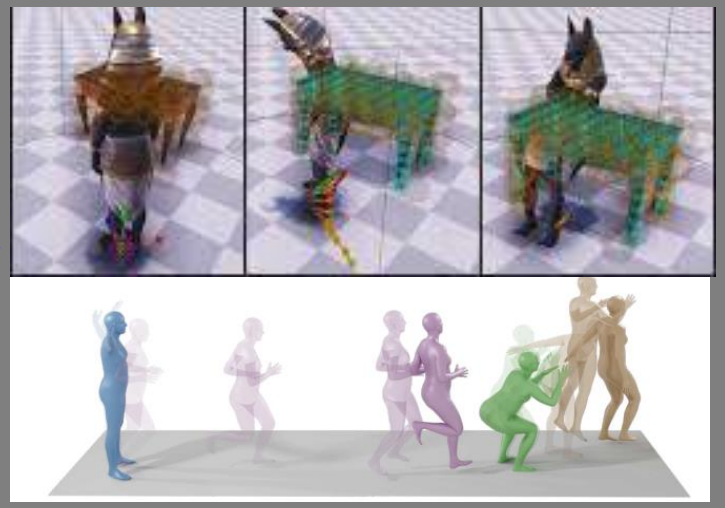


Human-Scene Interaction (HSI)

Section 3

Application

3D Generation



Outline

- Section 1: 2D HOI
  - General formulation
  - HOI detection
- Section 2: 3D HOI
  - 3D representation
  - 3D reconstruction from 2D image
  - From perception to generation
- Section 3: From HOI to HSI
  - Scene-conditioned human motion generation
  - Human-centric scene synthesis

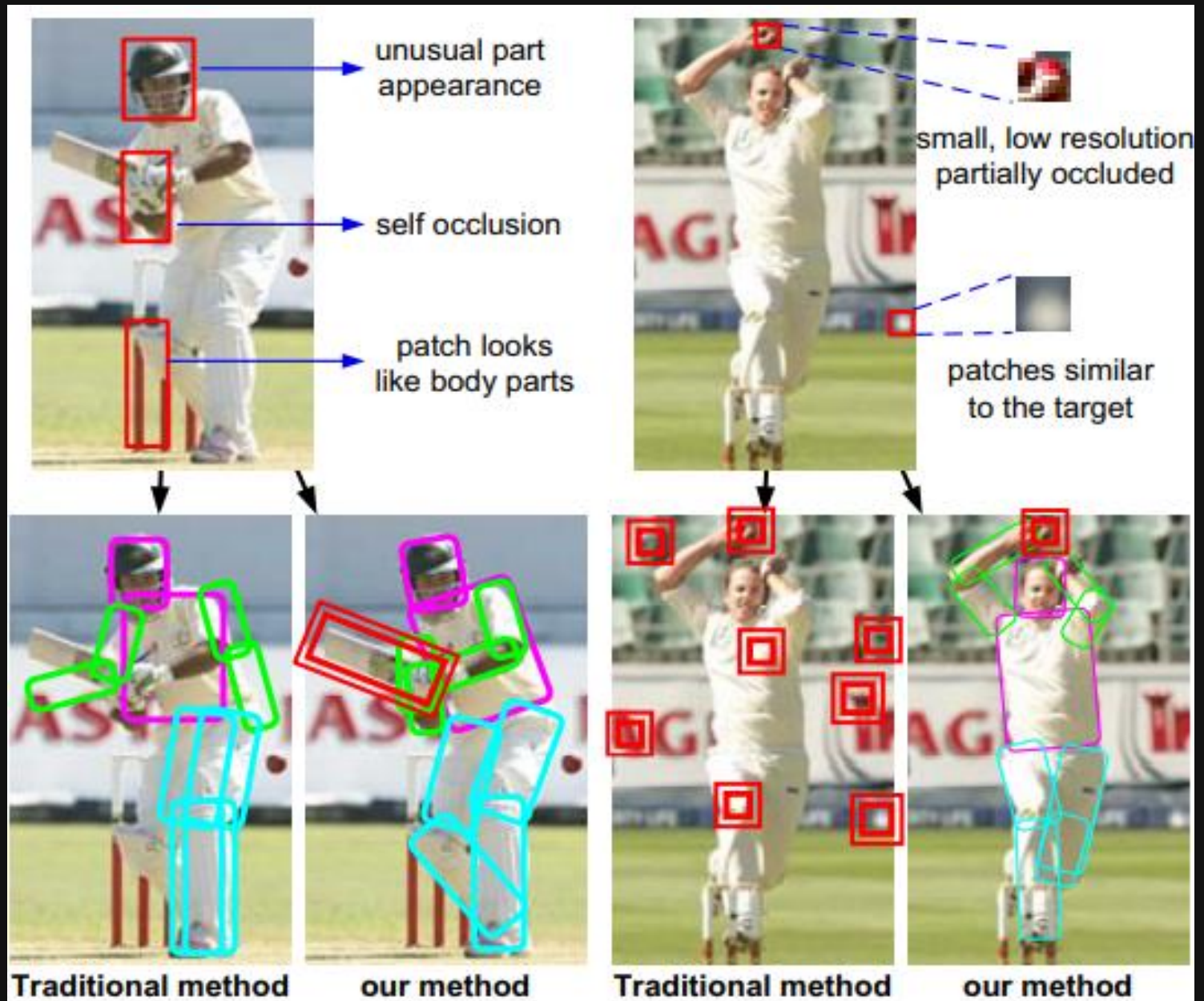
Outline

- Section 1: 2D HOI
  - General formulation
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- Section 2: 3D HOI
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A 2D Example


unusual part appearance
self occlusion
patch looks like body parts
small, low resolution partially occluded
patches similar to the target
Traditional method
our method
Traditional method
our method

A 2D Example

- Detect human

$$H = \arg\max_{H} P(H|I)$$

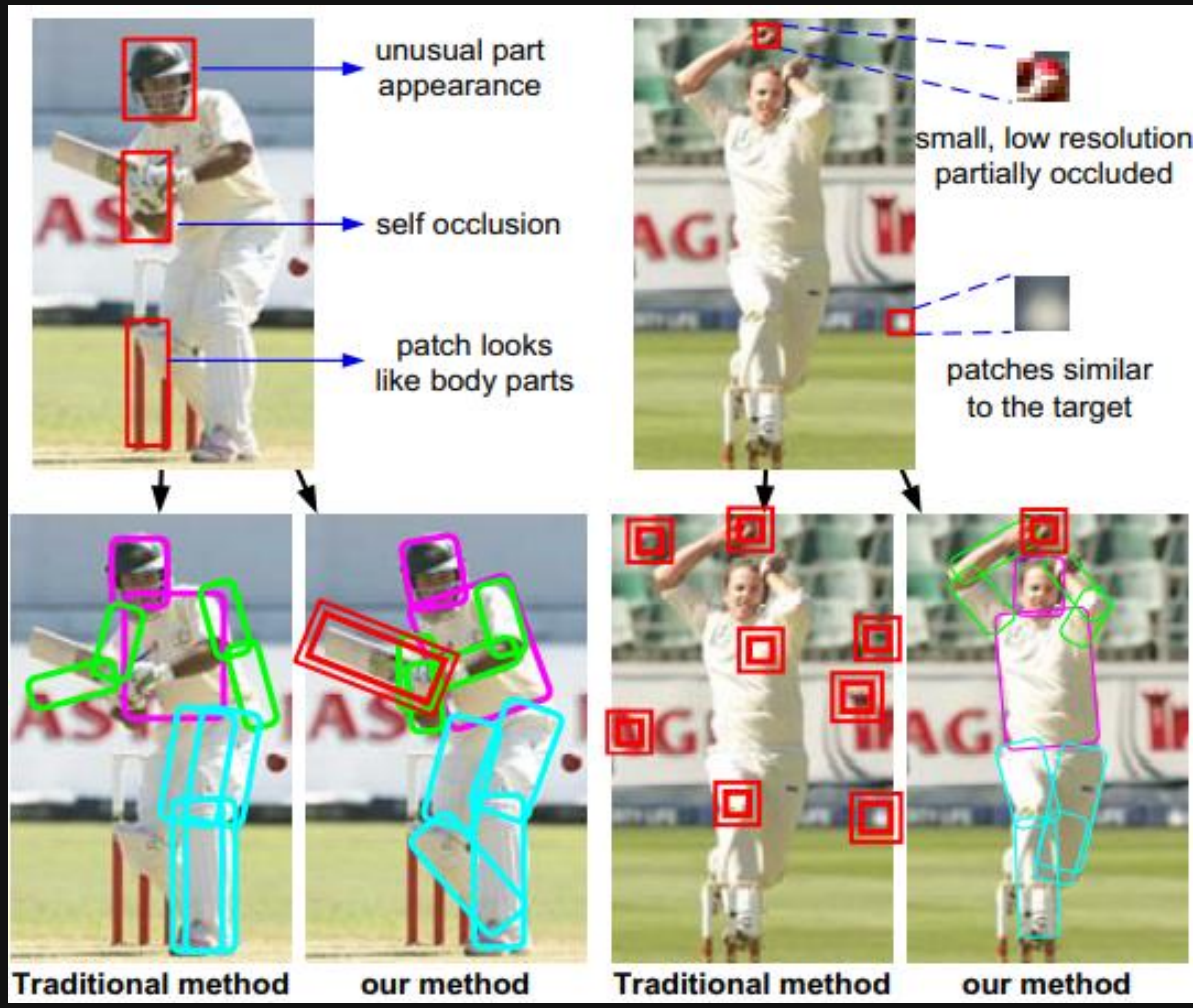
- Detect object

$$O = \arg\max_{O} P(O|I)$$

- Detect human + object

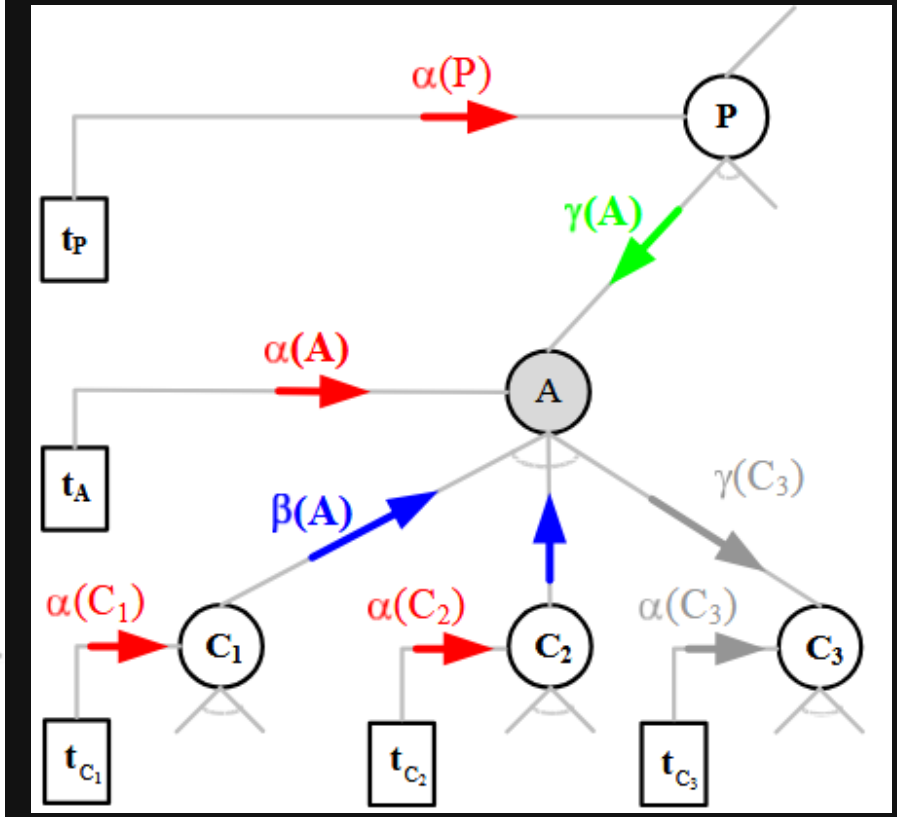
$$H, O = \arg\max_{H,O} P(H, O|I)$$

Which problem is the most challenging one?

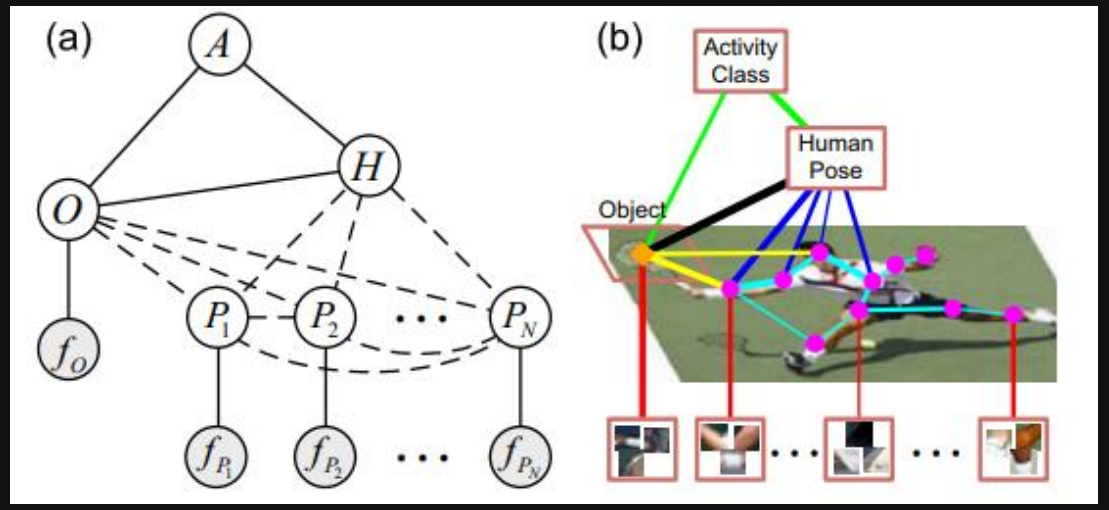


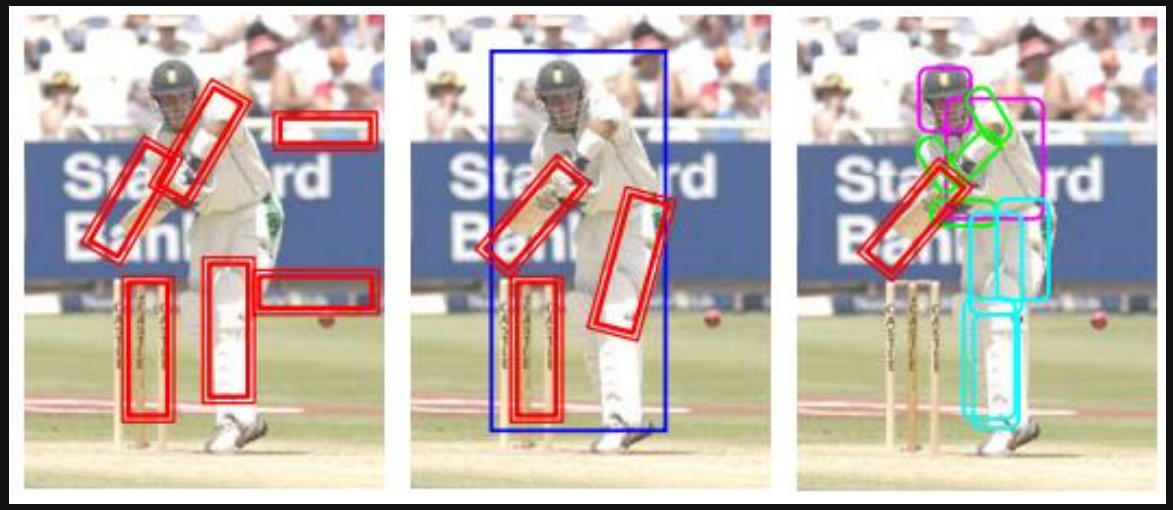
Counterintuitive!





Using HOIs to Improve Detection





A: Activity

O: Object H: Human P: Body Part

f: Patch Feature

Left: part detection

Middle: human detection with suppression

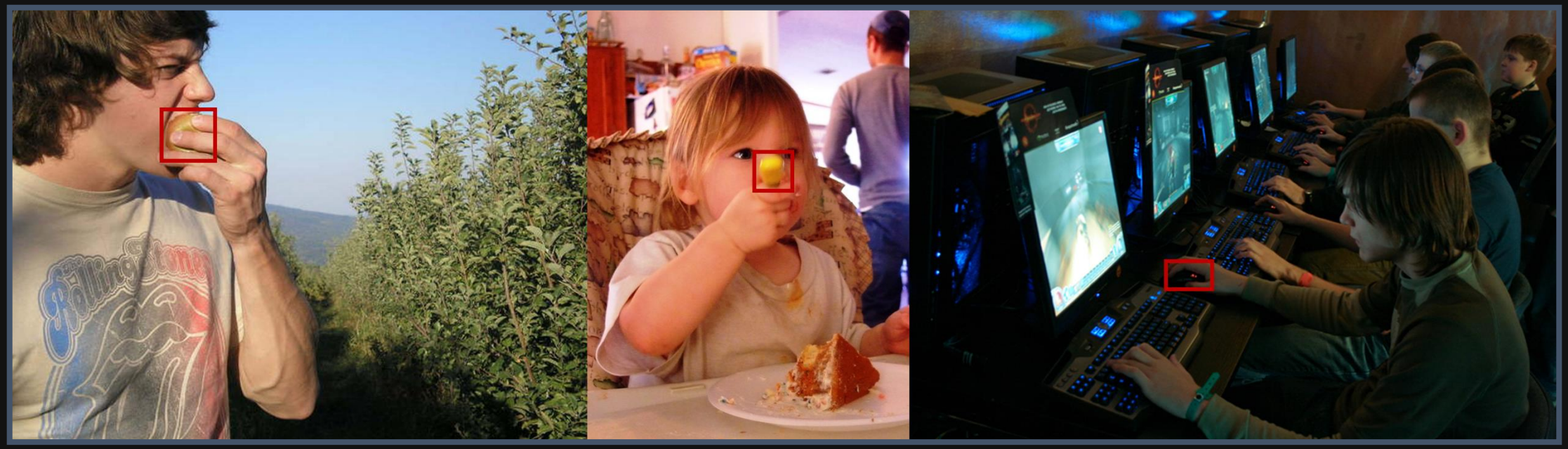
Right: part and human detection with HOIs

Let's see another example ...



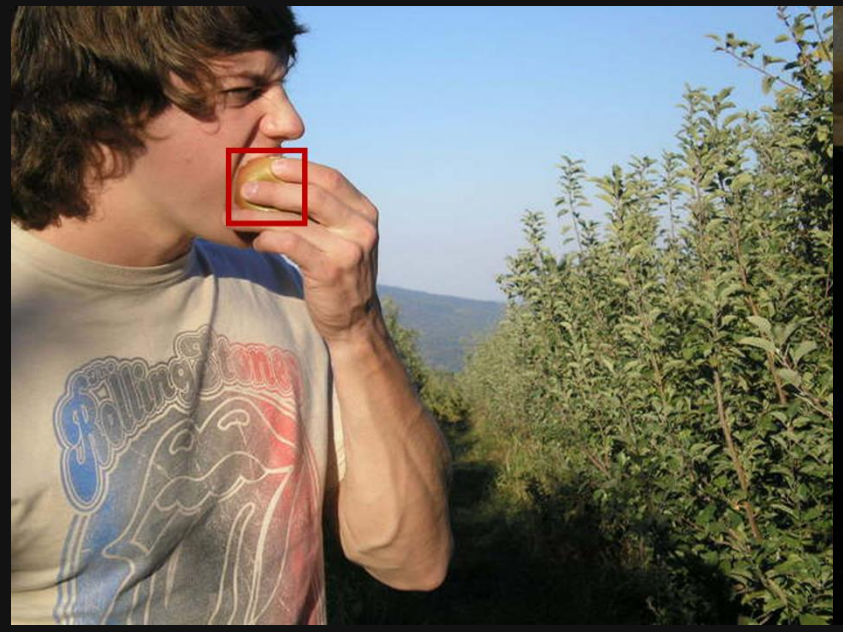
What are the objects in the three image patches?

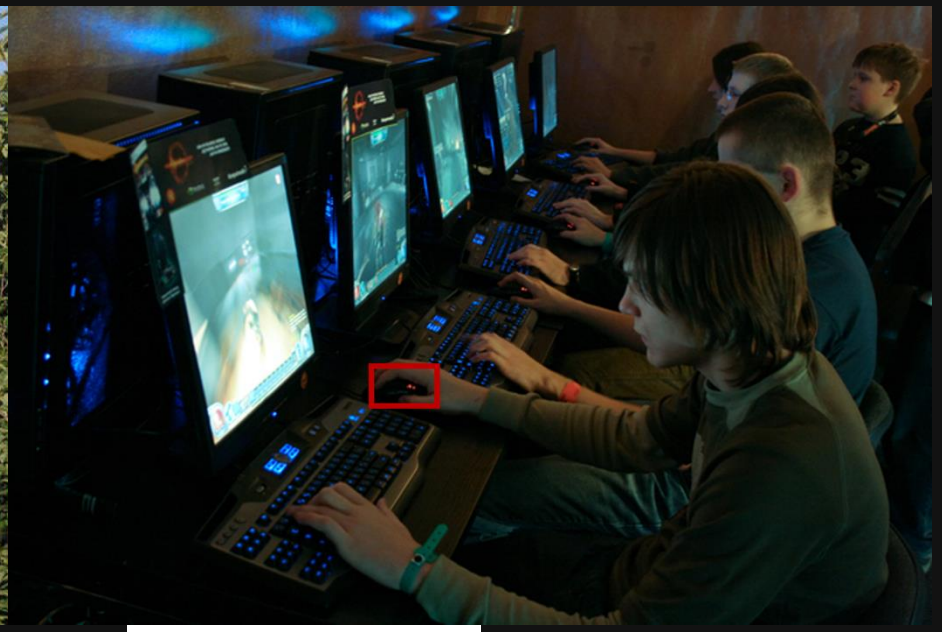
Let's see another example ...



Did you guess right?

Are Human-Objects Enough?















Are Human-Objects Enough?

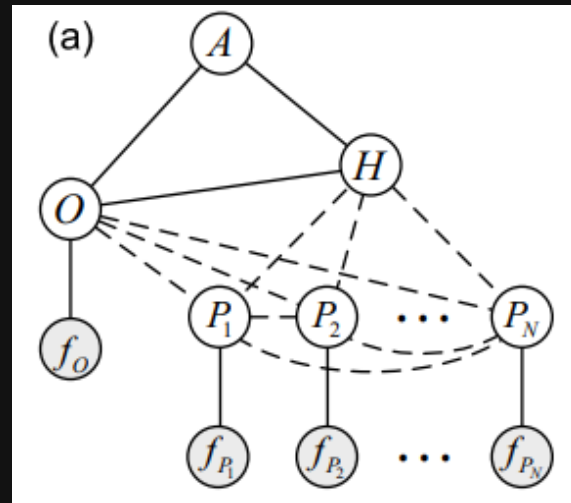


Humans, Objects, and Interactions/ Activities are strongly correlated

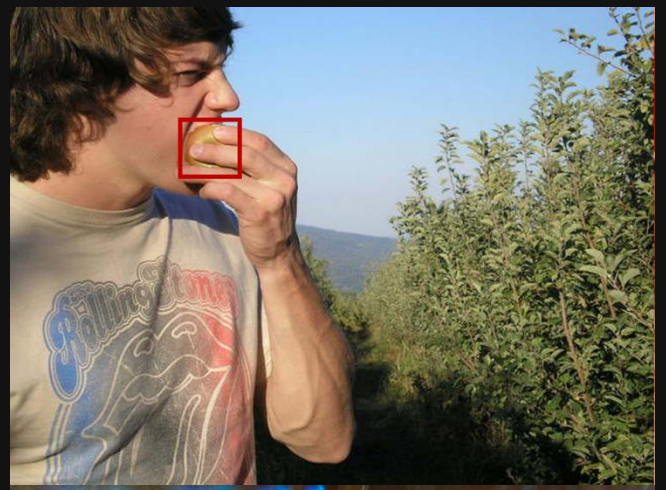
How to Compute?



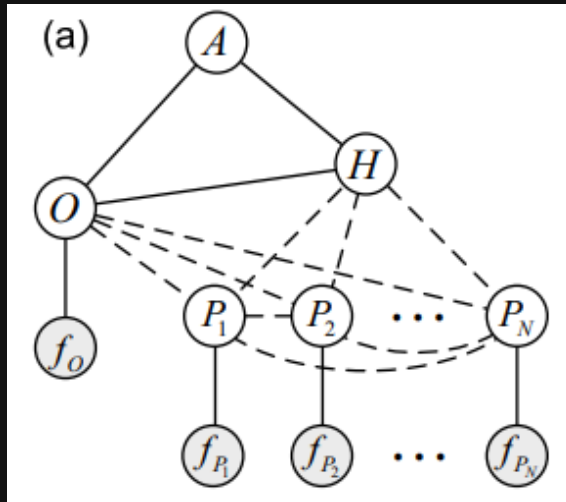
- $P(H, O, A)$ can be further decomposed
  - $P(H, O, A) = P(H, O|A)P(A)$ top down
  - $P(H, O, A) = P(A|H, O)P(H|O)P(O)$ bottom up
  - $P(H, O, A) = P(O|H, A)P(A|H)P(H)$
  - …



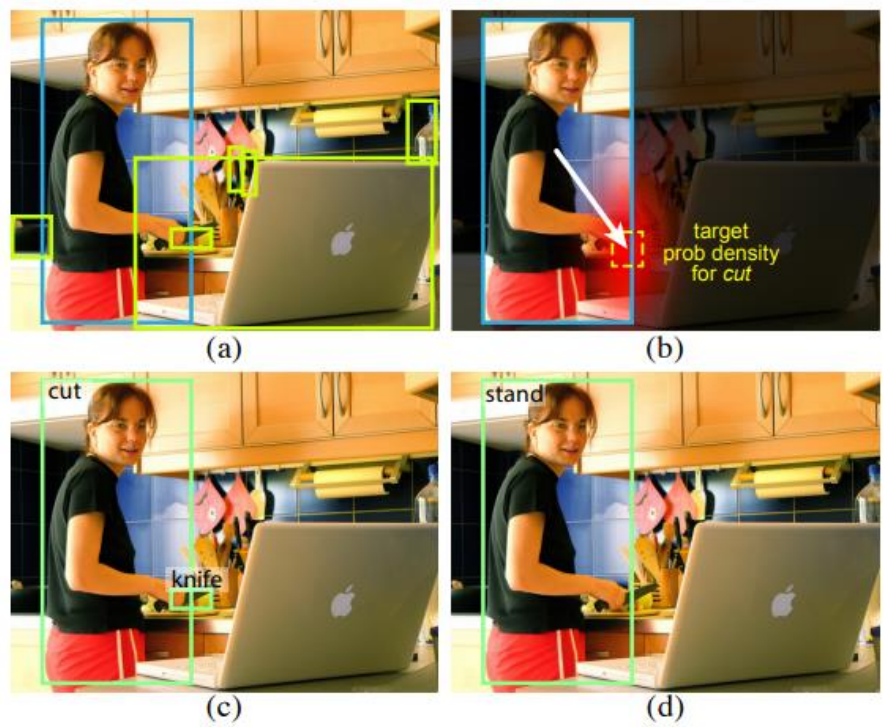
How to Compute?

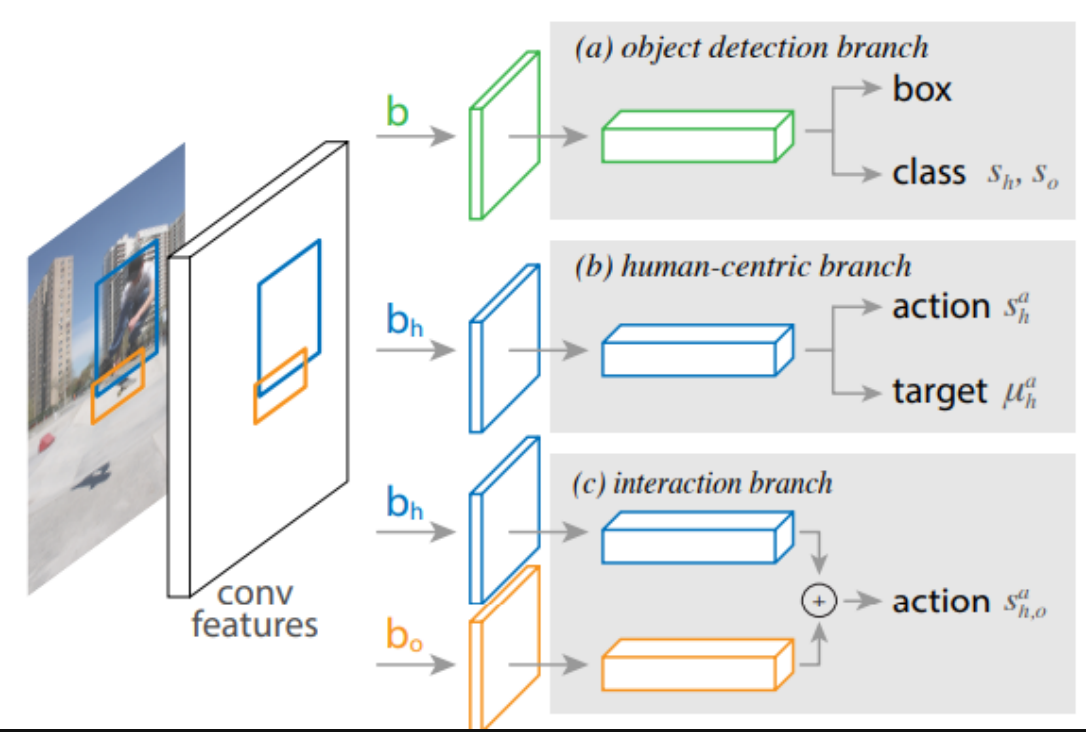


- $P(H, O, A)$ can be further decomposed
  - $P(H, O, A) = P(H, O|A)P(A)$ top down
  - $P(H, O, A) = P(A|H, O)P(H|O)P(O)$ bottom up
- How about combining top down and bottom up?



Obtain <human, verb, object> tuple from $P(A|H,O)$ $P(H,O|I)$





Human / Object Detection
$P(H|I), P(O|I)$

HOI Candidate Detection
$P(A_c, O_c|H)$

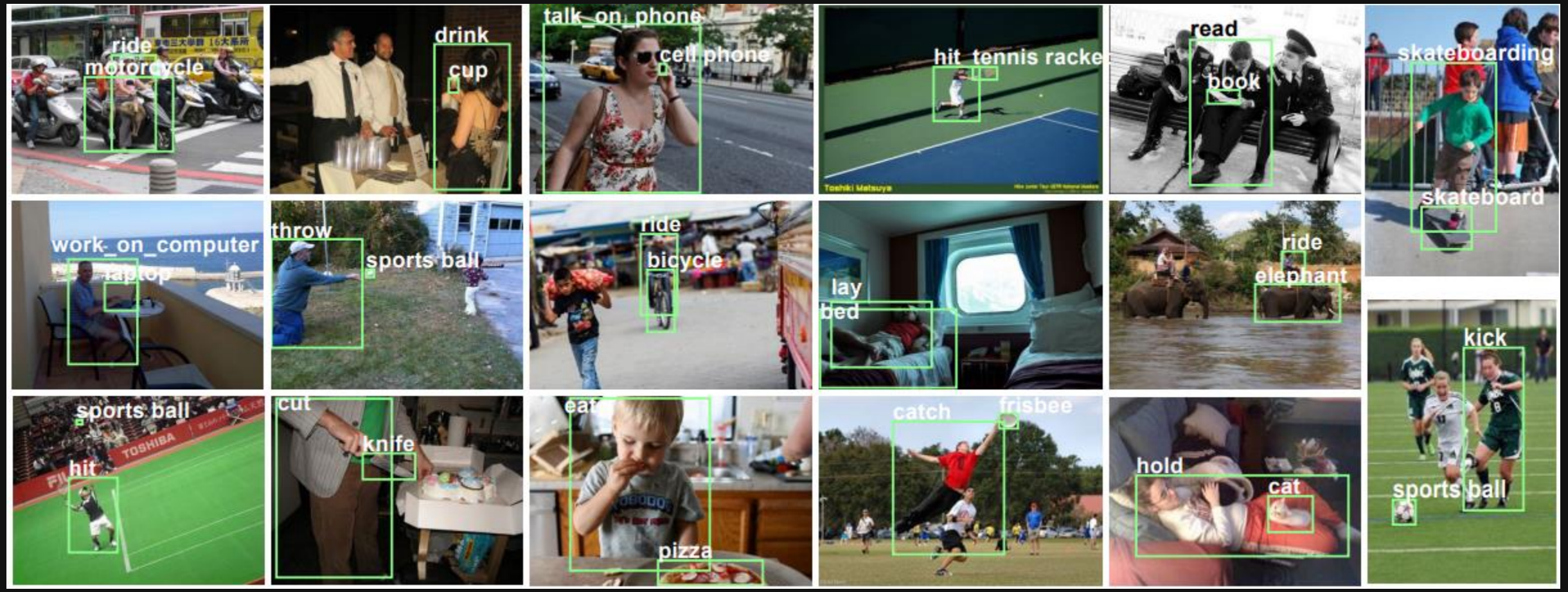
HOI detection
$P(A|A_c, O_c, H)$

I = Image, $(\cdot)_C$= candidate density map

(Gkioxari et al., 2018)

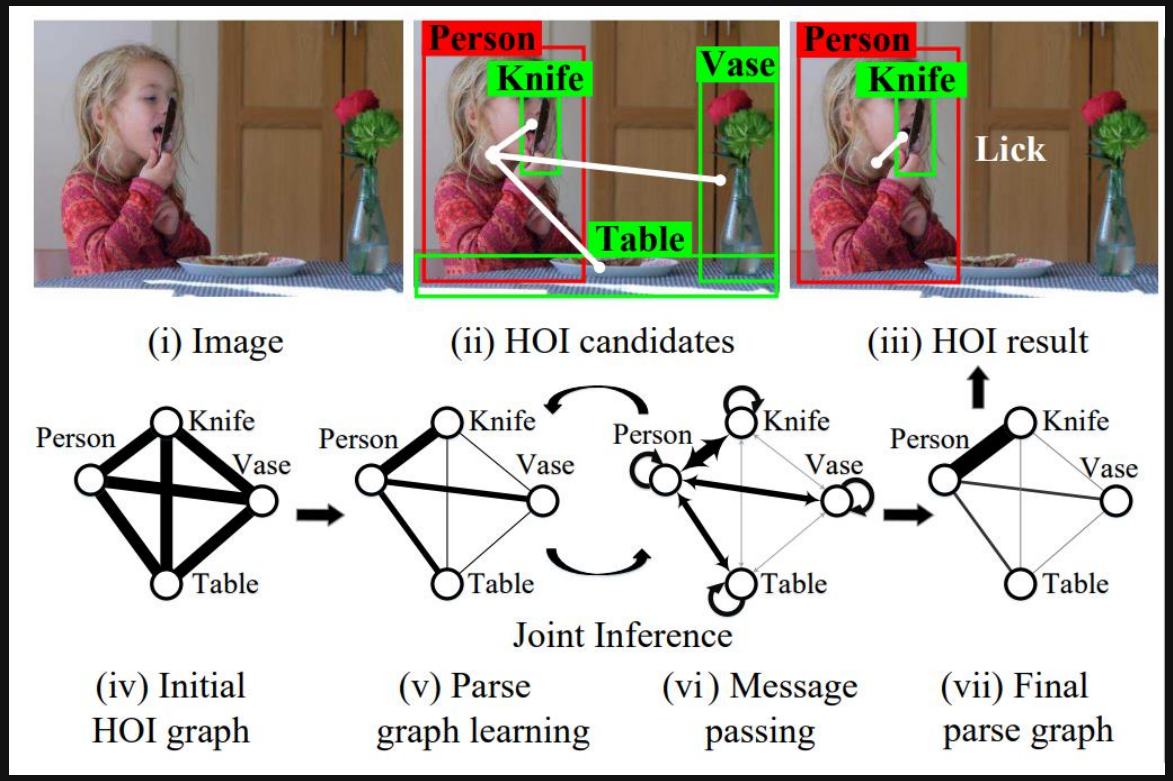
Detect the humans and objects first, then detect the actions

Obtain <human, verb, object> tuple from $P(A|H,O)$ $P(H,O|I)$



This paradigm can be applied to detect various types of HOI

Modeling $P(A|H,O)$ as a graph parsing neural network



Build a graph for better HOI modeling

image = graph

vertices = object + human, edges = interaction

Outline

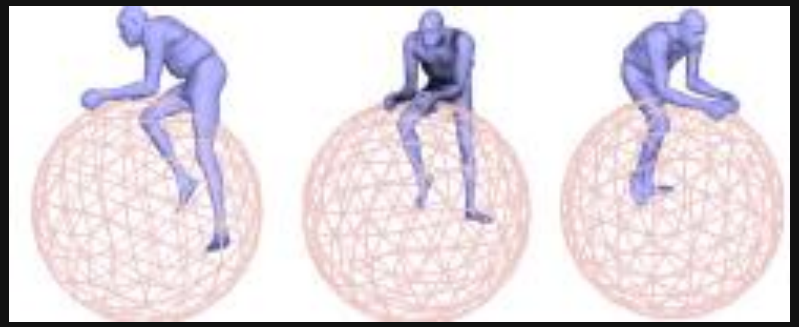
- Section 1: 2D HOI
  - General formulation
  - HOI detection

- Section 2: 3D HOI
  - 3D representation
  - 3D reconstruction from 2D image
  - From perception to generation

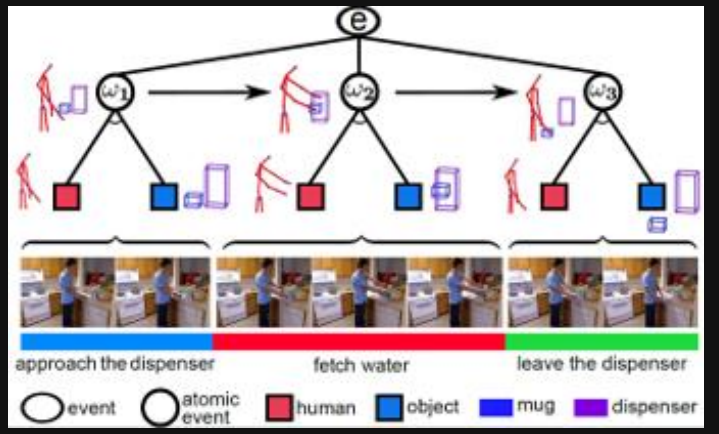
- Section 3: From HOI to HSI
  - Scene-conditioned human motion generation
  - Human-centric scene synthesis

From 2D to 3D

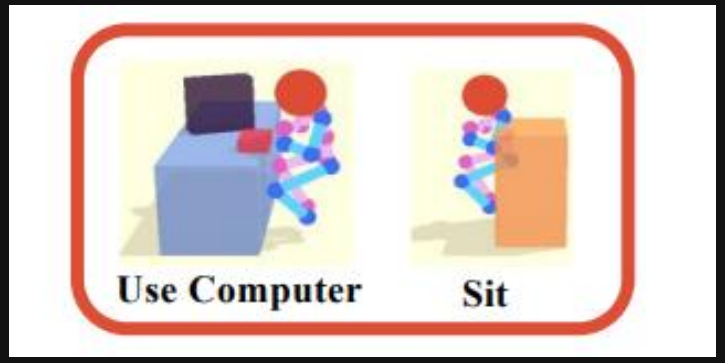
- Extending HOI to 3D and 4D



Infer 3D HOI from 2D observations



Recover 4D HOI with video



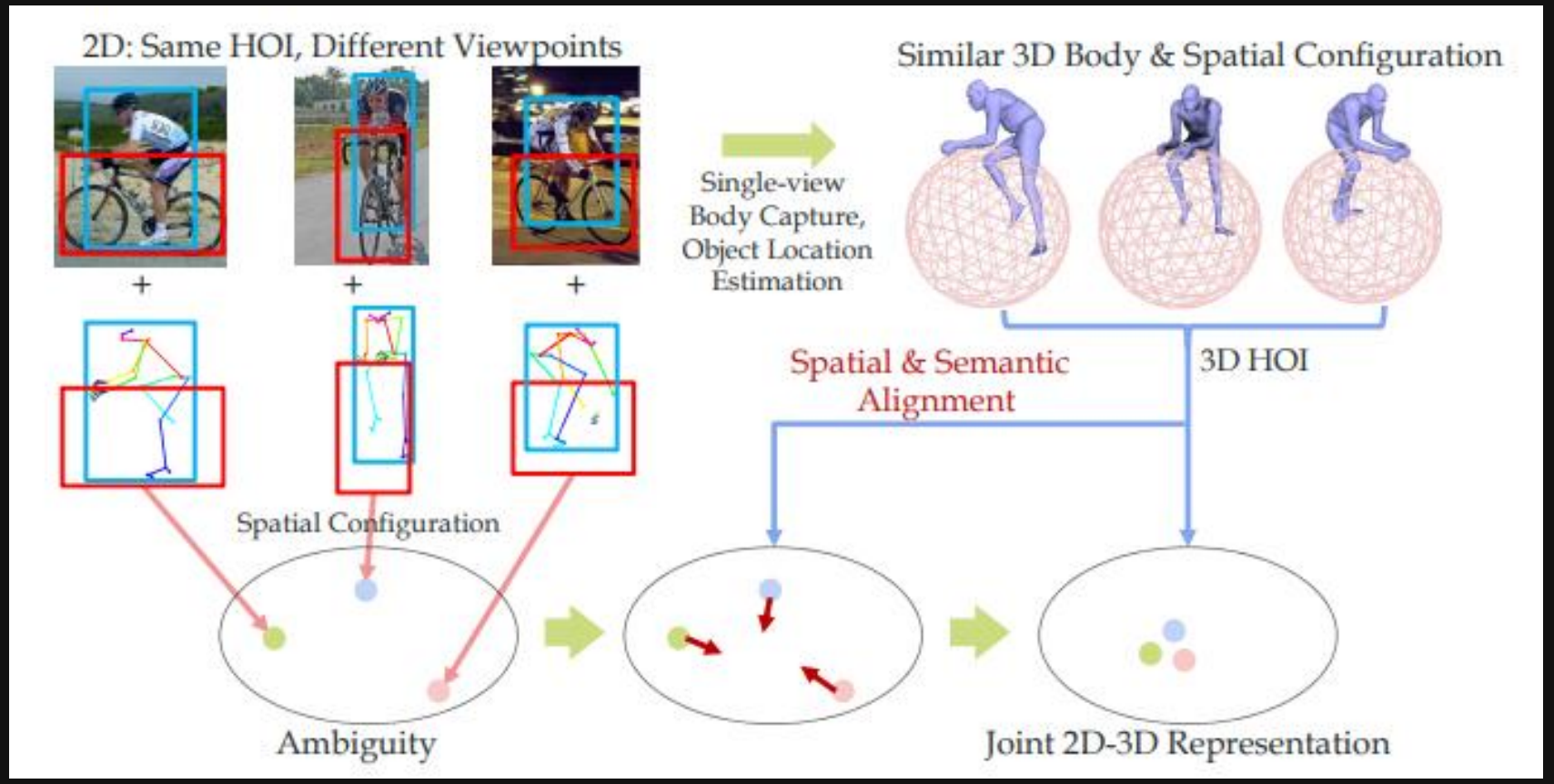
Reconstruct 3D scene using HOI as a prior



Generate 3D/4D HOI and HSI

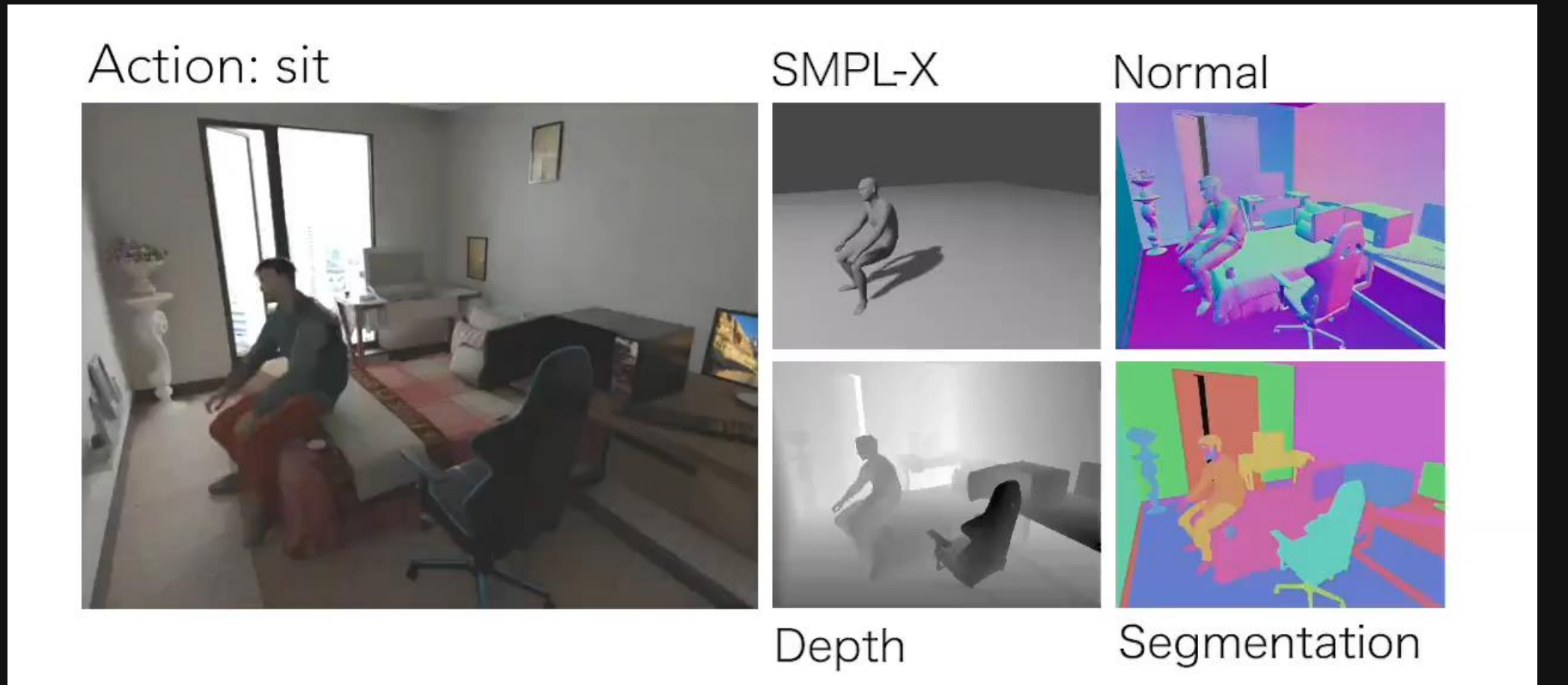
From 2D to 3D

Intuition: Relationship in 3D is much simpler than its projection on 2D



From 2D to 3D

Intuition: 3D information can be easily projected to 2D

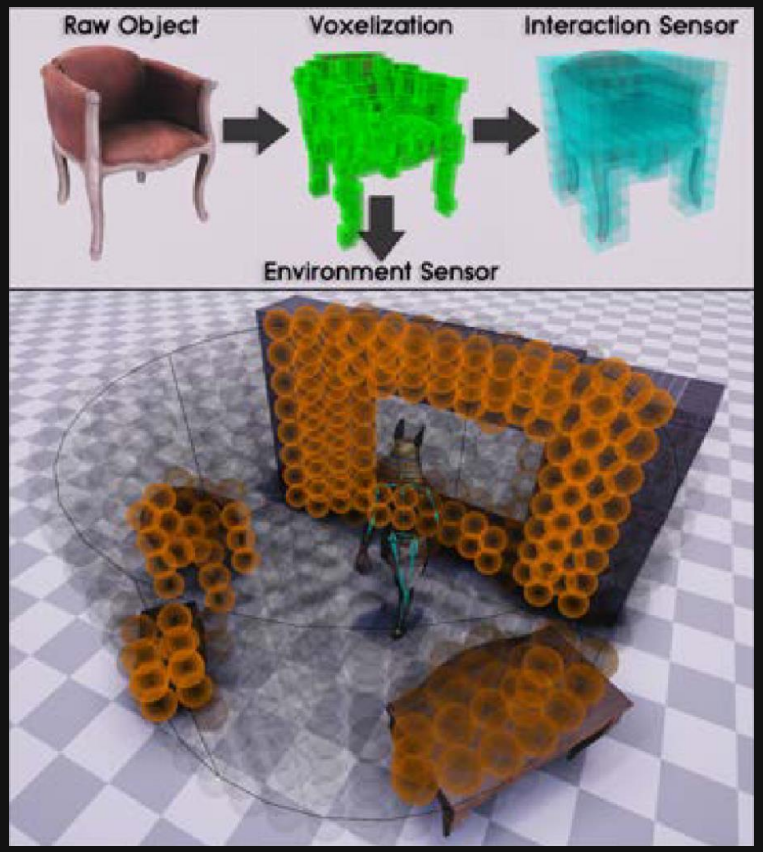


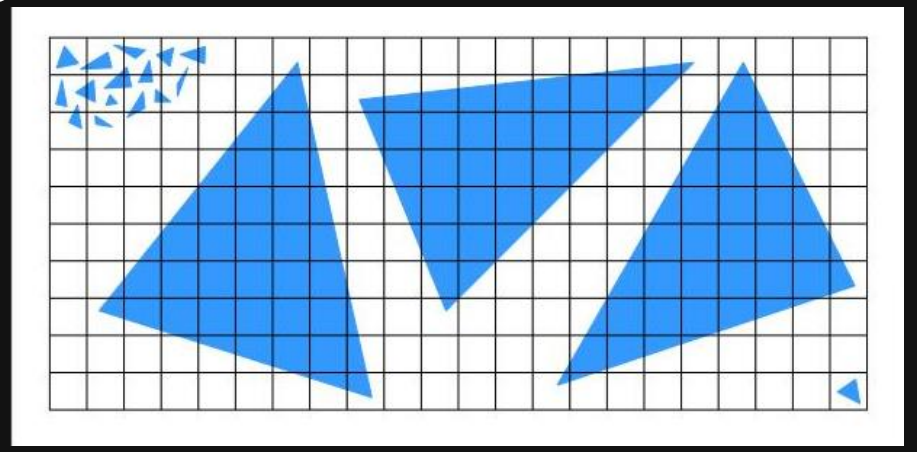
Challenge in 3D:
How to properly represent 3D objects and humans?



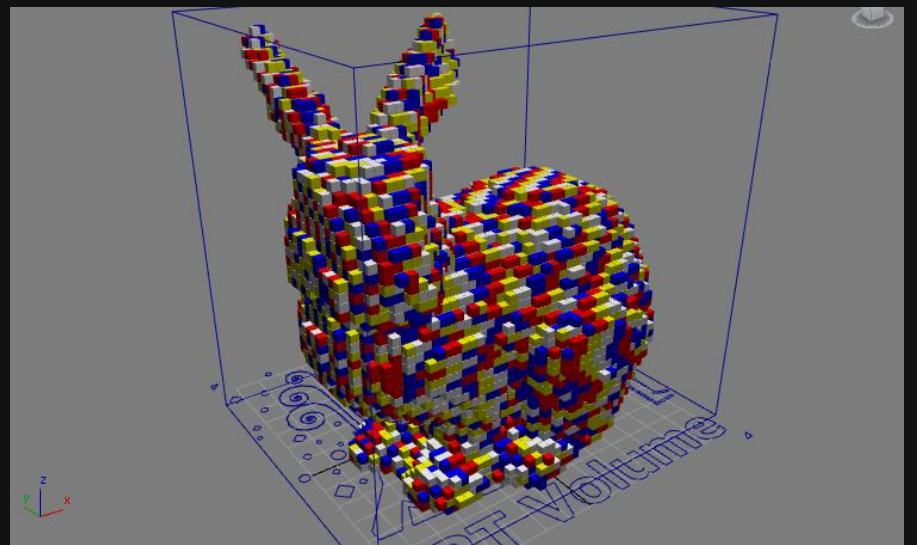
Representation of Object / Scene

- Common approach: Voxel grid





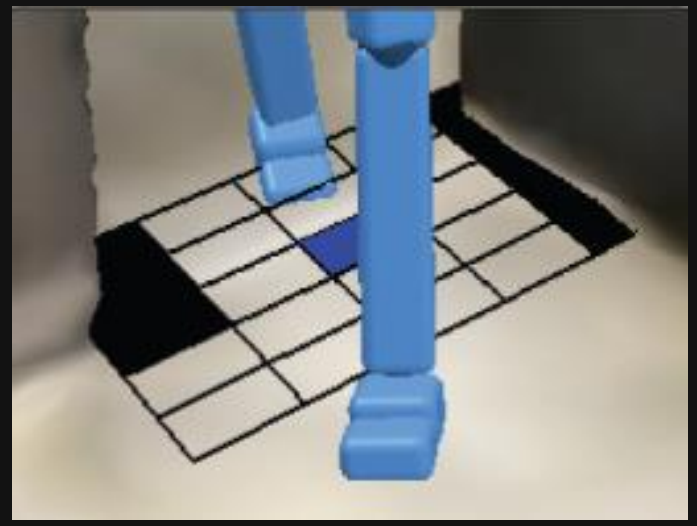
2D voxel grid

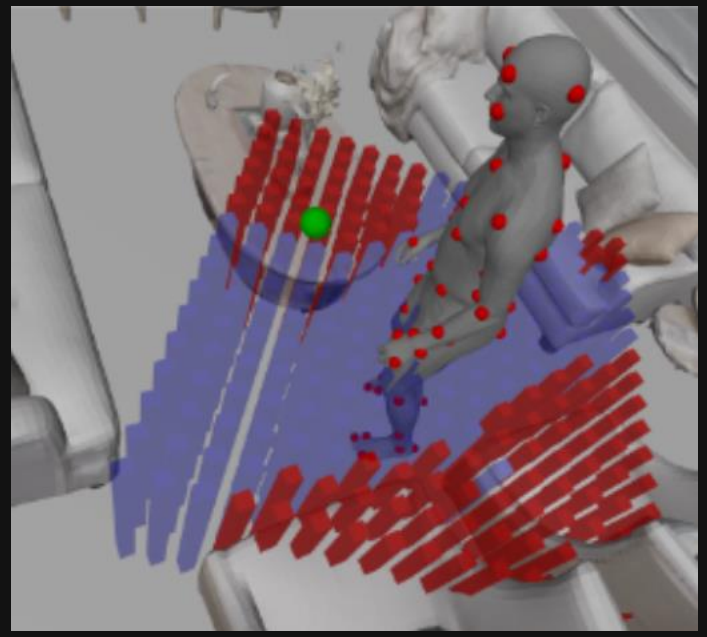


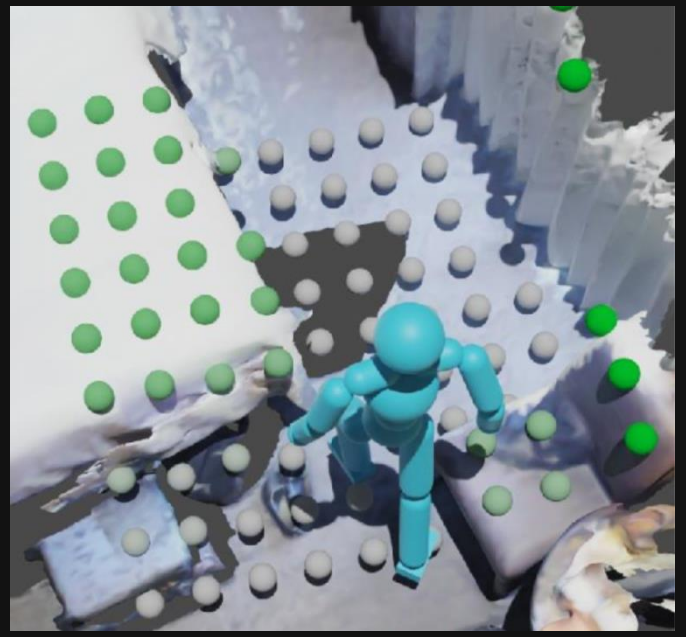
3D voxel grid

Representation of Object / Scene

Voxel grid can be also used to measure how much the space is occupied







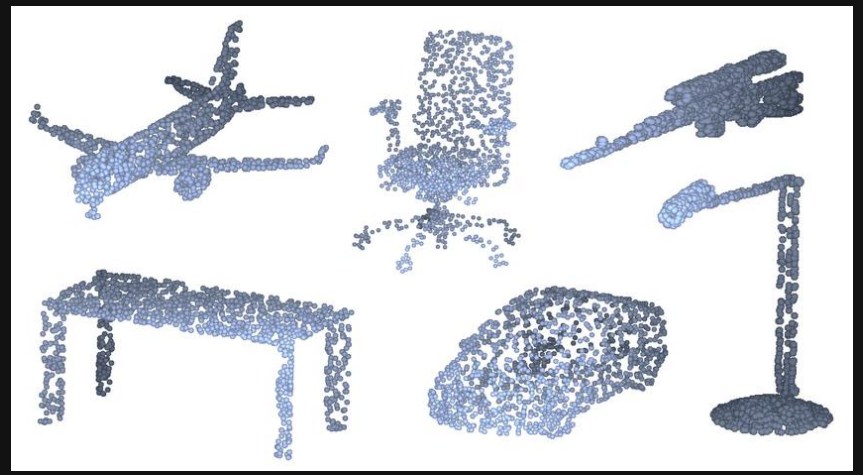
Pros and Cons of Voxel

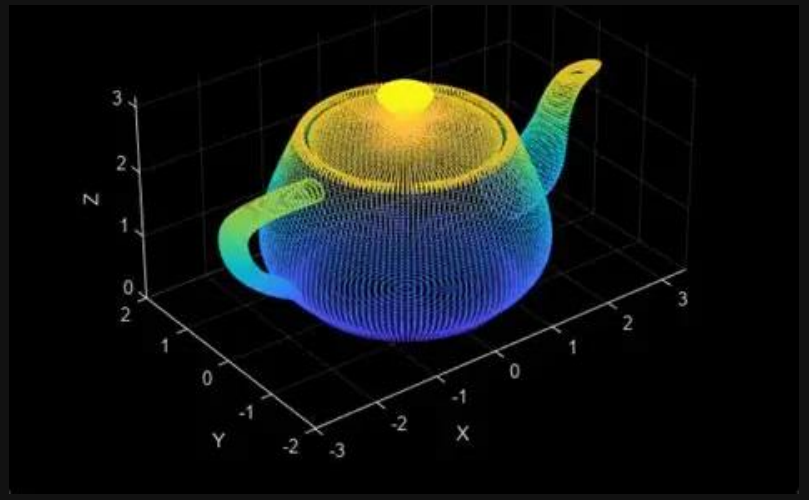
	Pros	Cons
Structure	• Regular 3D grid layout • Clear spatial relationships • Simple indexing	• Memory grows $O(n^3)$ with resolution • Inefficient for sparse data
Deep Learning	• Direct 3D CNN compatibility • Mature architectures available	• Limited resolution due to memory • Training becomes expensive
Processing	• Fast spatial queries O(1) • Efficient collision detection • GPU-friendly operations	• High memory bandwidth requirements • Redundant computation in empty regions
Representation	• Complete volume information • Good for interior structures • Easy boolean operations	• Quantization artifacts • Poor surface approximation • Fixed resolution limits detail
Applications	• Volume rendering • Medical imaging • Fluid simulation • Shape reconstruction	• Unsuitable for large scenes • Poor for thin structures • Limited real-time performance

Representation of Object / Scene

- Common approach: Point cloud







Point cloud is a set of 3D points

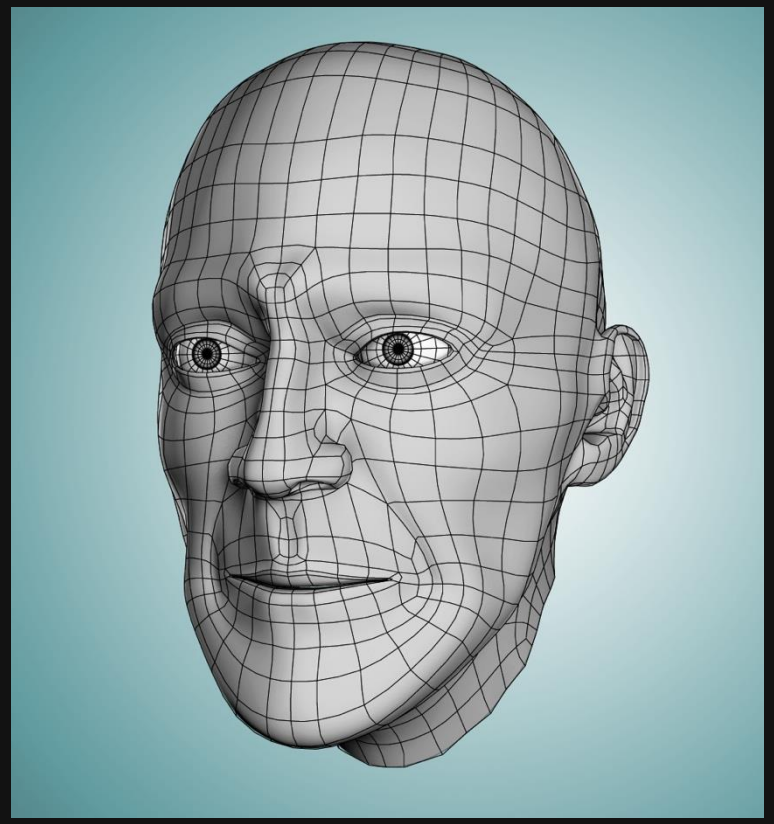


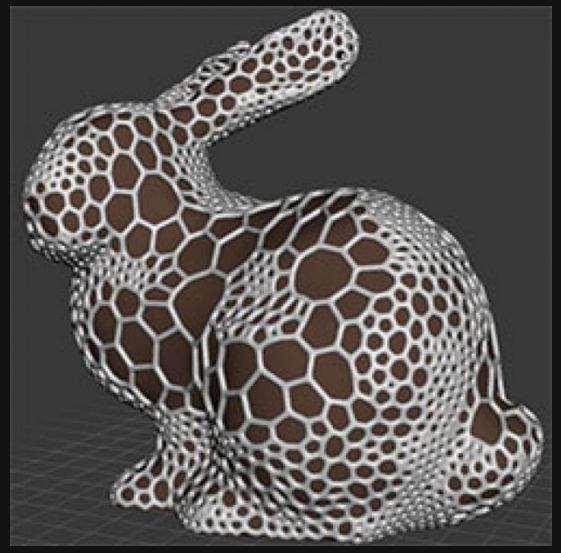
Pros and Cons of Point Clouds

	Pros	Cons
Structure	• Memory efficient for sparse data • Flexible point density • Direct sensor data capture	• Irregular topology structure • No fixed neighborhood relationships • Complex feature extraction
Deep Learning	• High fidelity representation • No discretization loss • Preserves original detail	• Requires specialized architectures (PointNet, PointNet++) • Permutation invariance challenges
Processing	• Scalable to large scenes • Adaptive resolution • Efficient for sparse geometry	• Expensive spatial queries • Requires auxiliary data structures (KD-tree/octree)
Representation	• Accurate surface modeling • Raw measurement preservation	• Limited volume information • Poor interior structure representation • Difficult boolean operations
Applications	• LiDAR scanning • 3D object detection • Surface reconstruction • Large-scale mapping	• Unsuitable for volume rendering • Complex collision detection • Limited fluid simulation support • No natural voxel-based operations

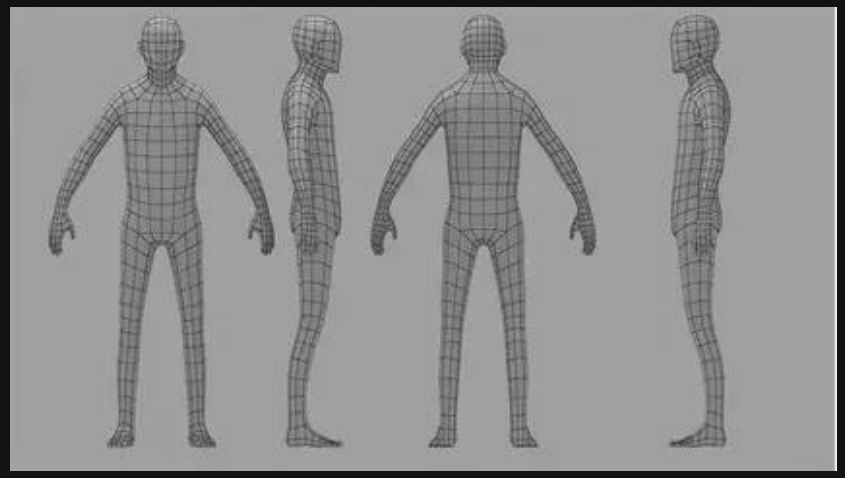
Representation of Object / Scene

- Common approach: Mesh





Mesh is a set of vertices and edges

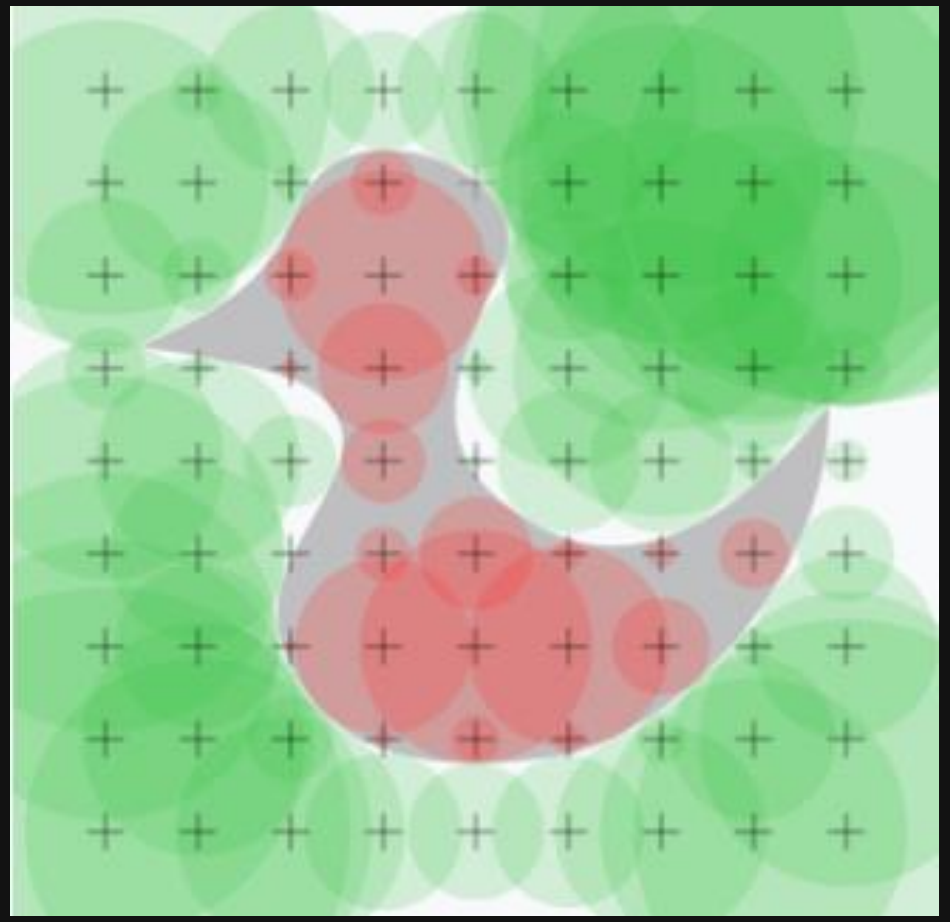


Pros and Cons of Mesh

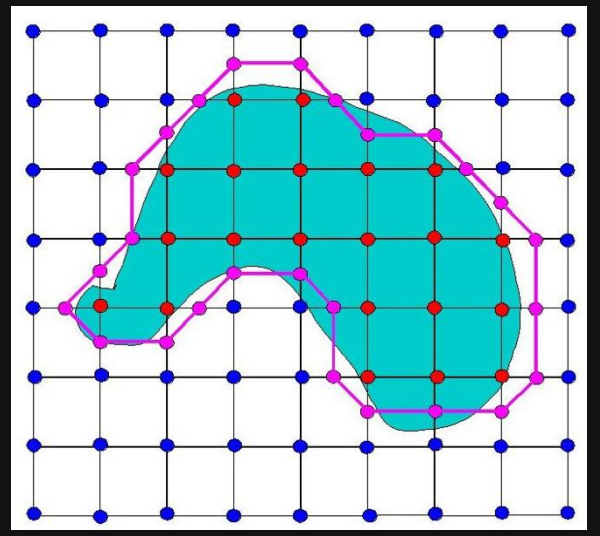
	Pros	Cons
Structure	• Efficient surface representation • Explicit vertex-edge-face relations • Compact storage for surfaces	• Complex topology maintenance • Irregular sampling distribution • Topology degeneracy issues
Deep Learning	• Rich geometric information • Continuous surface representation • Smooth interpolation capability	• Requires specialized Graph Neural Networks • No standard convolution operations • Complex irregular structure handling
Processing	• Mature ecosystem of algorithms • Efficient rendering pipelines • Advanced geometric operations	• Expensive boolean operations • Topology-sensitive transformations • Mesh quality maintenance overhead
Representation	• High-quality surface modeling • Adaptive resolution capability • Excellent visual fidelity	• Surface-only representation • No interior volume information • Limited for volumetric analysis
Applications	• Computer graphics rendering • CAD and 3D modeling • Animation and deformation • Surface analysis	• Unsuitable for volume-based simulation • Limited medical imaging applications

Representation of Object / Scene

- Common approach: Signed-Distance Field (SDF)



Compute the signed distance to the object surface for each location



Reconstructing object from SDF using marching cube

Pros and Cons of SDF

	Pros	Cons
Structure	• Continuous and differentiable • Topology-agnostic representation • Water-tight surface guarantee	• Indirect geometric access • Query-dependent evaluation overhead
Deep Learning	• Neural network friendly • MLP-based shape encoding • Gradient-based optimization • Implicit generative modeling	• Positive/negative point sampling needed • Requires dense training samples
Processing	• Smooth surface extraction • High-quality Marching Cubes • Robust boolean operations	• Expensive dense queries (e.g., marching cube) • Function evaluation bottleneck • Limited real-time performance
Representation	• Handles complex topology • Compact implicit encoding	• Geometry-only information • No texture/material encoding • Limited appearance modeling
Applications	• Neural shape synthesis • Shape optimization • 3D reconstruction • Generative modeling	• Poor for textured models • Limited graphics pipeline integration

Comparisons in one-sentence

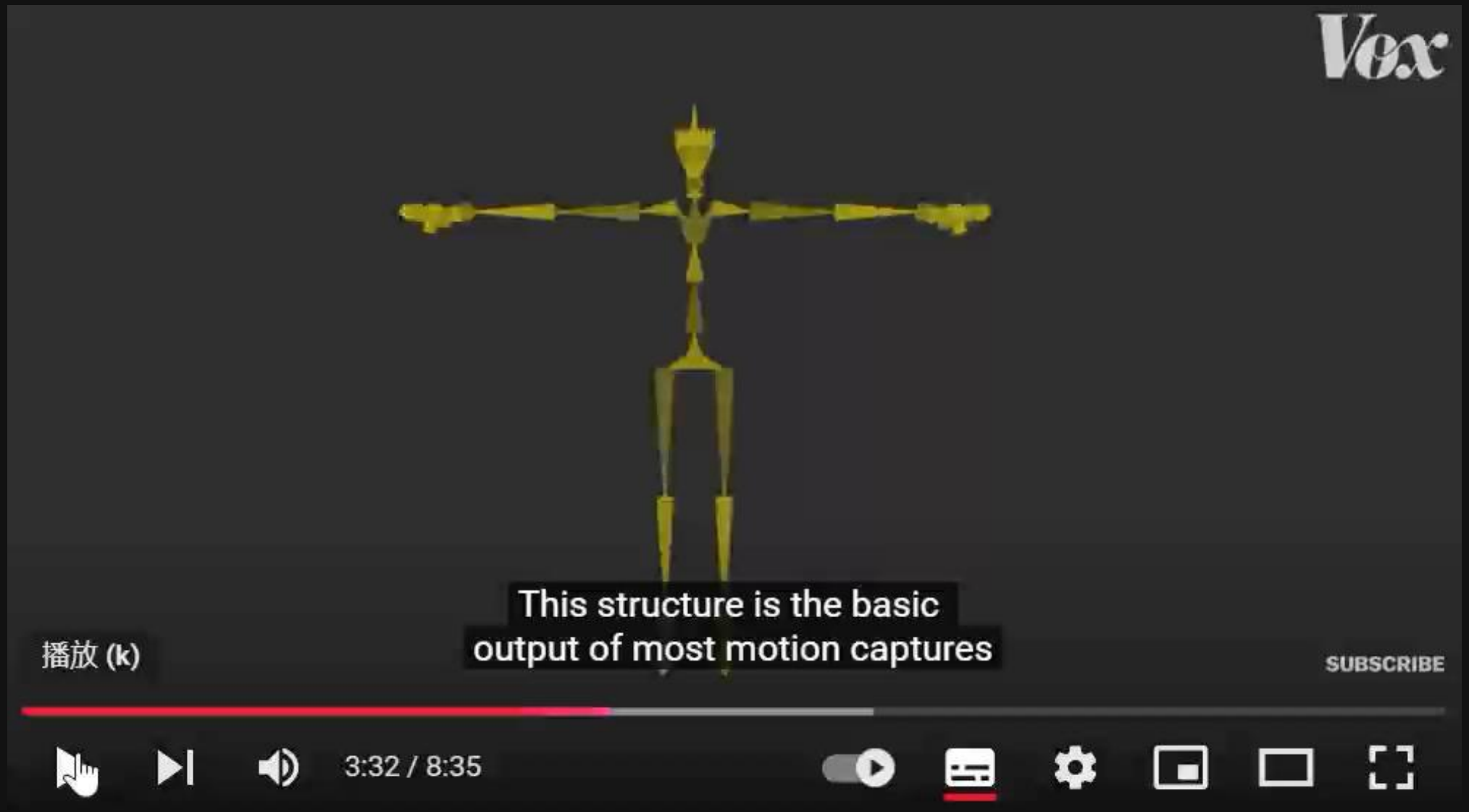
- Voxel Grid: Regular structure enables seamless 3D CNN processing and straightforward volumetric operations.
- Point Cloud: Raw sensor data with minimal processing overhead, ideal for sparse 3D scenes and direct LiDAR integration.
- Mesh: Industry-standard representation with mature graphics pipelines and exceptional rendering quality.
- SDF: Differentiable continuous fields that naturally handle topology changes and enable gradient-based optimization.

Representation	Differentiability	Memory Efficiency	Surface Quality	Volume Representation	Topology Handling	Neural Network Support	Continuous	Watertight
Point Cloud	No	High (sparse)	Discrete points	No	N/A	Specialized (PointNet)	No	No
Voxel Grid	No	Low (dense)	Blocky/aliased	Yes	Fixed grid	Standard 3D CNN	No	Yes
Mesh	Limited	Medium	High quality	No	Complex/fragile	Graph networks	Piecewise	No
SDF	Yes	Variable	Smooth/continuous	Yes	Flexible/robust	MLP-friendly	Yes	Yes

Representation	Real-time Rendering	Data Acquisition	Processing Complexity	Multi-resolution	Boolean Operations	Texture Support	Storage Format	Precision
Point Cloud	Fast	Direct (sensors)	Medium	Point density	Difficult	Yes	Efficient	Sensor-limited
Voxel Grid	Fast	Converted	Low	Fixed resolution	Easy	Yes	Space-intensive	Resolution-limited
Mesh	Fast	Reconstructed	High	Adaptive mesh	Complex	Yes	Compact	High
SDF	Slow (ray marching)	Learned/converted	Medium	Query-dependent	Natural	Requires extension	Function parameters	Function-limited

Representation of Human Pose / Motion

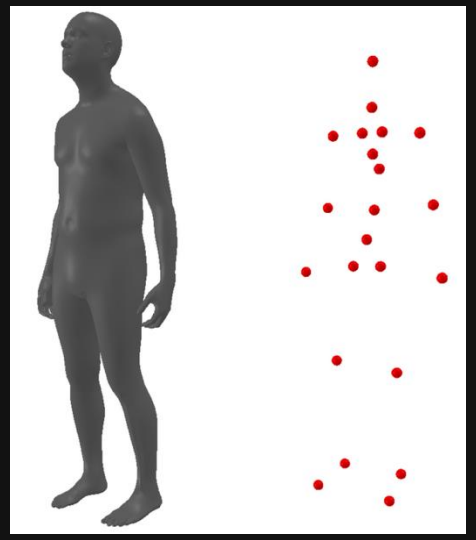
- Human skeletons



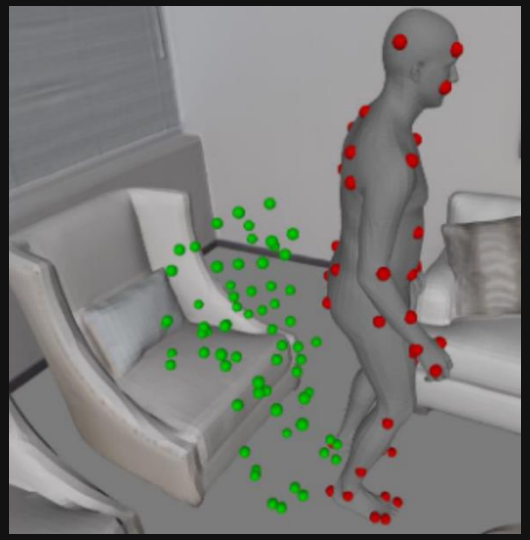
Representation of Human Pose / Motion

- SMPL parameters (for each frame):
  - Root global translation and orientation of human body
  - Local joint rotation

- Human 3D joints



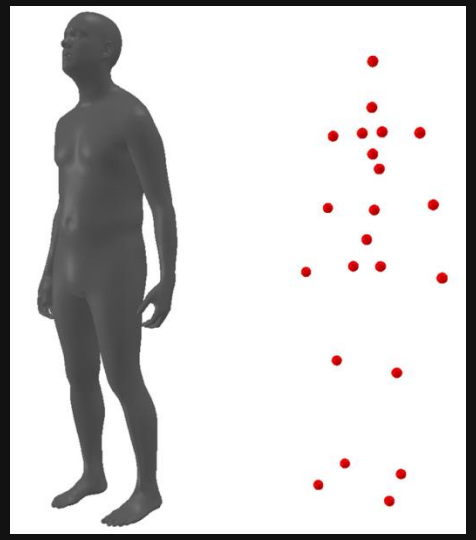
- Body surface markers



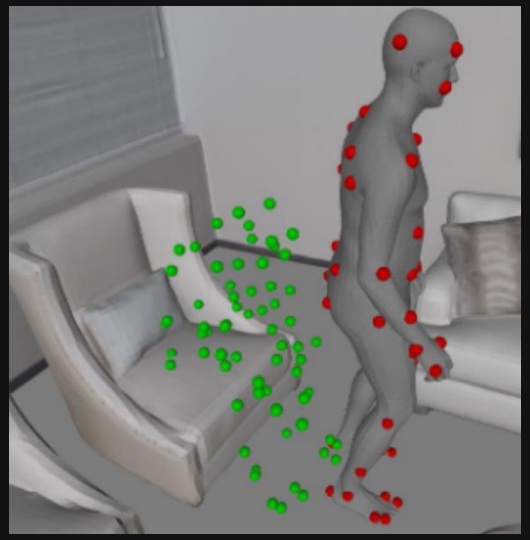
Representation of Human Pose / Motion

- SMPL parameters (for each frame):
  - Root global translation and orientation of human body
  - Local joint rotation

- Human 3D joints



- Body surface markers



Note: These representations can be converted to each other.

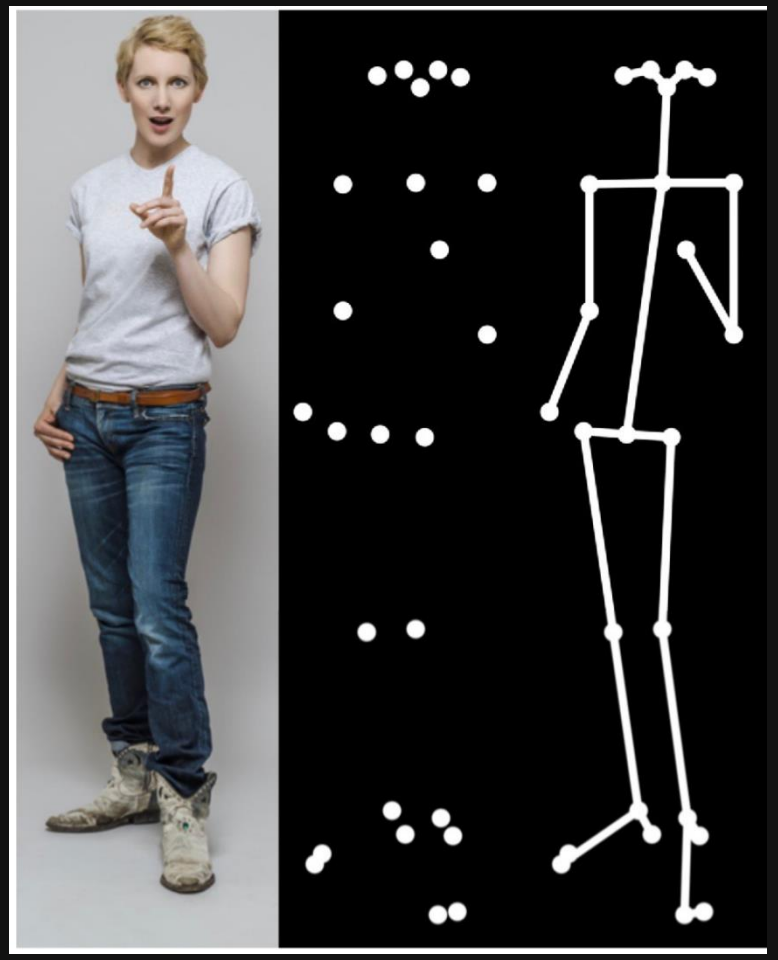
SMPL: Modern Way of Human Body Modeling

Before SMPL …



SMPL: Modern Way of Human Body Modeling

How to model a human body?



Essentials

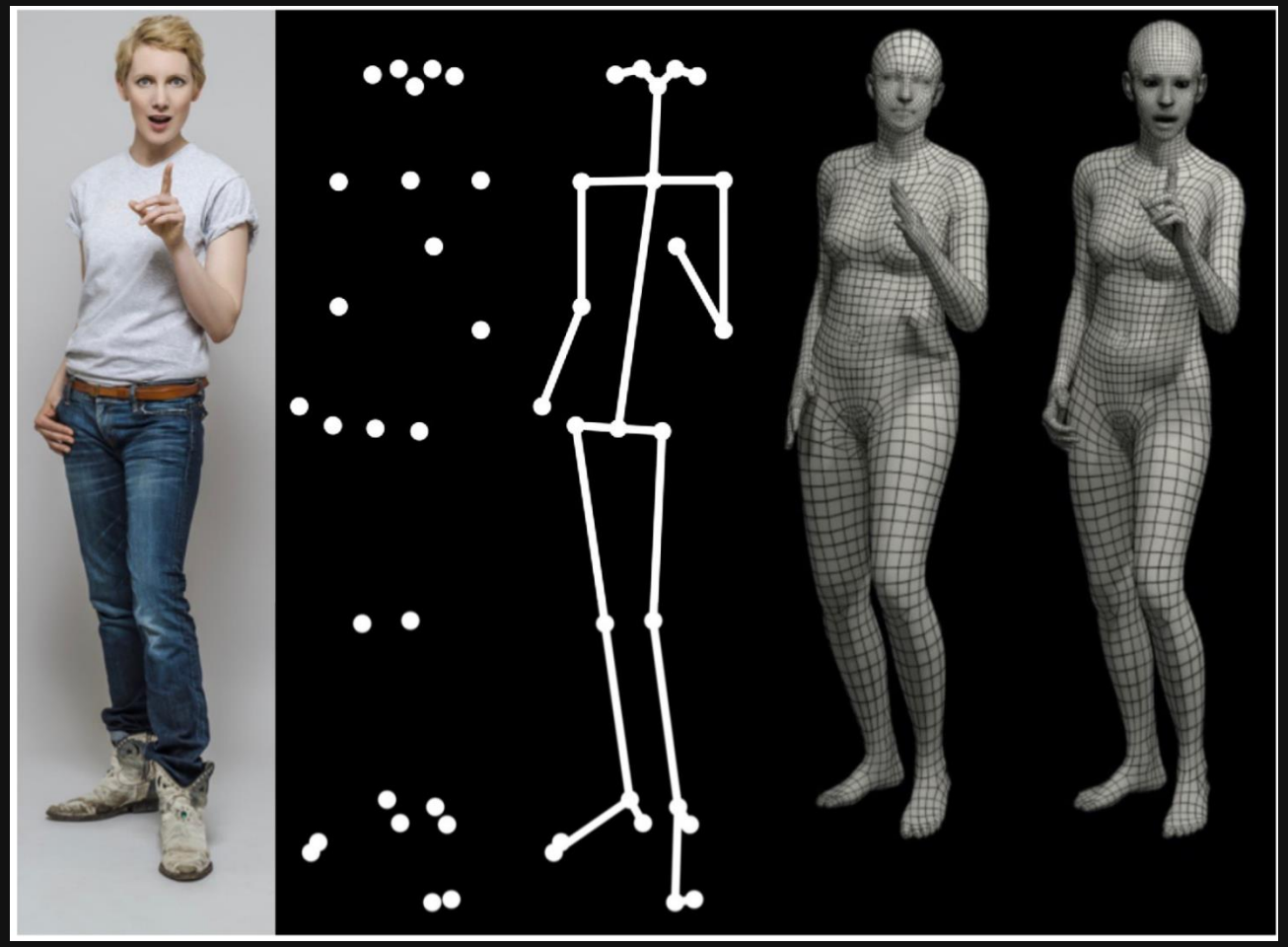
Skeleton

Body Pose

Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A. A., Tzionas, D., & Black, M. J. (2019). Expressive body capture: 3d hands, face, and body from a single image. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 10975-10985).

What is SMPL

How to model a human body expressively?



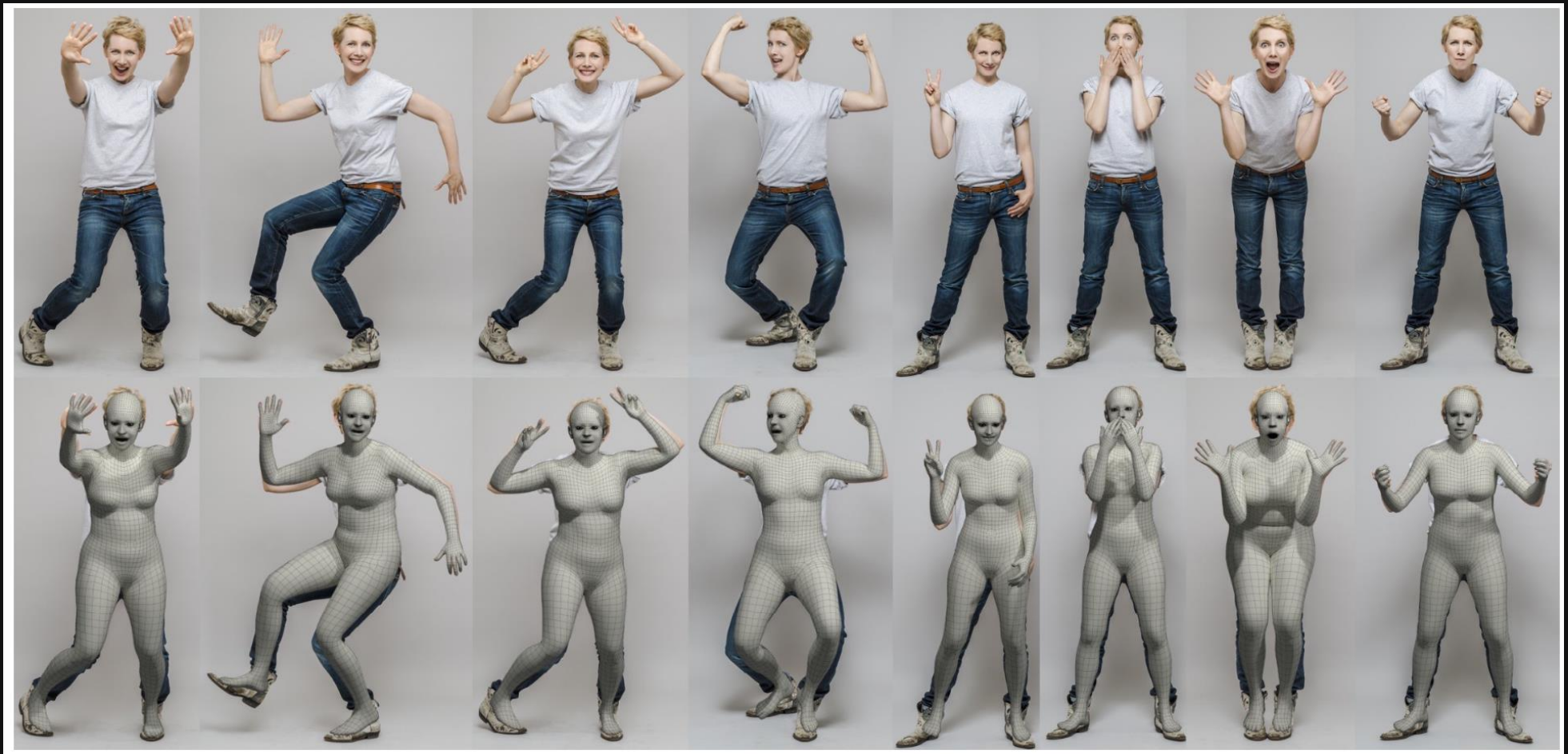
Essentials
Skeleton
Body Pose
Hand Pose
Skin

3D triangular mesh
6890 vertices for body
778 vertices for hand

Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A. A., Tzionas, D., & Black, M. J. (2019). Expressive body capture: 3d hands, face, and body from a single image. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 10975-10985).

What is SMPL

How to model a human body expressively?



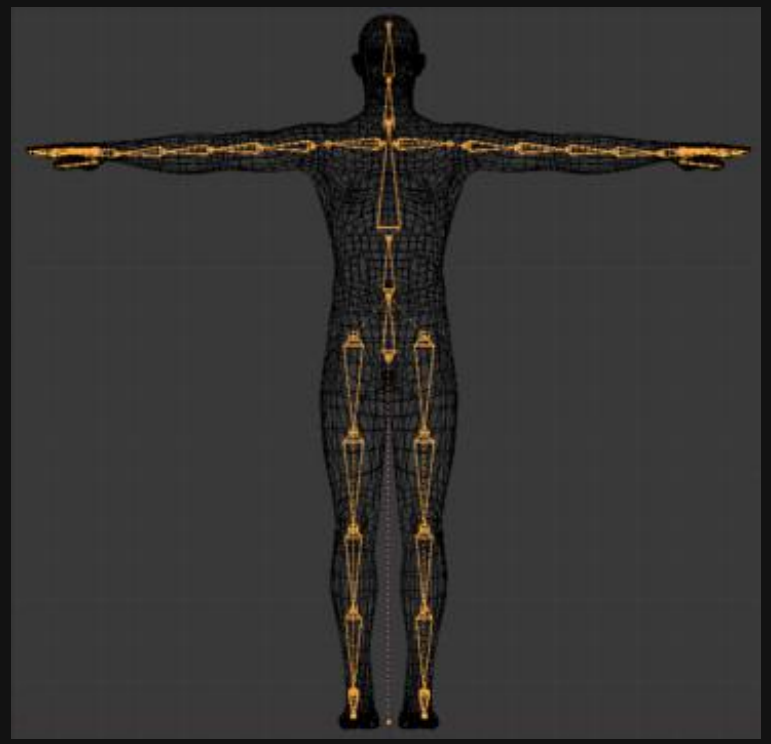
Essentials

Skeleton
Pose
Skin

3D triangular mesh
6890 vertices for body
778 vertices for hand
23 joint + 15 joints

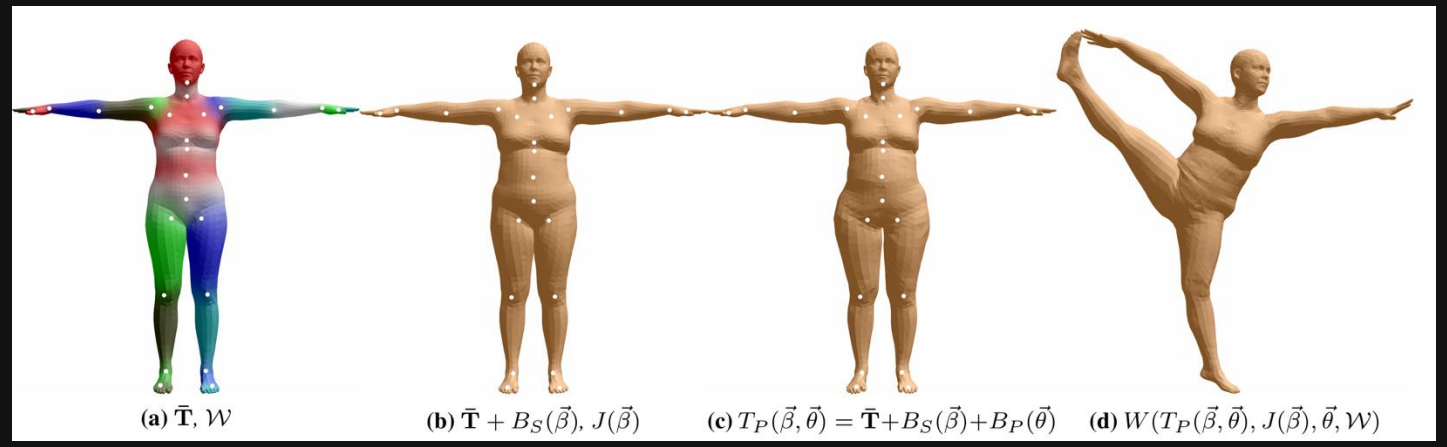
Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A. A., Tzionas, D., & Black, M. J. (2019). Expressive body capture: 3d hands, face, and body from a single image. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 10975-10985).

Preliminaries



Human model = Bones + Skin

- Bones = K x 3 vector, represents pose of the model
- Skin = 3D mesh
- We need to find a mapping between bones and skin



SMPL: A Skinned Multi-Person Linear Model

What is SMPL

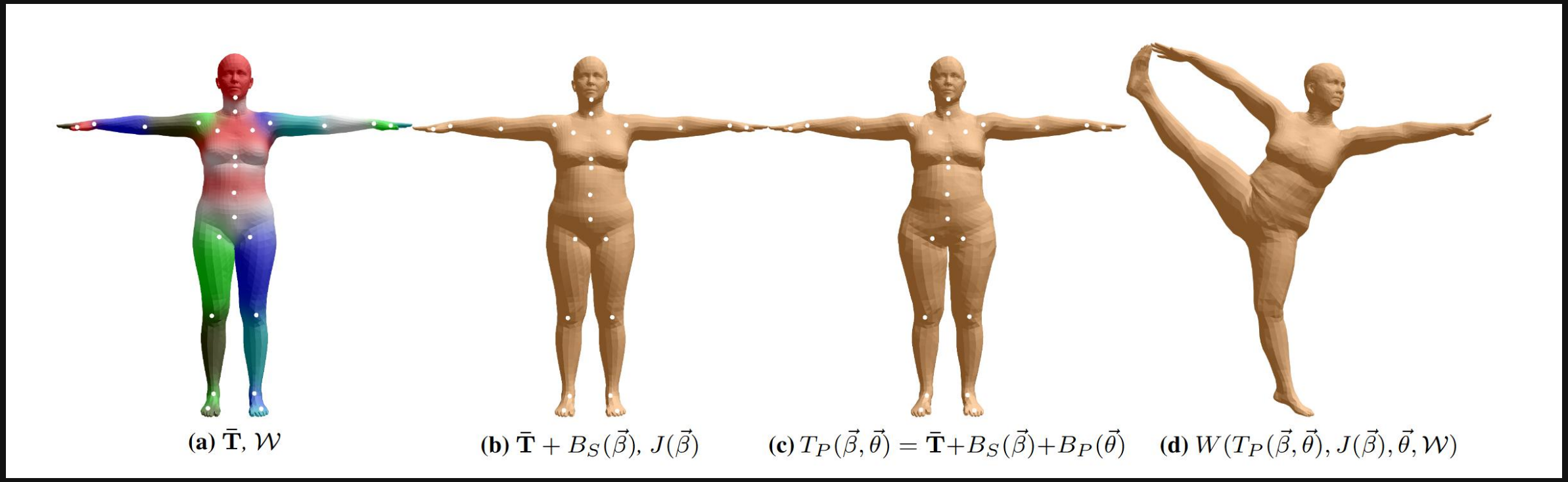
Dataset	Hours	MoCap	Human Rep.
GTA_IM [4]	9.3		skeleton
PiGraphs [37]	2.0		skeleton
PROX [16]	0.9		SMPL-X
GRAB [43]	3.8	✓	SMPL-X
SAMP [17]	1.7	✓	SMPL-X
RICH [20]	0.8		SMPL-X
BEHAVE [2]	4.2		SMPL
CHAIRS [22]	17.3	✓	SMPL-X
COUCH [59]	3.0	✓	SMPL
iReplica [15]	0.8	✓	SMPL
CIRCLE [1]	10.0	✓	SMPL-X
TRUMANS	15.0	✓	SMPL-X

What is SMPL

How to model a human body with shape and pose?

Essentials

Skeleton + Pose + Skin



Loper, M., Mahmood, N., Romero, J., Pons-Moll, G., & Black, M. J. (2023). SMPL: A skinned multi-person linear model. In Seminal Graphics Papers: Pushing the Boundaries, Volume 2 (pp. 851-866).

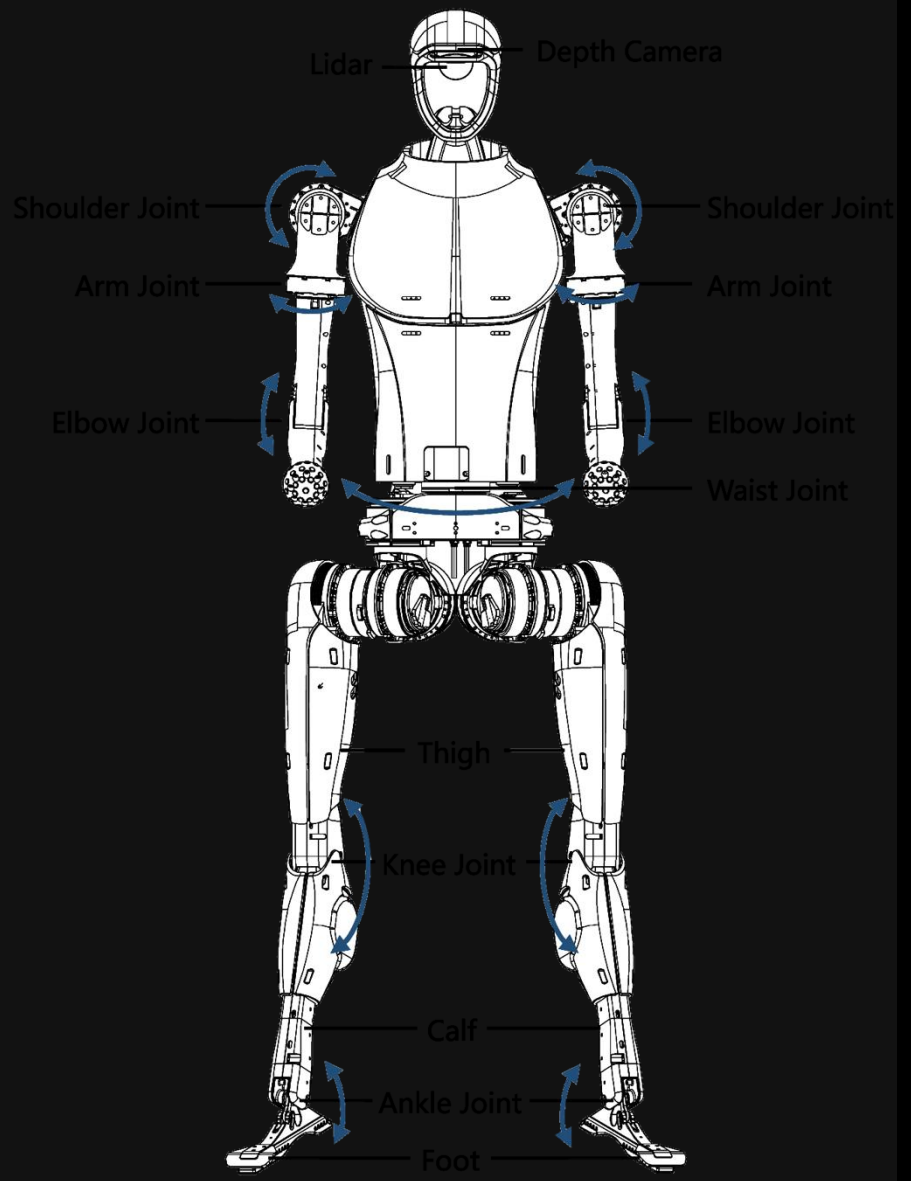
What is SMPL

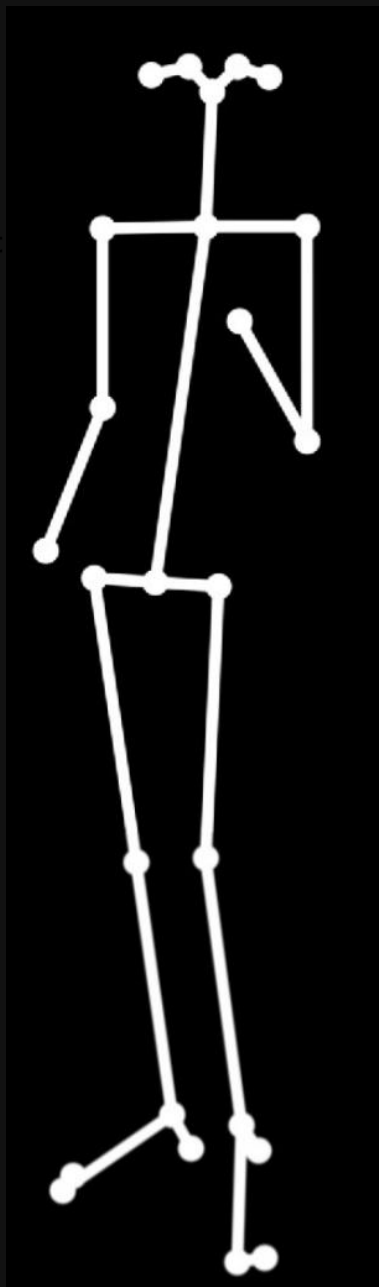
To represent K joint locations:

- $T_i^j \in \mathrm{SE}(3)$ represents the transformation between two linked joints i, j, with 3DoF in rotation ($\omega_k \in \mathbb{R}^3$) and fixed link position j_j (corresponding to the bone)
- The pose parameter: $\theta \in \mathbb{R}^{3K+3}$
- All joint locations $\mathrm{J} \in \mathbb{R}^{3K}$

The forward kinematics:

$$G_k(\vec{\theta}, \mathbf{J}) = \prod_{j \in A(k)} \left[\begin{array}{c|c} \exp(\vec{\omega}_j) & \mathbf{j}_j \\ \hline \vec{0} & 1 \end{array} \right]$$



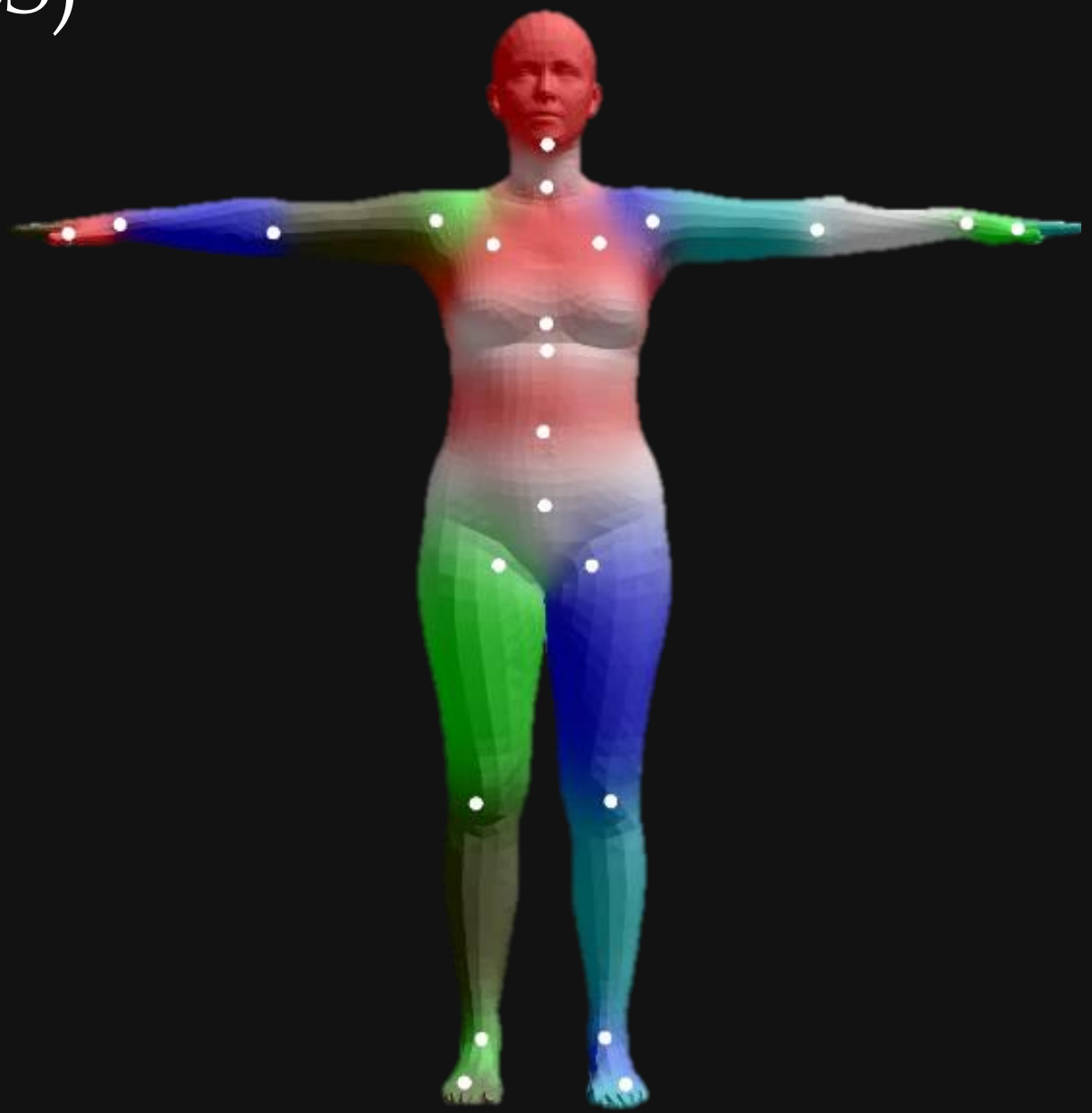


Linear Blend Skinning (LBS)

A skinning template

- N vertices $\overline{\mathrm{T}} \in \mathbb{R}^{3N}$, with faces (→ triangular mesh)
- Linear blending weight matrix $\mathcal{W} \in \mathbb{R}^{N \times K}$ describes how the location of each joint contributes to the location of each vertex
- Under the template, the vertex i has the position:

$$\bar{\mathbf{t}}_i = \sum_{i=1}^{K} w_{k,i} \mathbf{J}_k$$



Linear Blend Skinning (LBS)

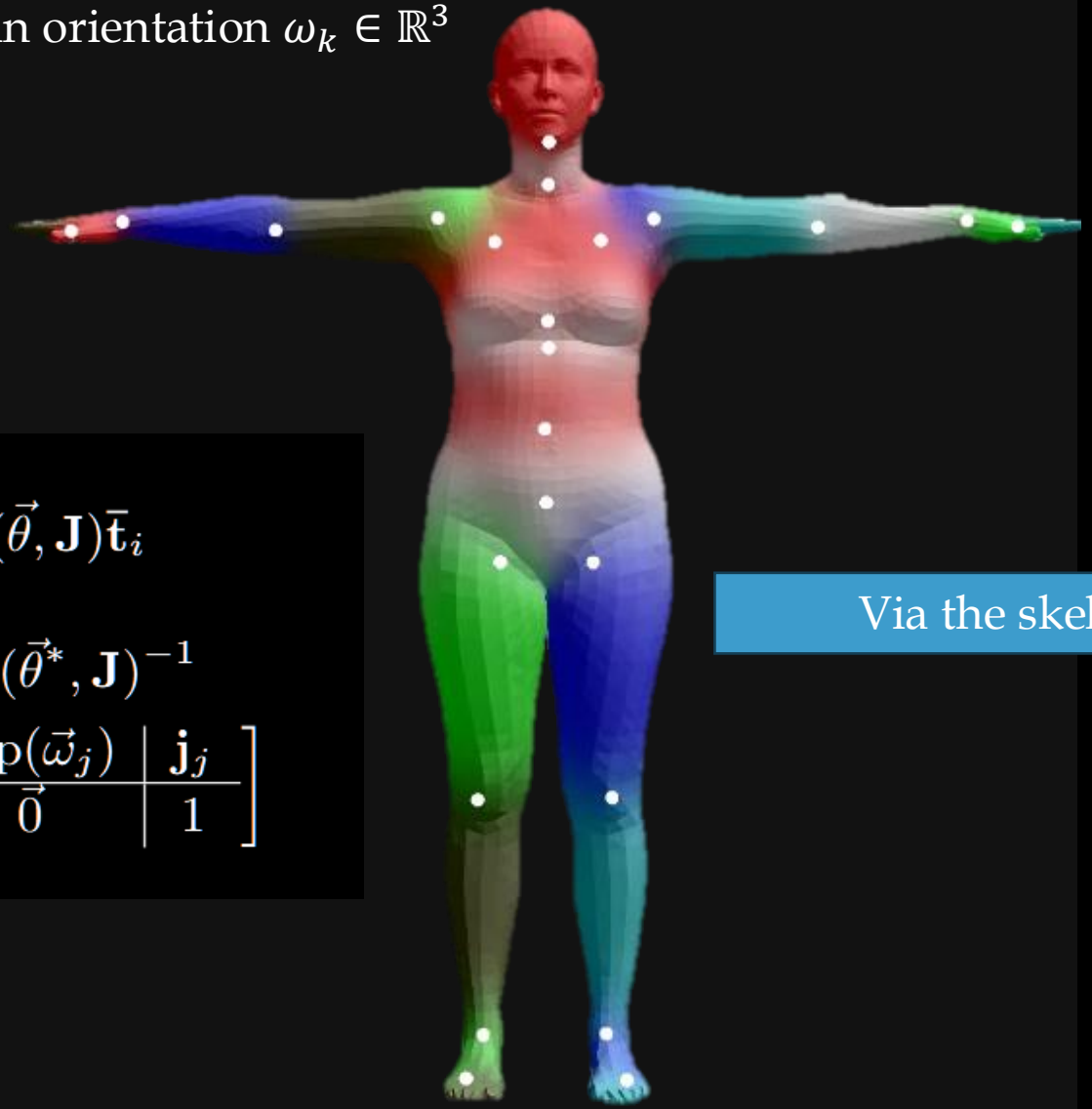
Deforming the template with the pose

- Current pose θ, with joint k in orientation $\omega_k \in \mathbb{R}^3$
- All joint locations $J \in \mathbb{R}^{3K}$

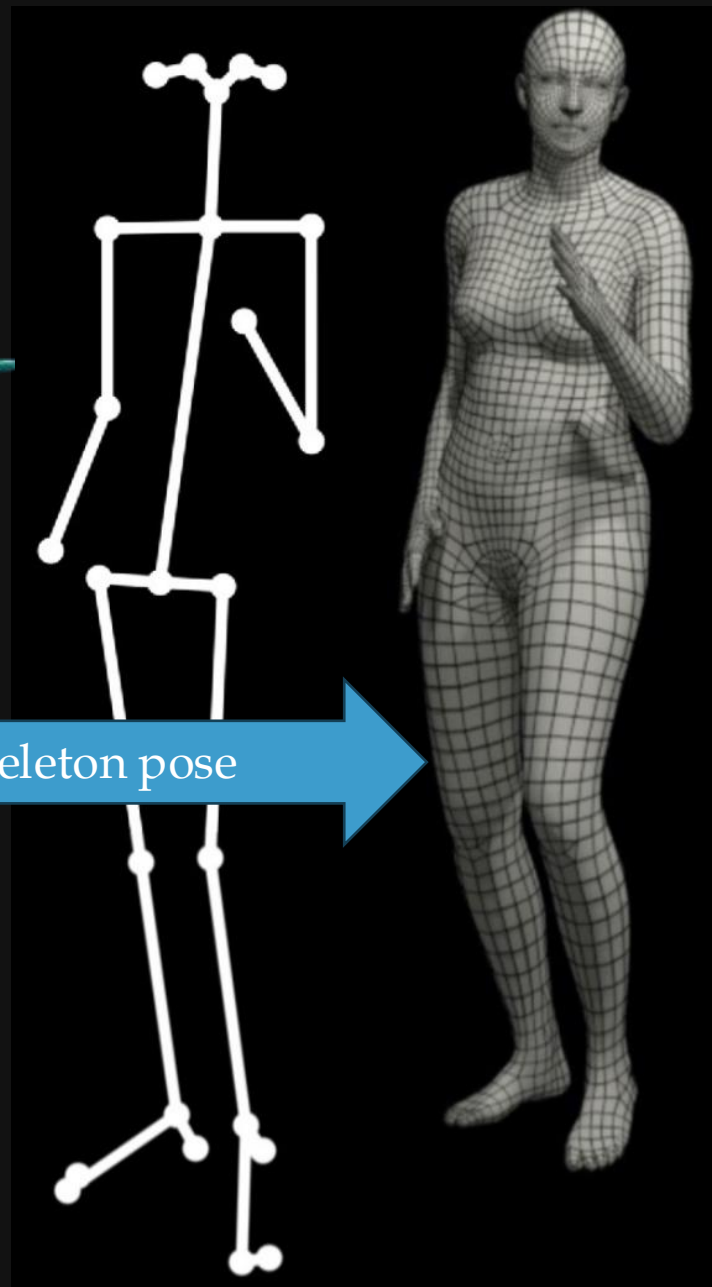
$$\bar{\mathbf{t}}'_i = \sum_{k=1}^{K} w_{k,i} G'_k(\vec{\theta}, \mathbf{J}) \bar{\mathbf{t}}_i$$

$$G'_k(\vec{\theta}, \mathbf{J}) = G_k(\vec{\theta}, \mathbf{J}) G_k(\vec{\theta}^*, \mathbf{J})^{-1}$$

$$G_k(\vec{\theta}, \mathbf{J}) = \prod_{j \in A(k)} \left[\begin{array}{c|c} \exp(\vec{\omega}_j) & \mathbf{j}_j \\ \hline \vec{0} & 1 \end{array} \right]$$



Via the skeleton pose



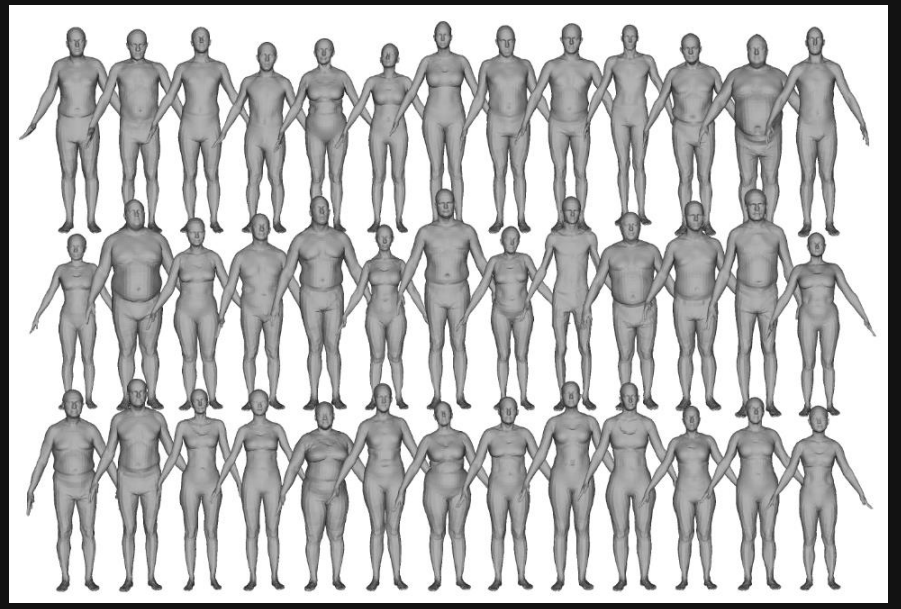
Personalized Body Shape

- Observations: Different people have different
  - Skeleton (shapes and joints)
  - Skin (shapes)
  - Skin-skeleton relations

$$\bar{\mathbf{t}}'_i = \sum_{k=1}^{K} w_{k,i} G'_k(\vec{\theta}, \mathbf{J}) \bar{\mathbf{t}}_i$$

$$G'_k(\vec{\theta}, \mathbf{J}) = G_k(\vec{\theta}, \mathbf{J}) G_k(\vec{\theta}^*, \mathbf{J})^{-1}$$

$$G_k(\vec{\theta}, \mathbf{J}) = \prod_{j \in A(k)} \begin{bmatrix} \exp(\vec{\omega}_j) & \mathbf{j}_j \\ \hline \mathbf{0} & 1 \end{bmatrix}$$

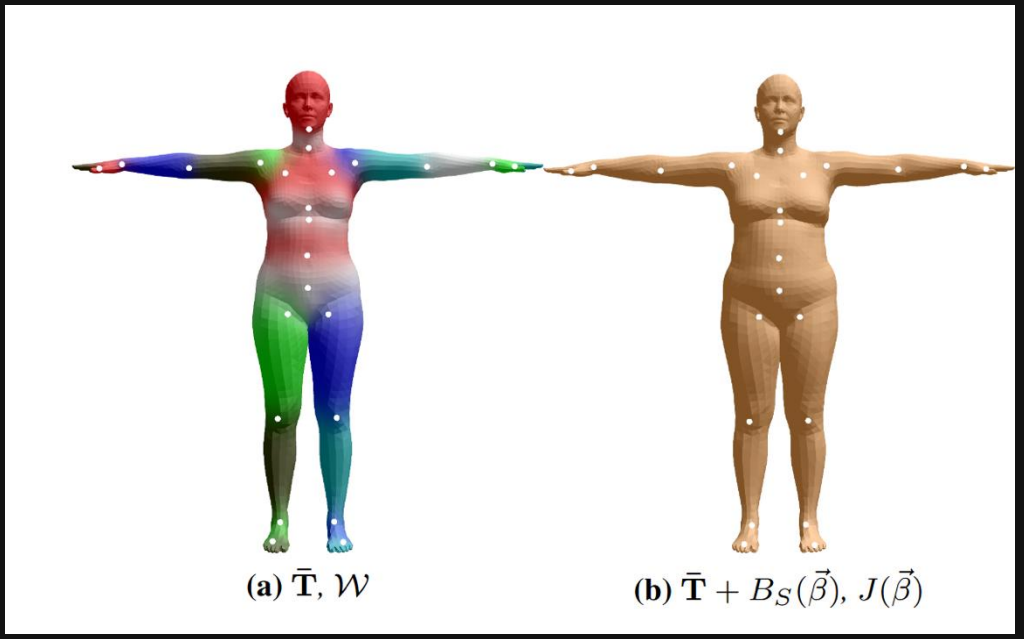


Personalized Body Shape

Personalized parameters β

Shape blending function $B_S(\beta)$

$$M(\vec{\beta},\vec{\theta}) = W(T_P(\vec{\beta},\vec{\theta}), J(\vec{\beta}), \vec{\theta}, \mathcal{W})$$
$$T_P(\vec{\beta},\vec{\theta}) = \bar{\mathbf{T}} + B_S(\vec{\beta})$$

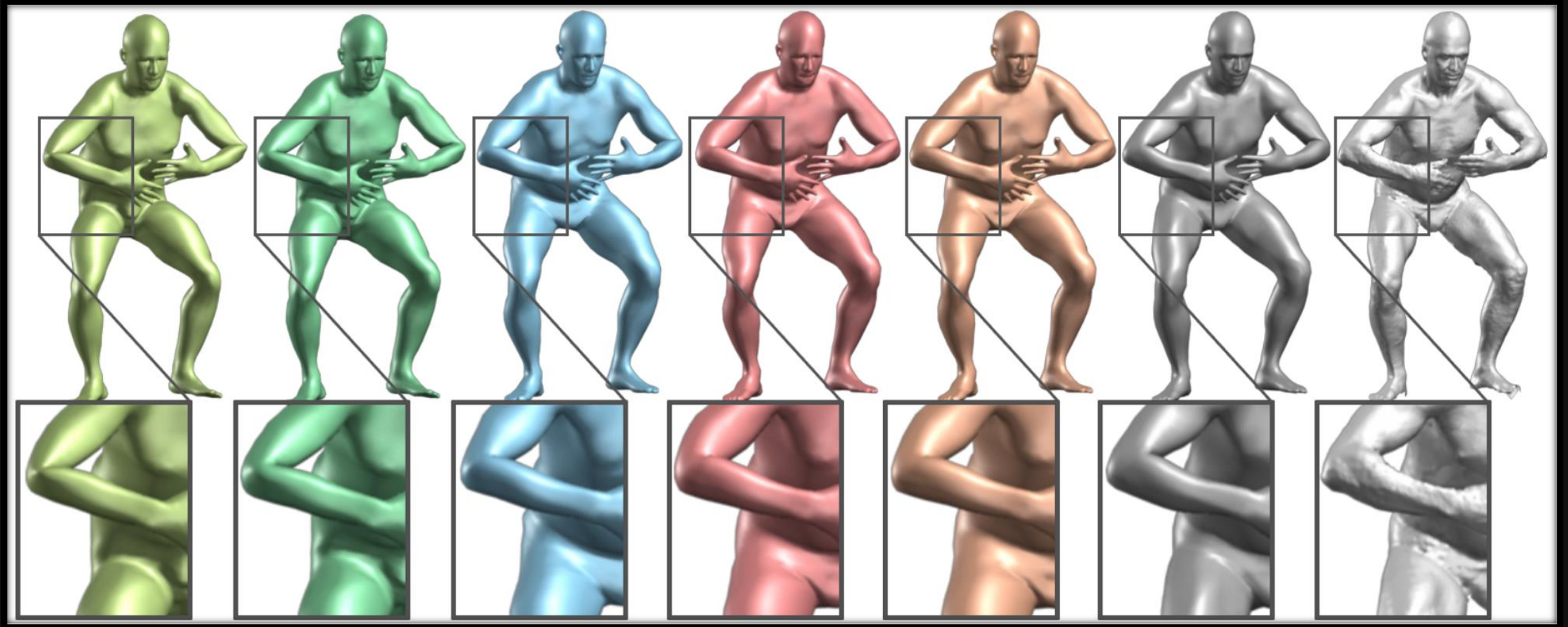


(a) $\bar{\mathbf{T}}$, $\mathcal{W}$ (b) $\bar{\mathbf{T}} + B_S(\vec{\beta})$, $J(\vec{\beta})$

More Natural Shapes

Personalized parameters β + body pose θ

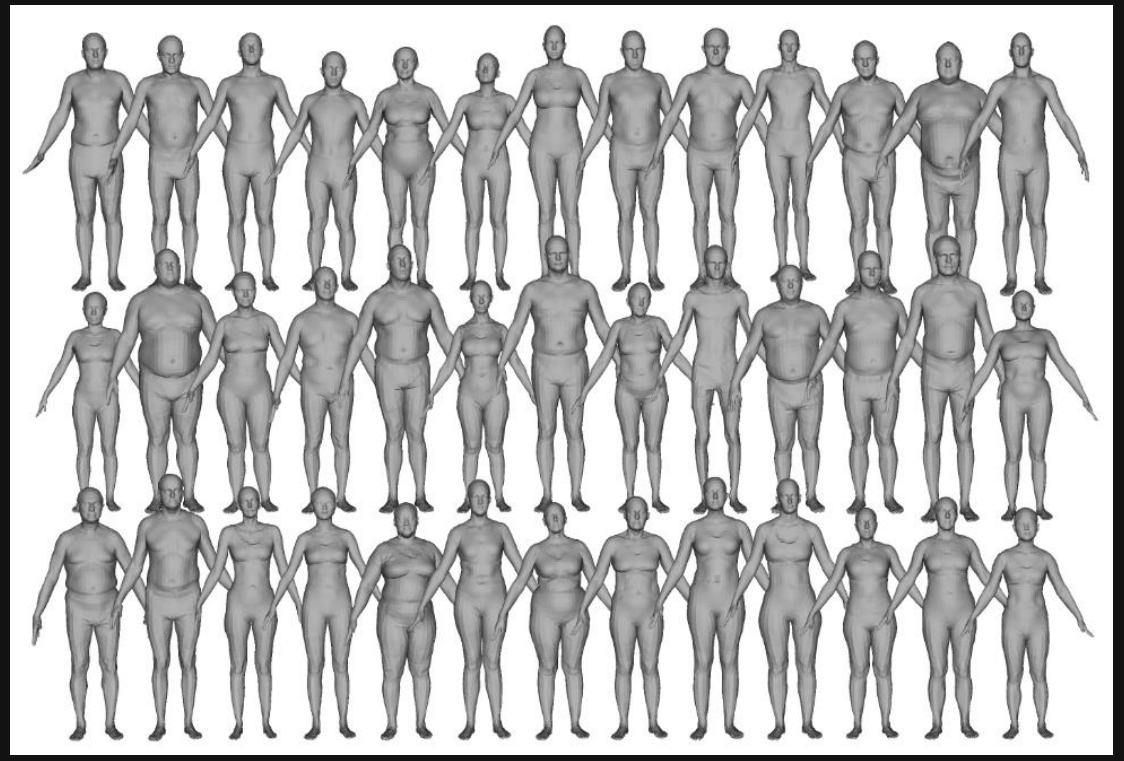
Shape blending function $B_S(\beta)$ + Pose blending function $B_P(\theta)$



Registering SMPL from Data

- How to determine the parameters?
  - Template mesh vertices $\overline{\mathbf{T}}$
  - Joint-to-skin blending weights $\mathcal{W}$
  - Shape blending parameters $\mathcal{S}$
  - Pose-to-joint forward kinematics $\mathcal{J}$
  - Pose blending parameters $\mathcal{P}$
  - (Personalized joint parameters β and pose θ)

Optimization from body scans!



The MultiShape Dataset (Robinette et al., 2002)
1786 Aligned body scans of 40 individuals

Registering SMPL from Data

Optimization from body scans!

Key objectives:

Vertex alignment:

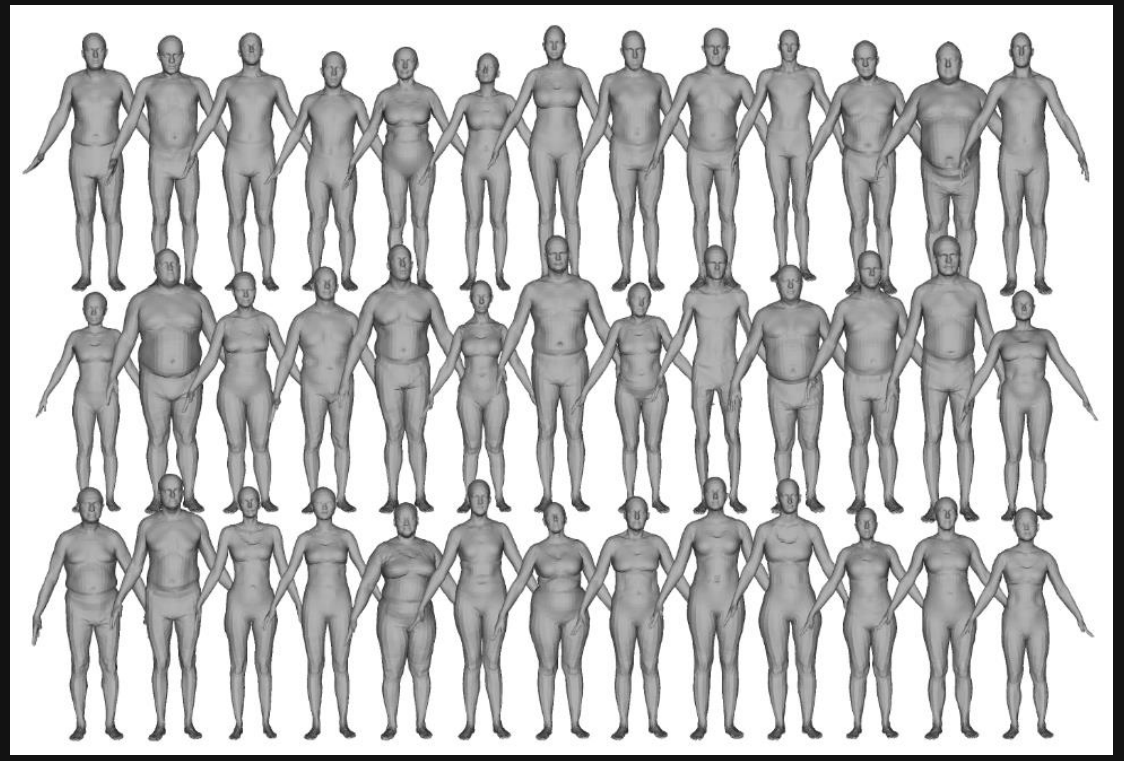
$$E_D(\hat{\mathbf{T}}^P, \hat{\mathbf{J}}^P, \mathcal{W}, \mathcal{P}, \Theta) = \sum_{j=1}^{P_{\text{reg}}} ||\mathbf{V}_j^P - W(\hat{\mathbf{T}}_{s(j)}^P + B_P(\vec{\theta}_j; \mathcal{P}), \hat{\mathbf{J}}_{s(j)}^P, \vec{\theta}_j, \mathcal{W})||^2$$

Symmetry regularization for the template:

$$E_Y(\hat{\mathbf{J}}^P, \hat{\mathbf{T}}^P) = \sum_{i=1}^{P_{\text{subj}}} \lambda_U ||\hat{\mathbf{J}}_i^P - U(\hat{\mathbf{J}}_i^P)||^2 + ||\hat{\mathbf{T}}_i^P - U(\hat{\mathbf{T}}_i^P)||^2,$$

Blending weight regularization:

$$E_W(\mathcal{W}) = ||\mathcal{W} - \mathcal{W}_I||_F^2.$$



The MultiShape Dataset (Robinette et al., 2002)
1786 Aligned body scans of 40 individuals

MANO

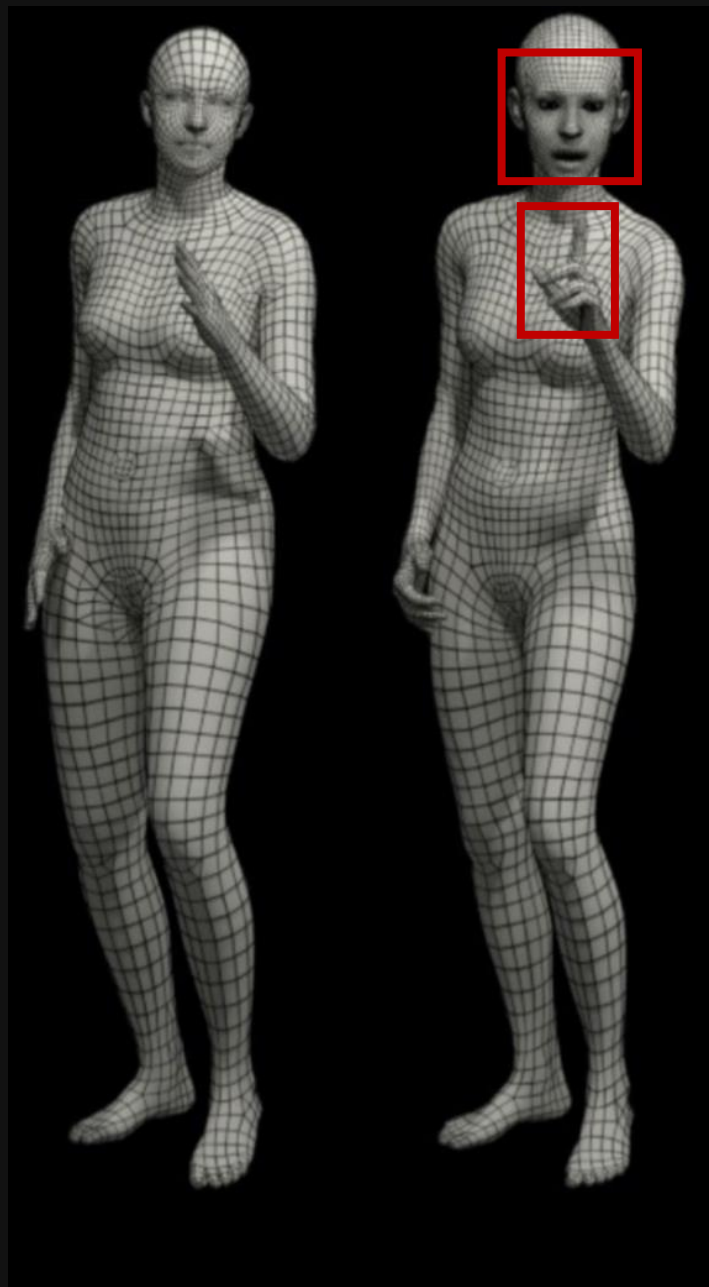
Also for the hand!



Fig. 6. Examples of the captured poses for a single subject. The *top* row shows some of the captured *scans*, while the *bottom* row shows the corresponding mesh *alignments* with *pink* color.



Fig. 7. Examples of the captured *"flat-hand"* pose for various subjects. The *top* row shows some of the captured *scans*, while the *bottom* row shows the corresponding mesh *alignments* with *pink* color.

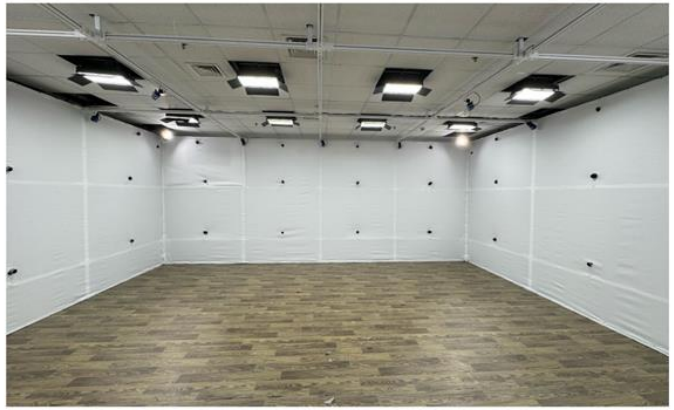


Where dose the 3D data come from…

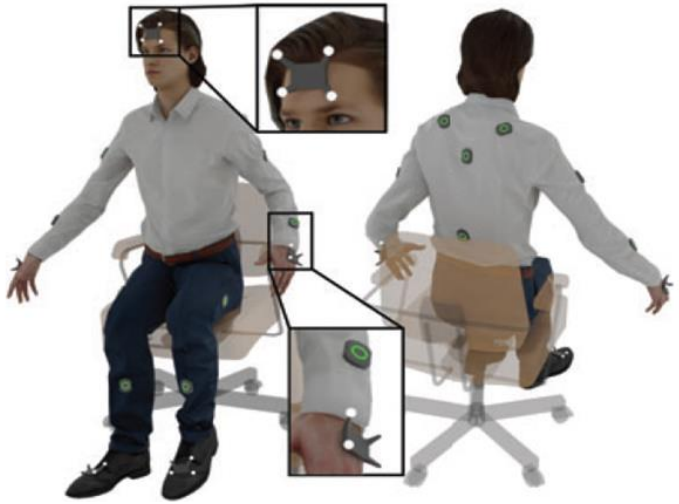
From Motion Capture (MoCap)



What is MoCap













MoCap Data is VERY Expensive

A GitHub issue in a MoCap dataset repo

Is there an explanation to why it is so painful to setup this dataset ? #116

Closed



valentin-fngr opened on Jan 2, 2024

Hi,

I am new to the field and desires to work on that solving motion to text problems.

I am probably asking a very trivial question : Why is it so painful to setup everything ? What can be done to simplify this process ? It seems like a lot of instructions are counte



EricGuo5513 on Jan 3, 2024 Owner

Hi, I understand that it's quite cumbersome to go through these pipelines. That's because humanml3D uses the motion data from various datasets which are under different distribution license. Please note capturing 3d motion data is an VERY expensive and laborious process. Every dataset creator pays a lot of effort on this. We can not distribute them as our own data, even SMPL-H/DMPL models. Therefore, we request everyone to download the motion data and body models from their source website, and comply the license.

9

One of the user complained that MoCap data pre-processing pipeline is painful

Because he has to gather MoCap data from different data sources

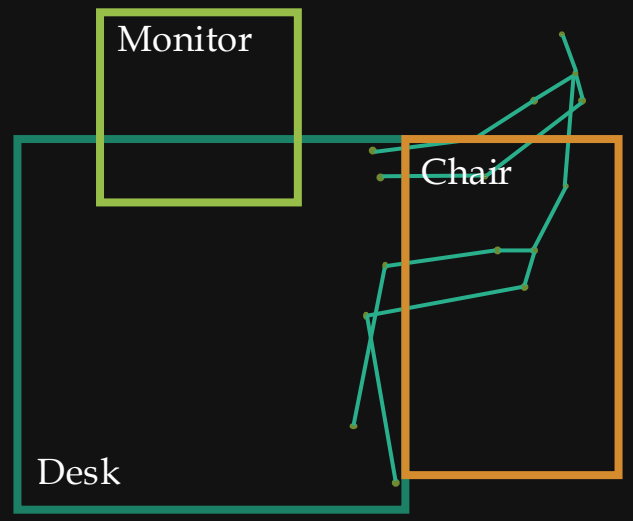
Outline

- Section 1: 2D HOI
  - General formulation
  - HOI detection

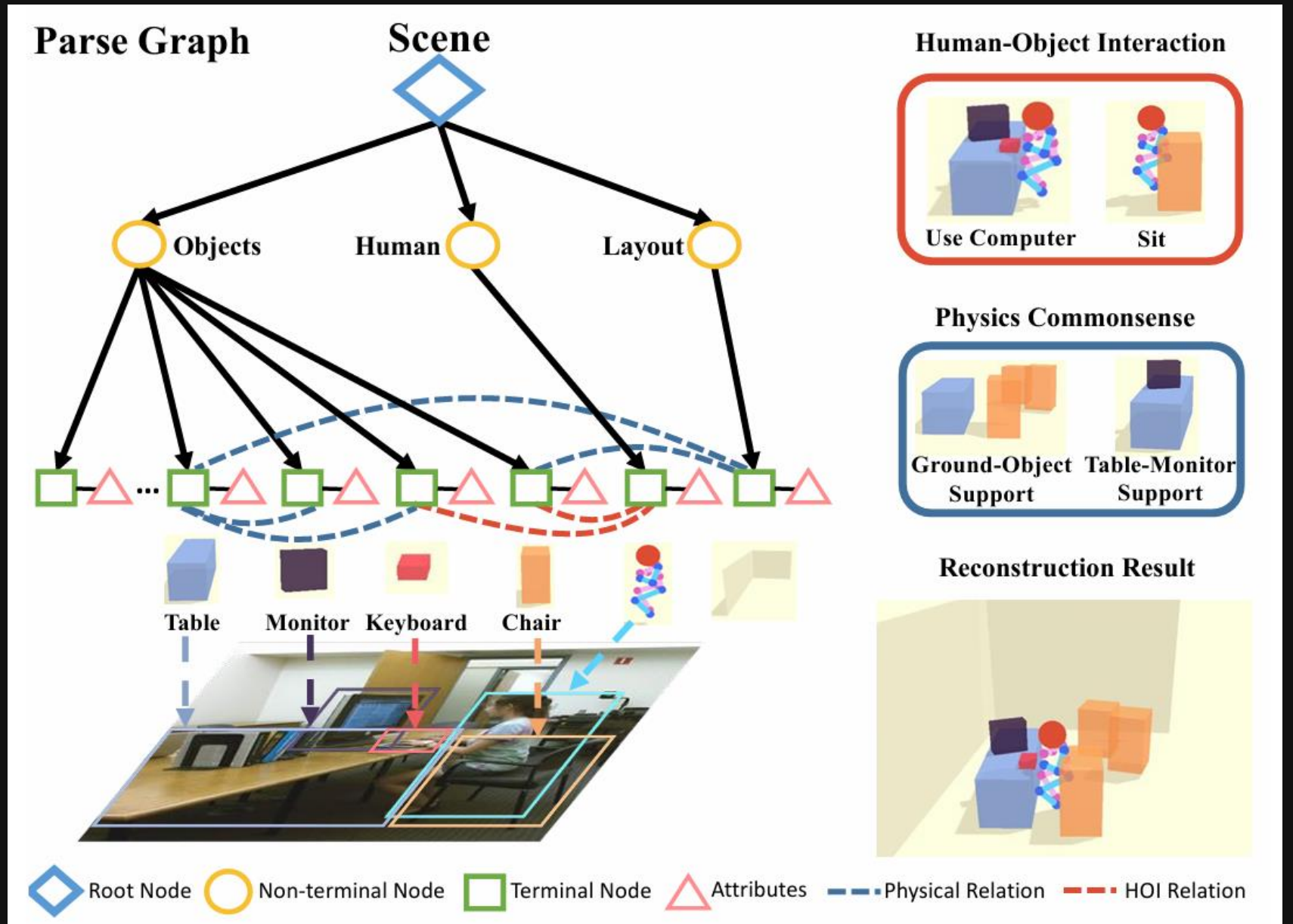
- Section 2: 3D HOI
  - 3D representation
  - 3D reconstruction from 2D image
  - From perception to generation

- Section 3: From HOI to HSI
  - Scene-conditioned human motion generation
  - Human-centric scene synthesis

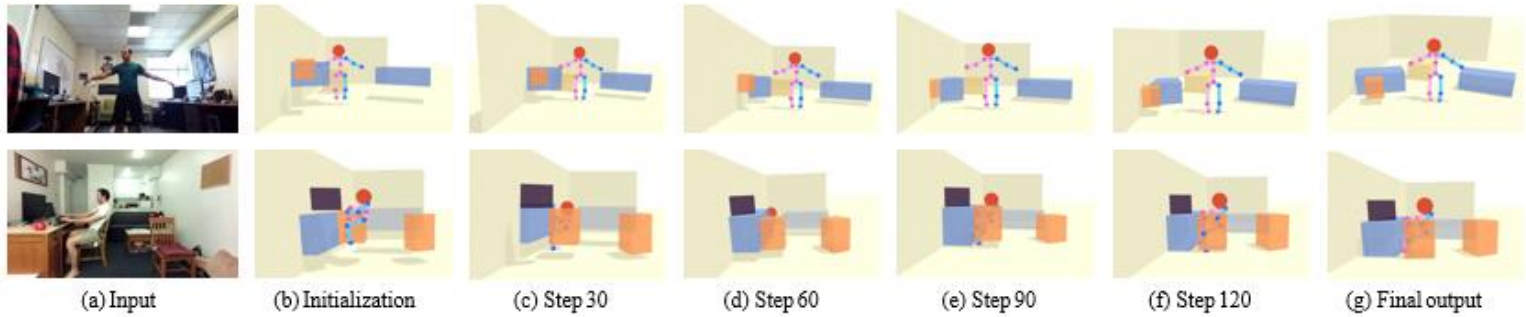
Joint Parsing



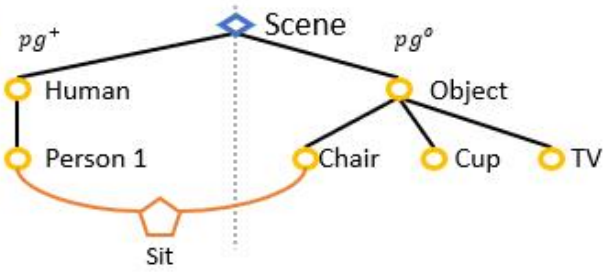
Single Image Joint Parsing

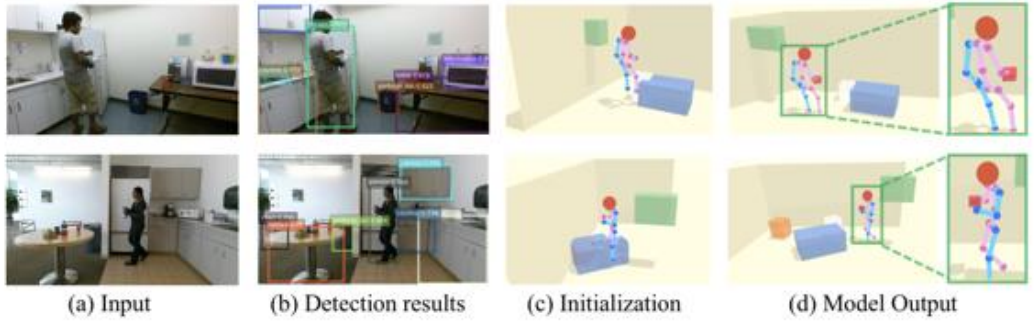


Reconstruct 3D Scene Using HOI as a Prior



Error Correction





Top-down inference of small object

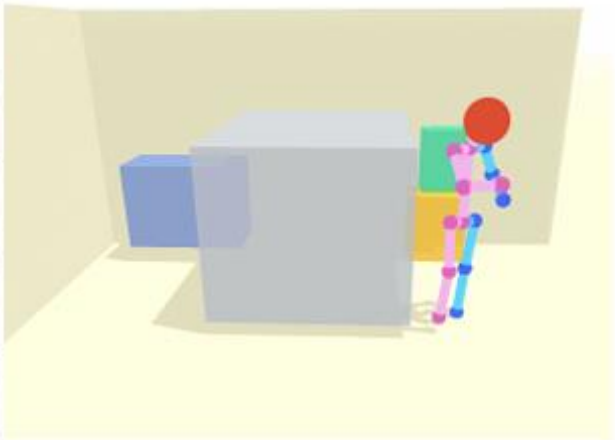
Methods	*w/o hoi*			*Full model*		
HOI Type	Object ↑	Pose ↓	MR ↓	Object ↑	Pose ↓	MR ↓
Sit	26.9	0.59	15.2	**27.8**	**0.581**	**13.1**
Hold	17.4	0.537	65.3	**17.6**	**0.470**	**41.9**
Use Laptop	14.1	0.544	58.8	**15.0**	**0.534**	**43.3**
Read	**14.5**	0.453	78.9	14.3	**0.466**	**54.6**

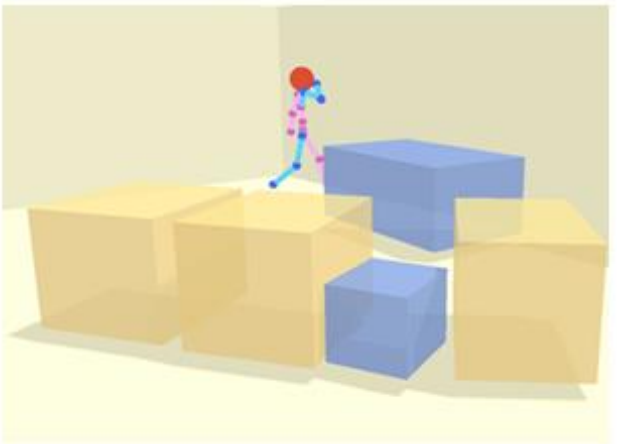
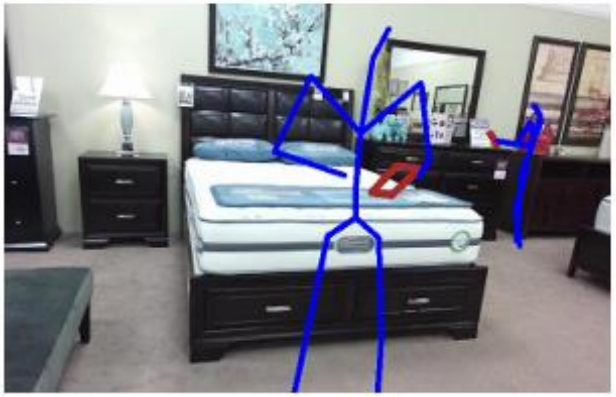
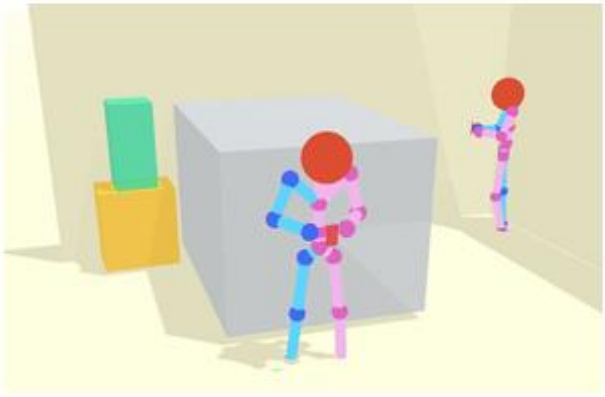
Ablative result of HOI on 3D object IoU(%), 3D pose estimation error (m), and miss-detection rate (MR, %)

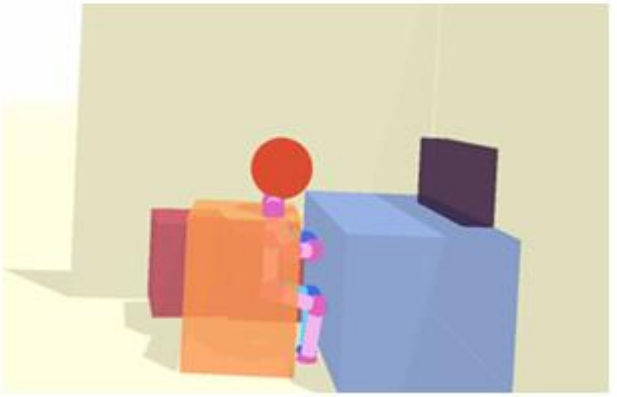
$$pg^* = \arg\max_{pg} p(pg|I) = \arg\max_{pg} p(pg) \cdot p(I|pg)$$

$$= \arg\max_{pg} \frac{1}{Z} \exp\{-\mathcal{E}_{phy}(pg) - \mathcal{E}_{hoi}(pg) - \mathcal{E}(I|pg)\}. \quad (1)$$

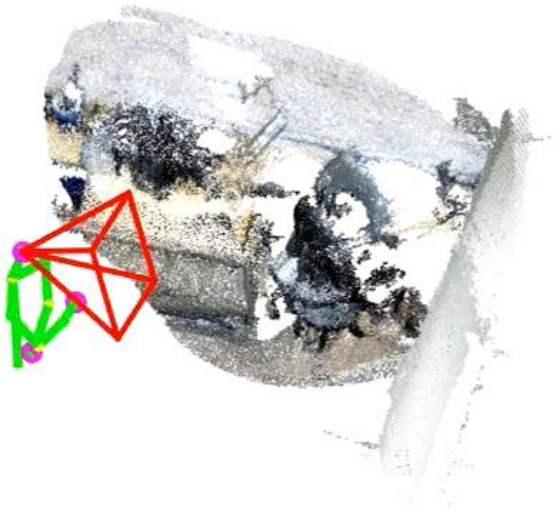
Single Image Joint Parsing

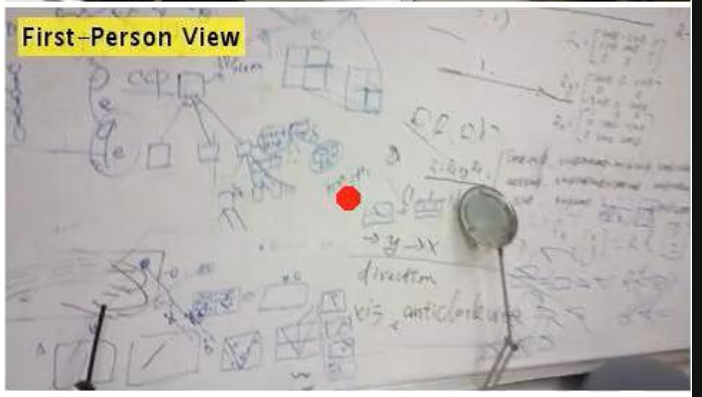






HOIs and 3D Reconstruction







Locating Objects

Directing Hand Motion

Guiding the Approach

Checking Status

Idle

Frame #

A Restricted Turing Test on Understanding Object, Scene, and Event

3 Areas, 30+ cameras (ground, tower, mobile), 3,000,000 frames (1 TB).
Ontology: Objects, attributes, scenes, actions, group activities, spatial-temporal relations.

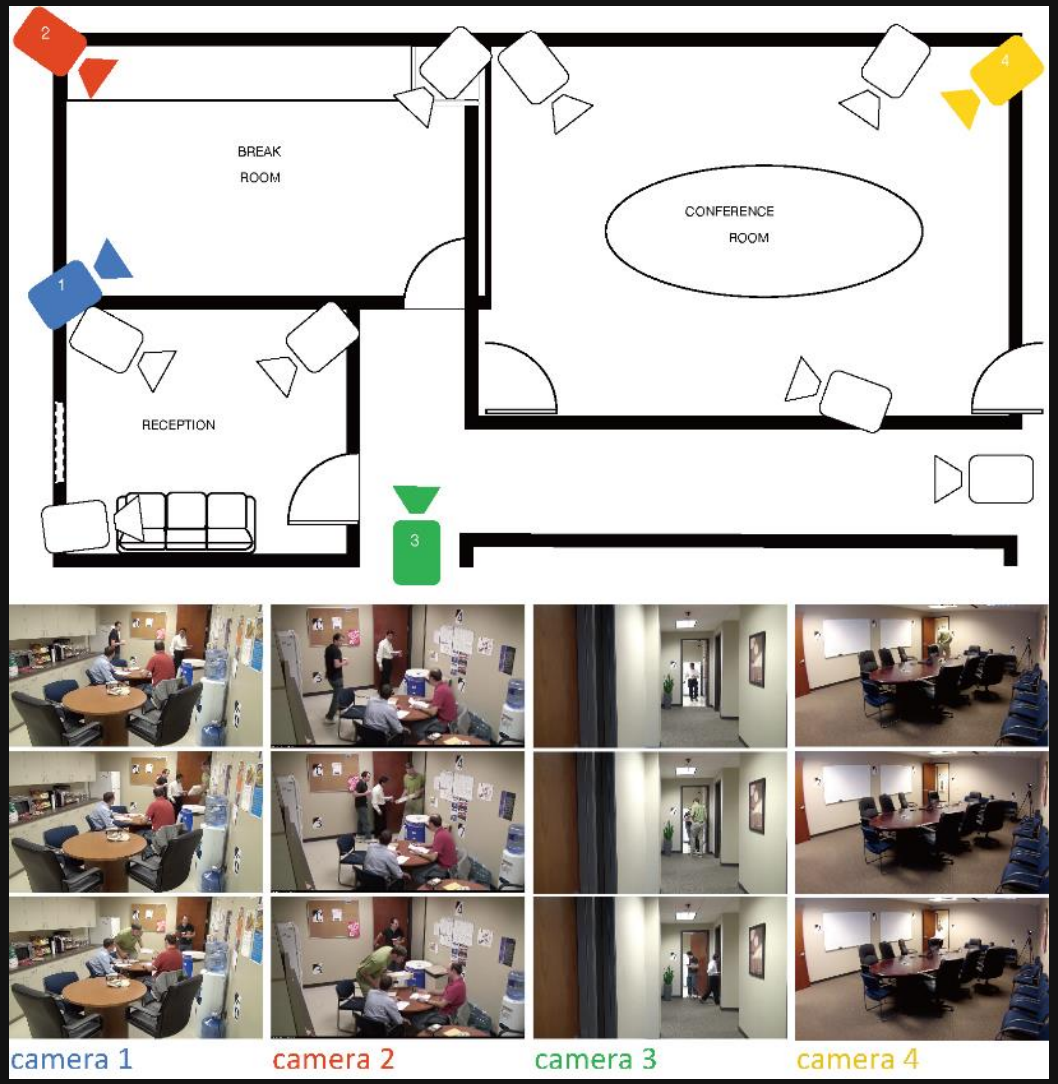








Video Analytics: Rich Indoor and Outdoor Activities





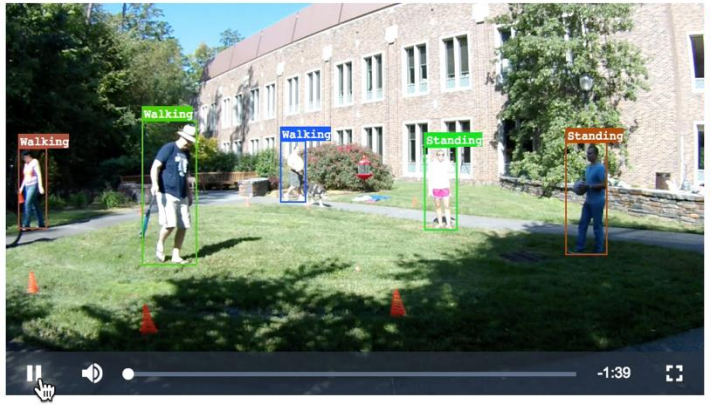




Camera 1 Camera 2 Camera 3 Camera 4

3D Scene and Event Parsing and Understanding with Rich HOIs

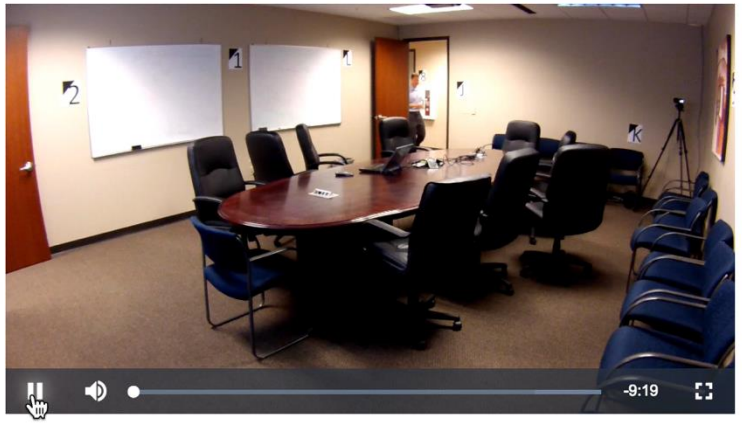




Female ShortHair Glasses Sleeves Shorts
Male ShortHair Tshirt Shorts
Male ShortHair Hat Tshirt Shorts
Female LongHair Sleeves Pants
Female LongHair Glasses Tshirt Pants

#FRAME: 5, TIME: 0:00



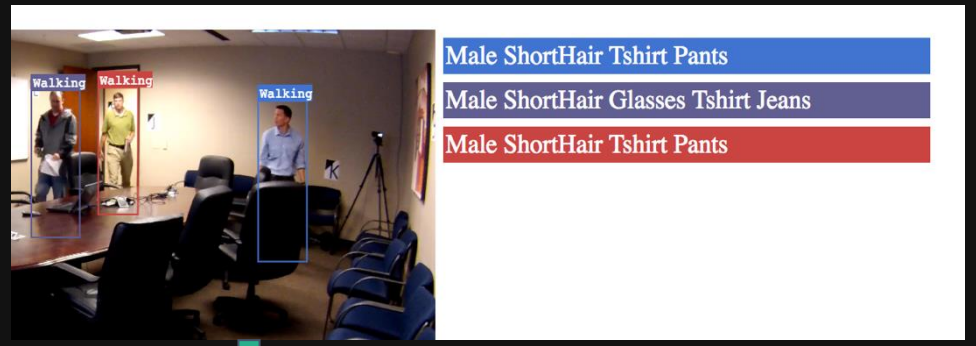


#FRAME: 14, TIME: 0:01

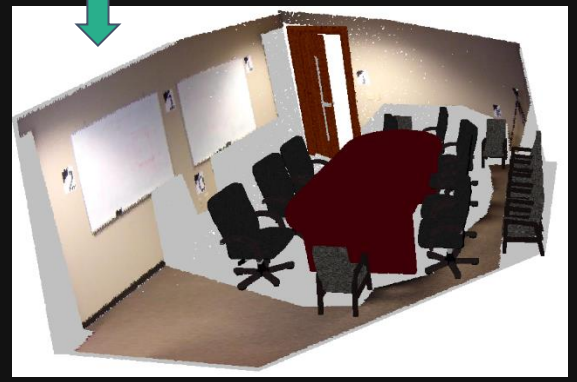
Video Analytics Framework



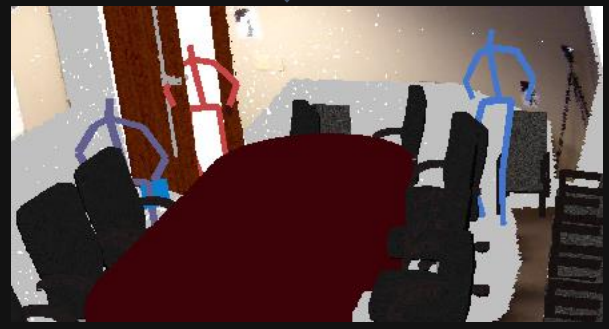
Detection, Tracking,
Attribute Estimation



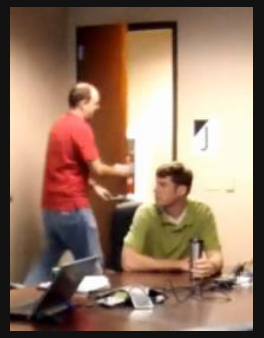
Scene Reconstruction



People 3D Pose Estimation

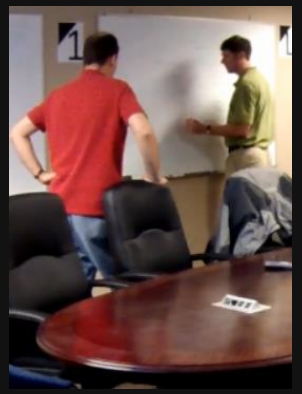






Human-Object Interaction
(physics, causality)

Discussing

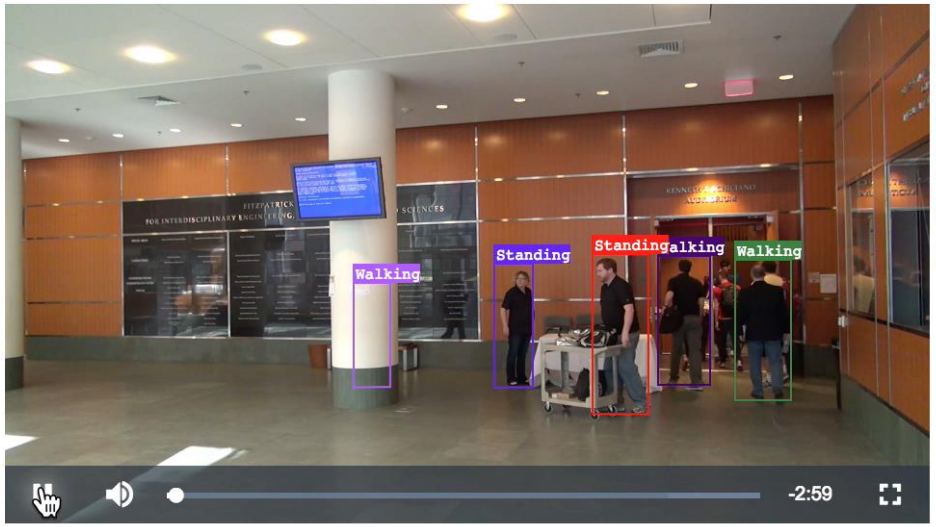


Following



Social Interactions





Female ShortHair Sleeves Pants

Male ShortHair Sleeves Jeans

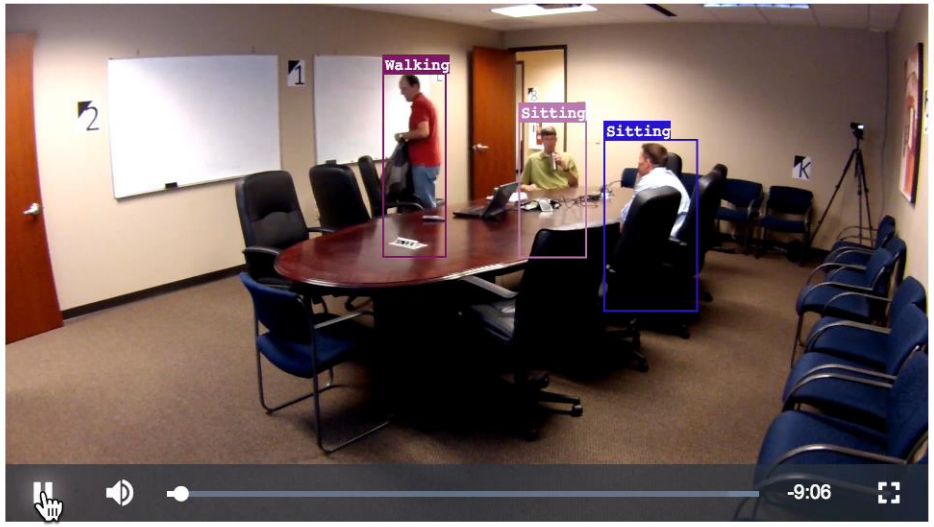
Male ShortHair Tshirt Pants

Male ShortHair Tshirt Jeans

Female ShortHair Glasses Tshirt Jeans

#FRAME: 90, TIME: 0:03





Male ShortHair Tshirt Pants

Male ShortHair Glasses Tshirt Jeans

Male ShortHair Tshirt Pants

#FRAME: 343, TIME: 0:14

Playback

World Clock

15:05:00.000

SMPTE Timecode

00:00:00:00

Current Time

0.000000

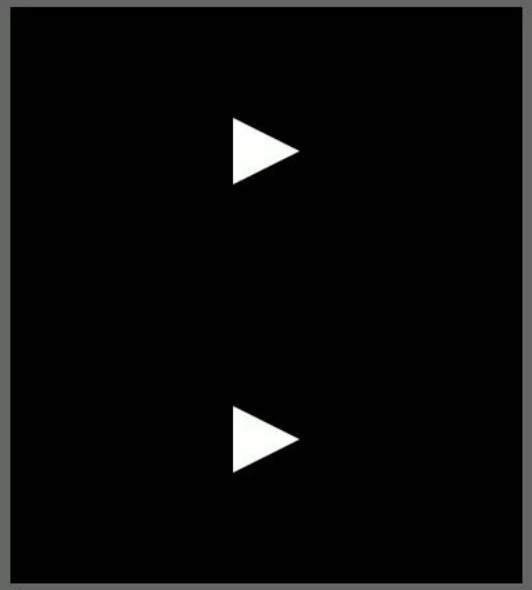
Current Frame 25 fps

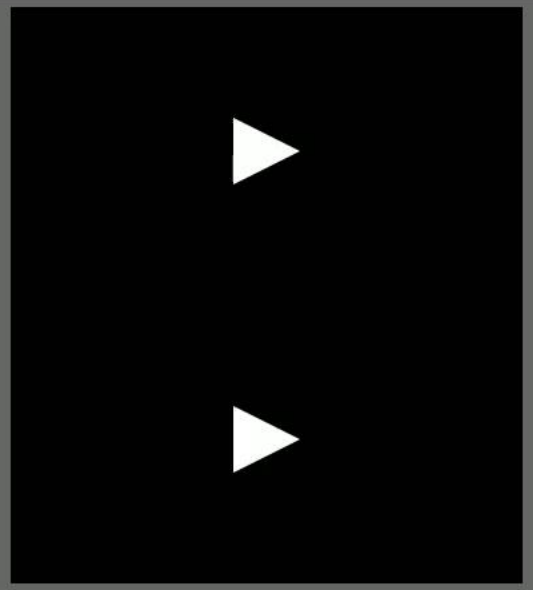
0 / 3000

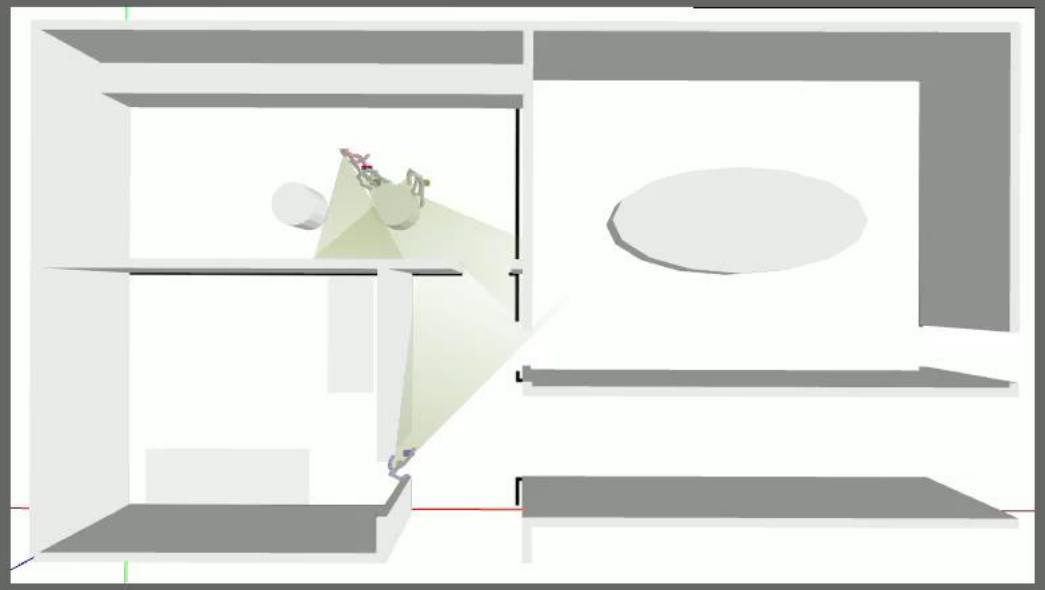
Visualization Options

Status: ....

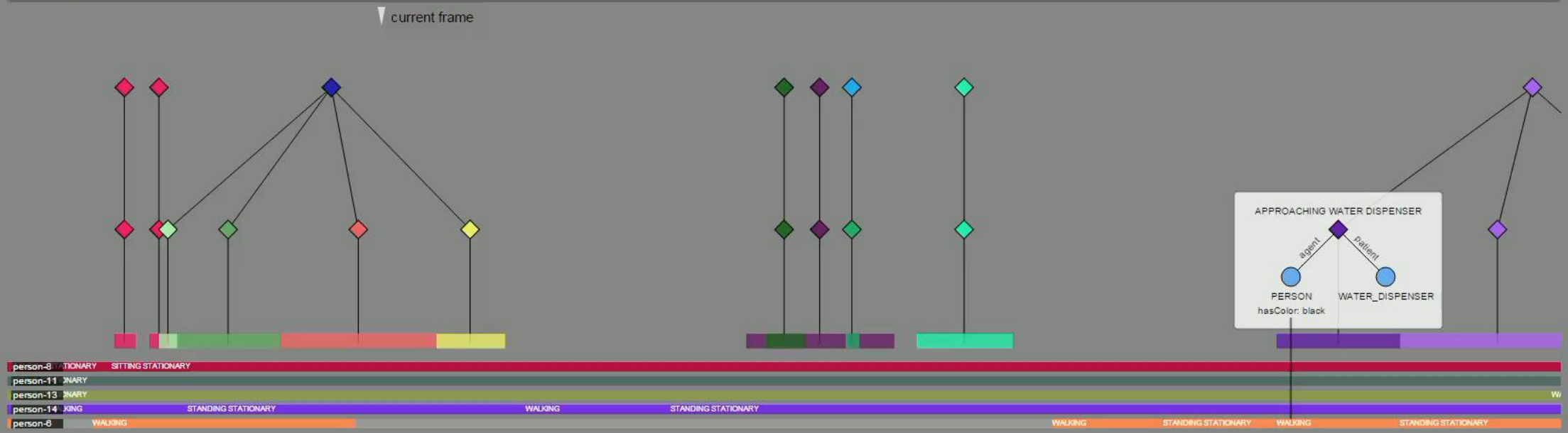
- [x] Tracks
- [x] Human attributes
- [x] Behavior labels
- [x] Text Generation
- [ ] Text-to-Speech
- [ ] Group labels







Text Generation:



© 2014. Center for Vision, Cognition, Learning and Art. University of California, Los Angeles.

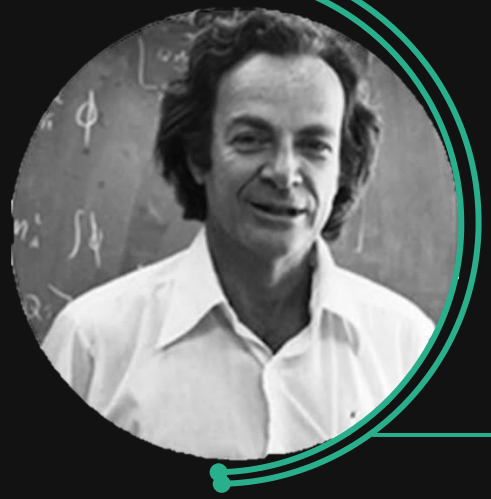
Outline

- Section 1: 2D HOI
  - General formulation
  - HOI detection

- Section 2: 3D HOI
  - 3D representation
  - 3D reconstruction from 2D image
  - From perception to generation

- Section 3: From HOI to HSI
  - Scene-conditioned human motion generation
  - Human-centric scene synthesis

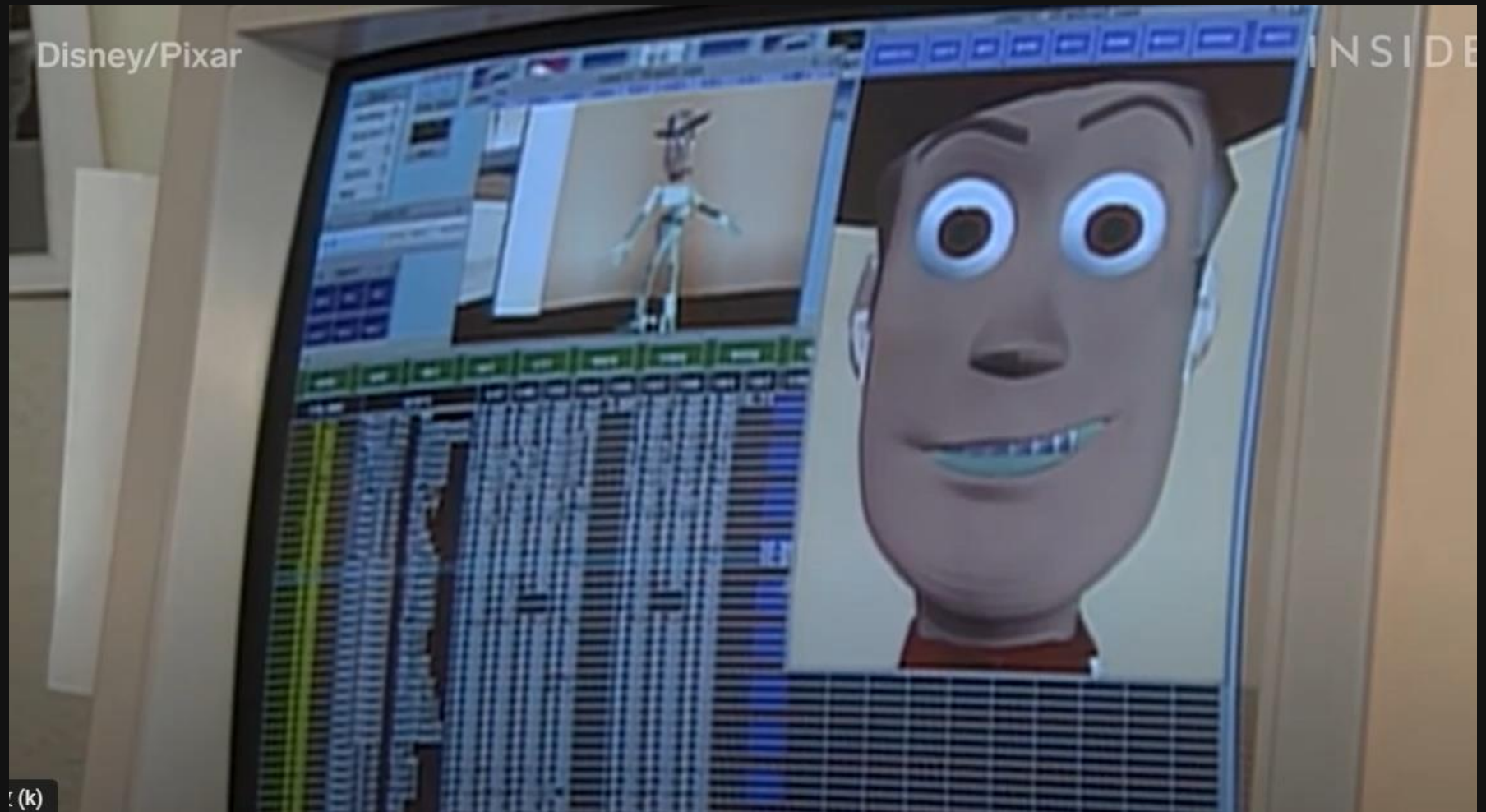
Why Generation



> What I cannot create, I do not understand.

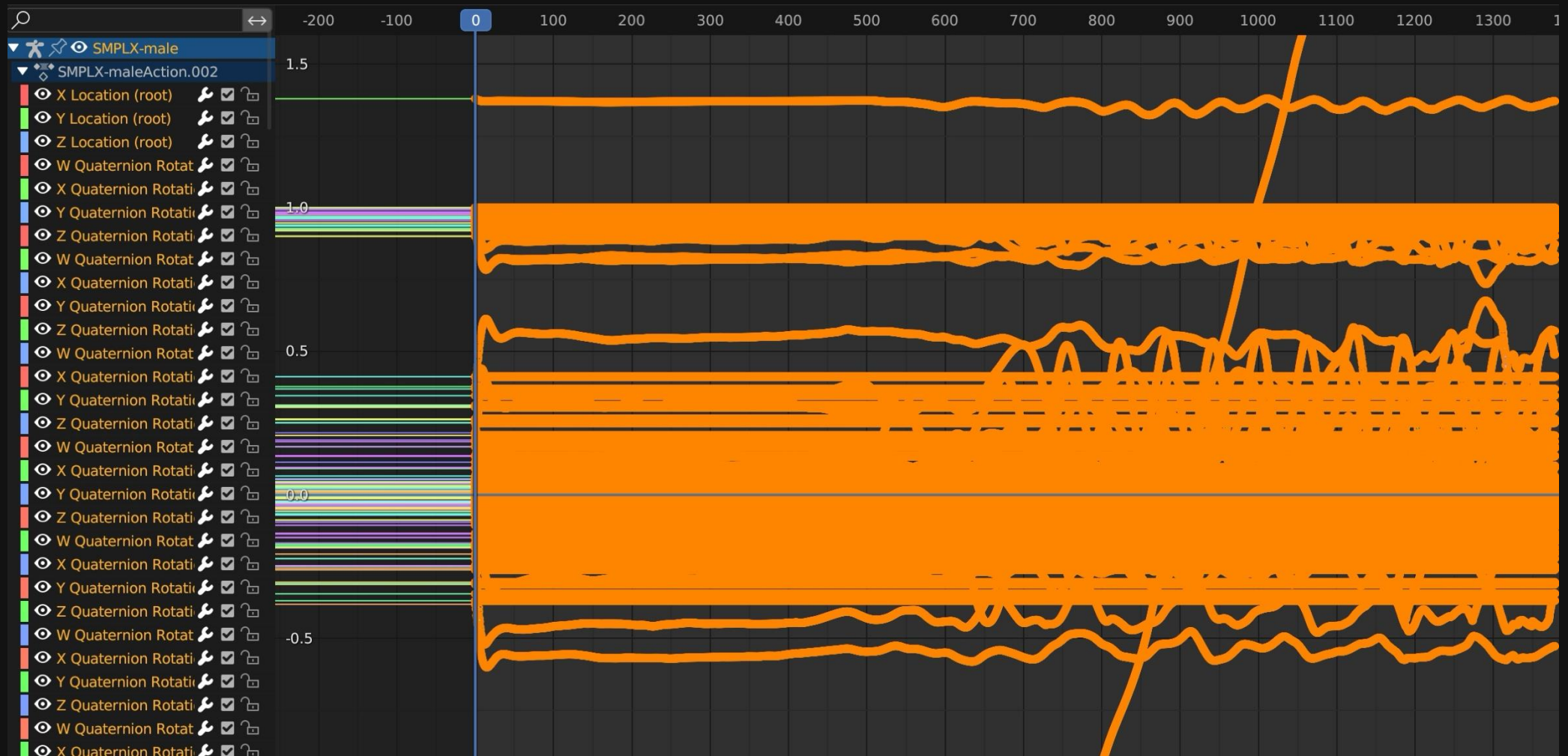
Keyframing (2000s)

- Generate motion from human efforts
- Main idea: Set the key-points positions in each frame



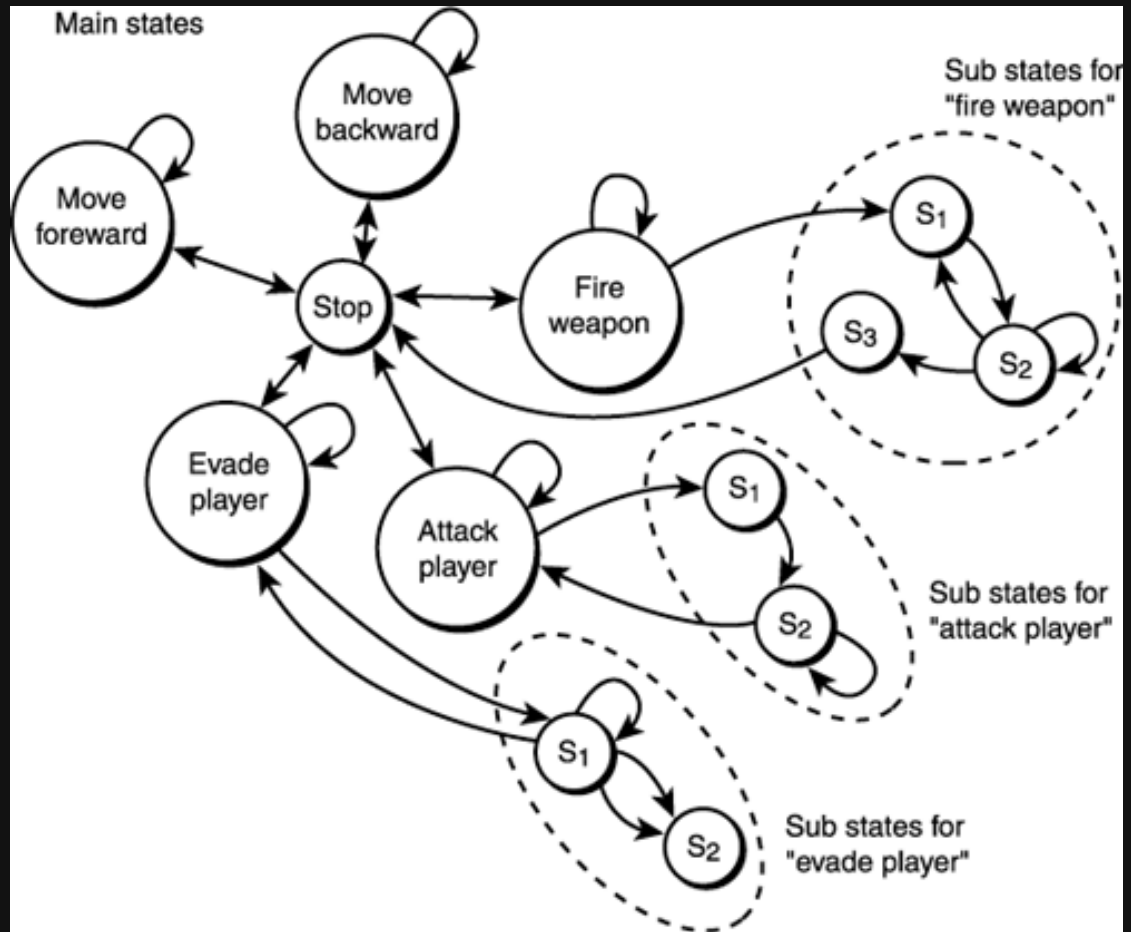
Keyframing (2000s)

- Keyframing requires huge amount of human efforts
  - Animators need to change these f-curves frame by frame



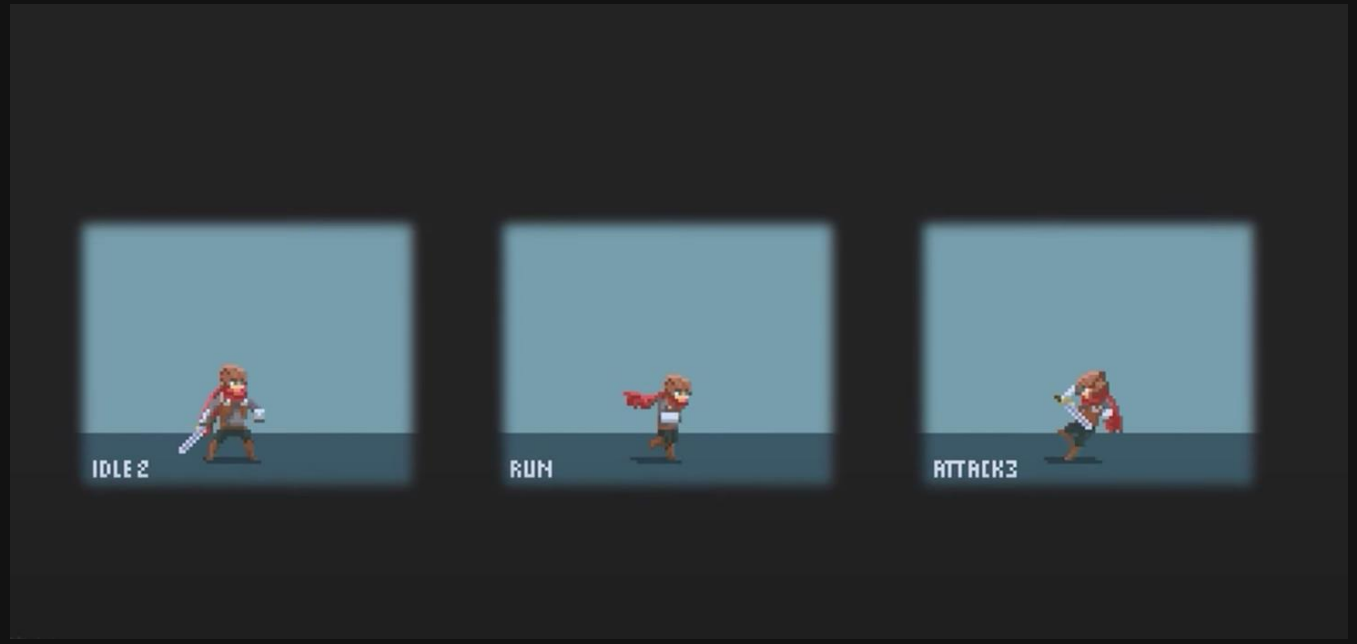
Animation State Machine (2000s)

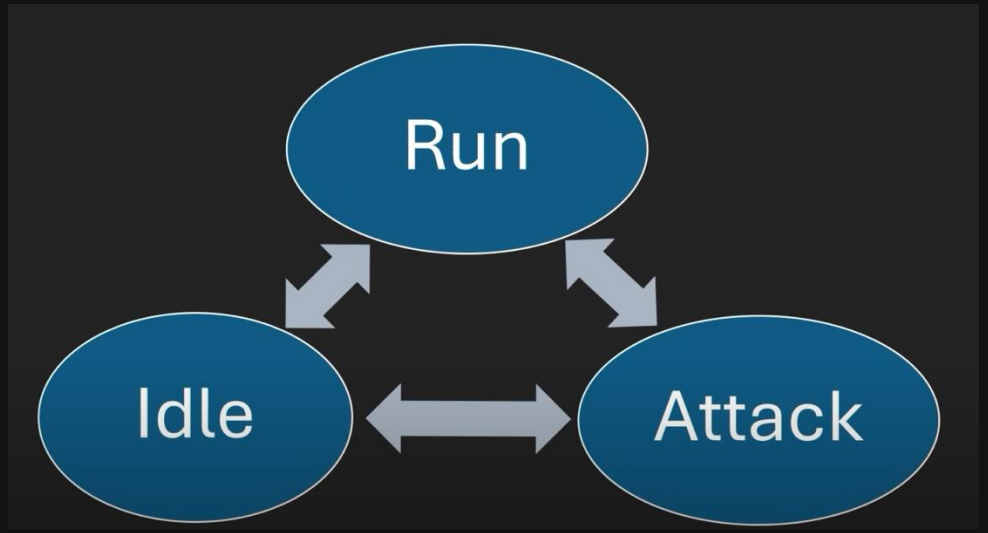
- Generate motion from user control
- Main idea: Prepare a fixed motion states set, and define human motion as transitions between different states



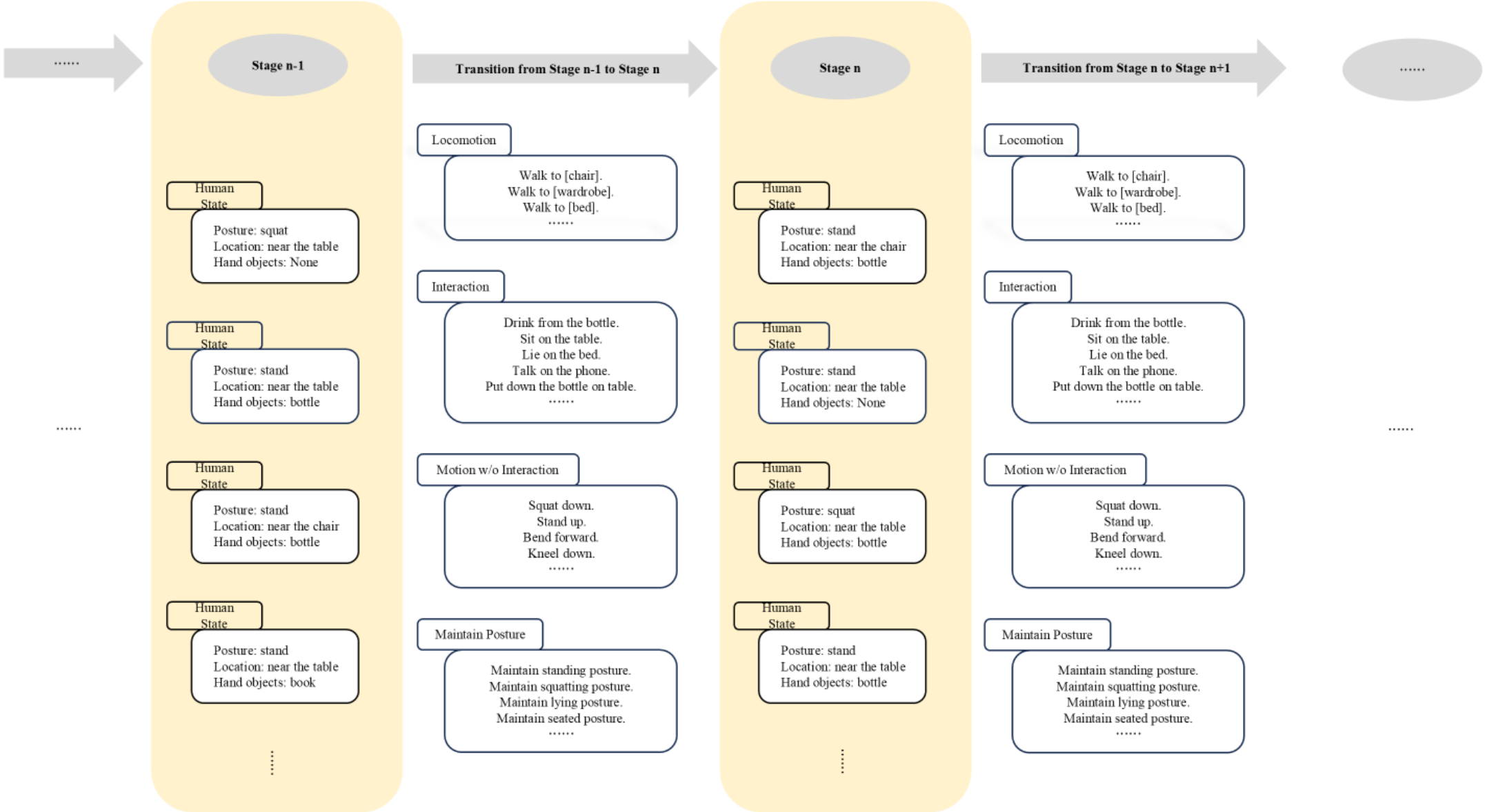
Animation State Machine (2000s)

- Main idea: Prepare a fixed motion states set, and define human motion as transitions between different states

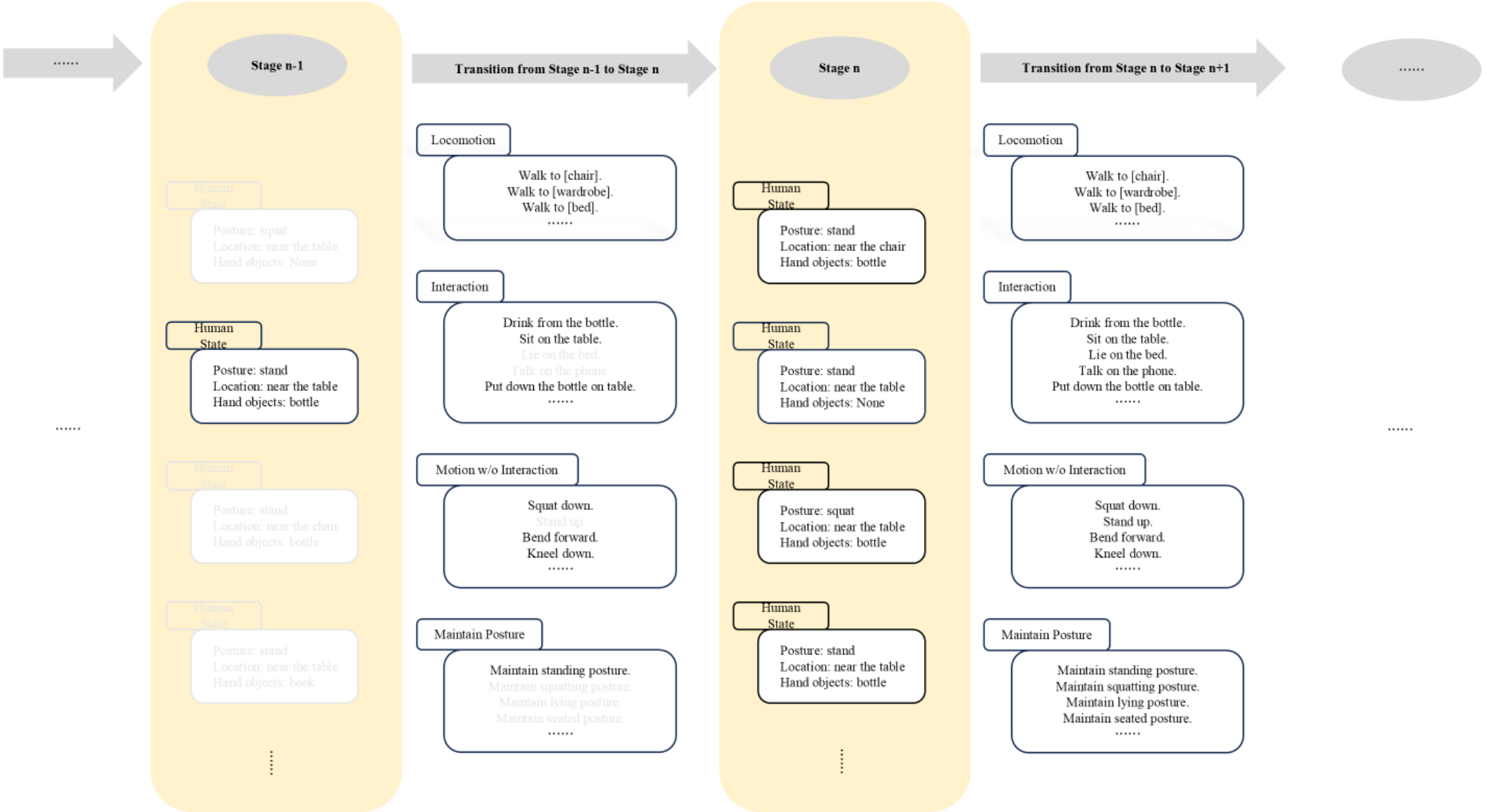




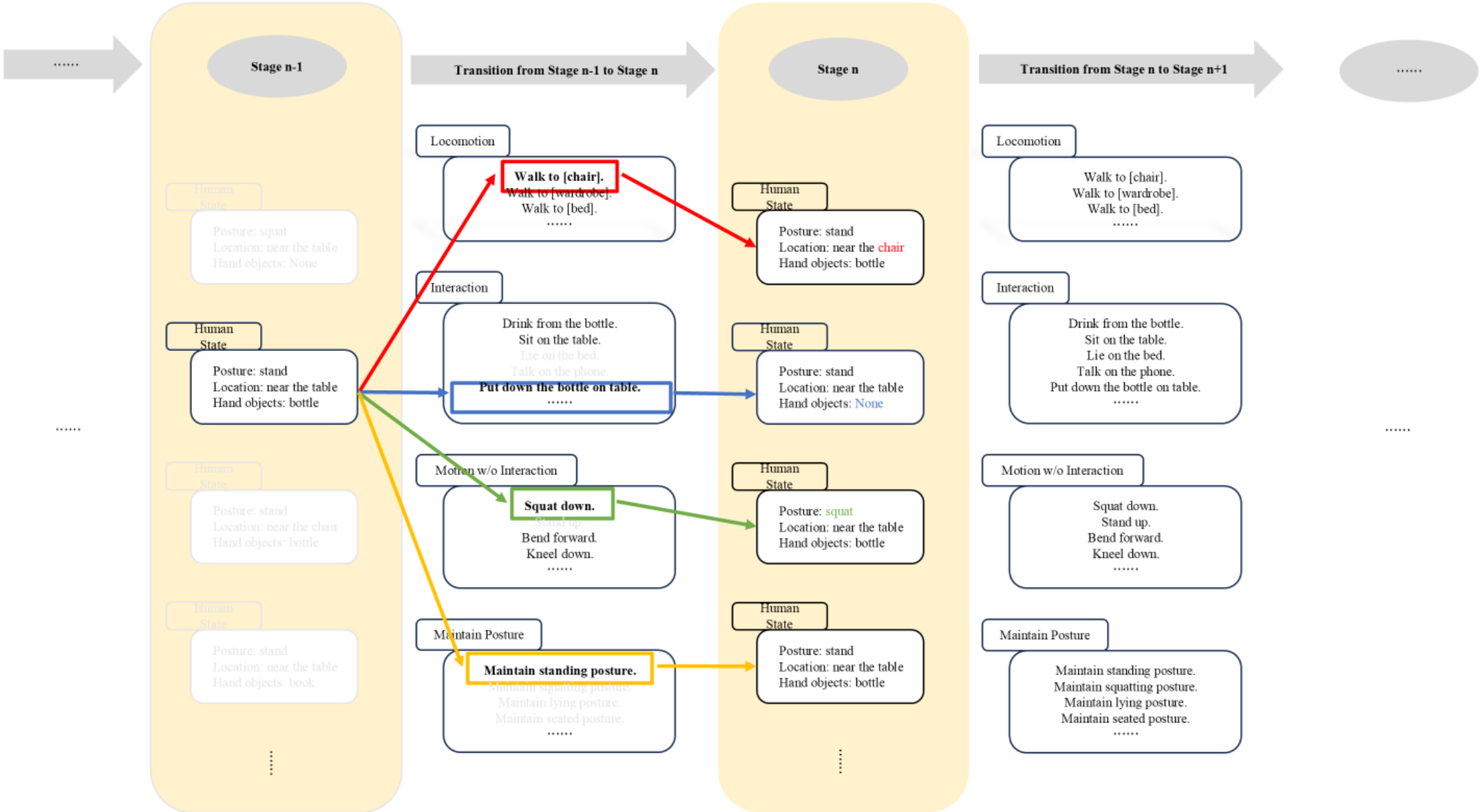
A State Transition Example



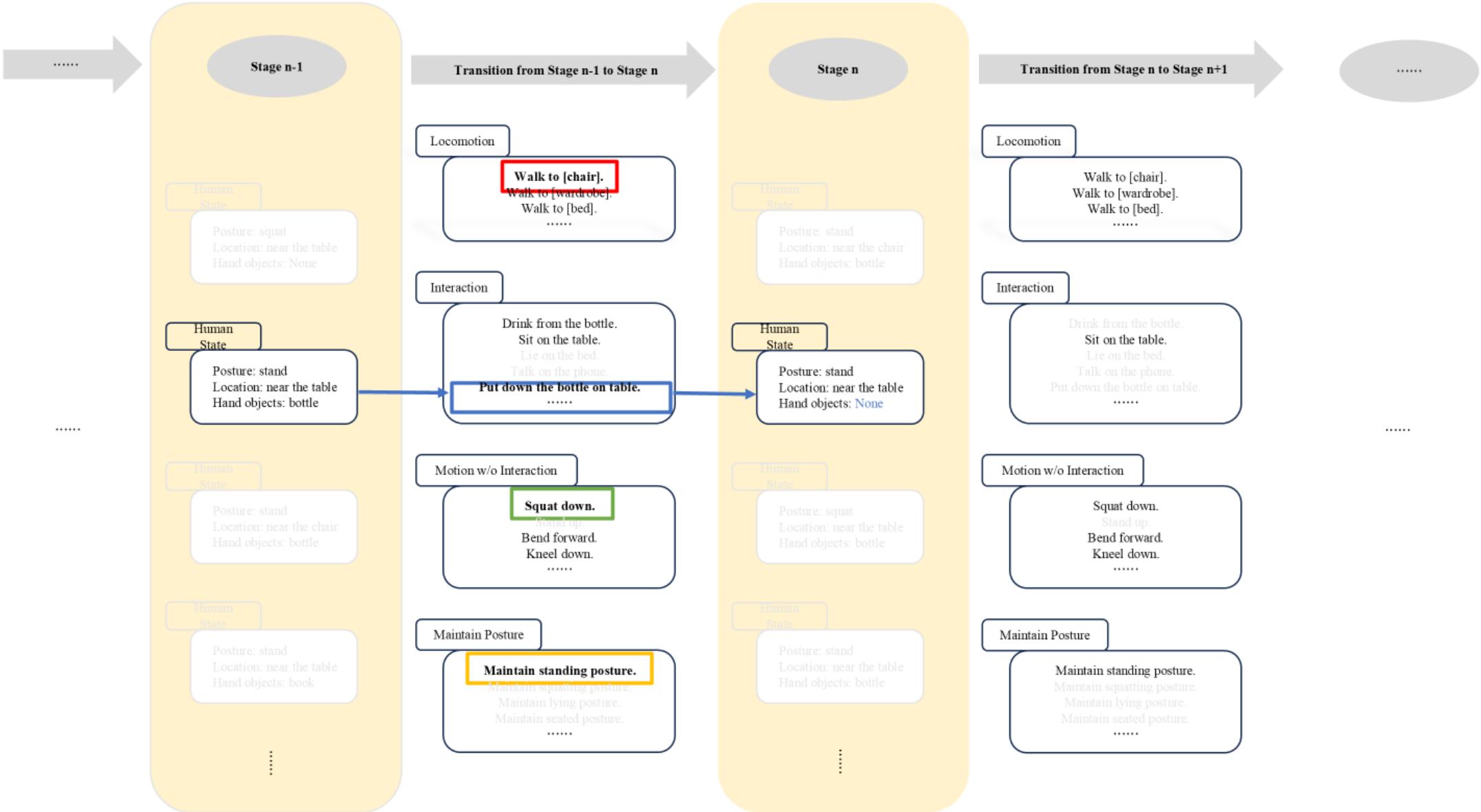
A State Transition Example



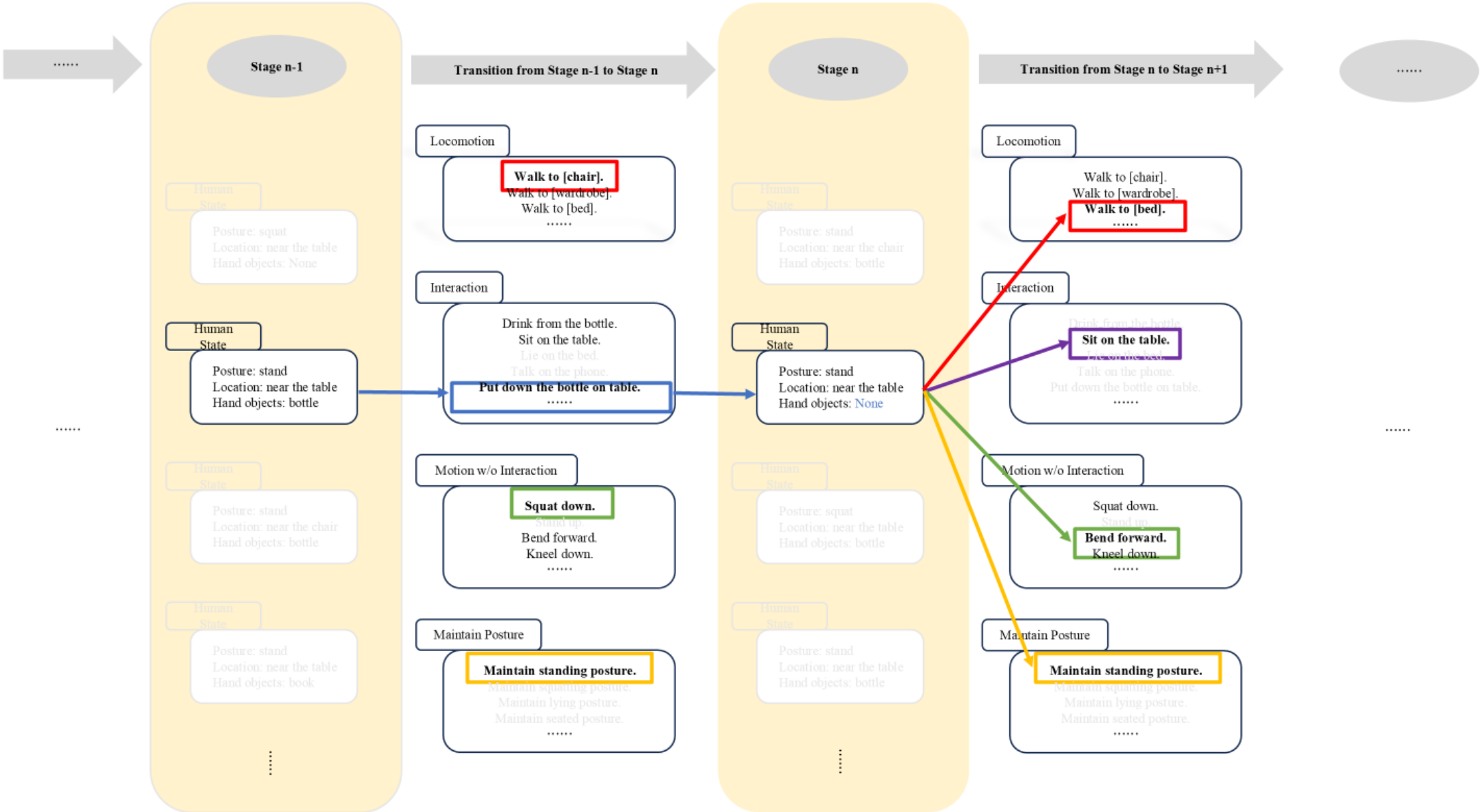
A State Transition Example



A State Transition Example



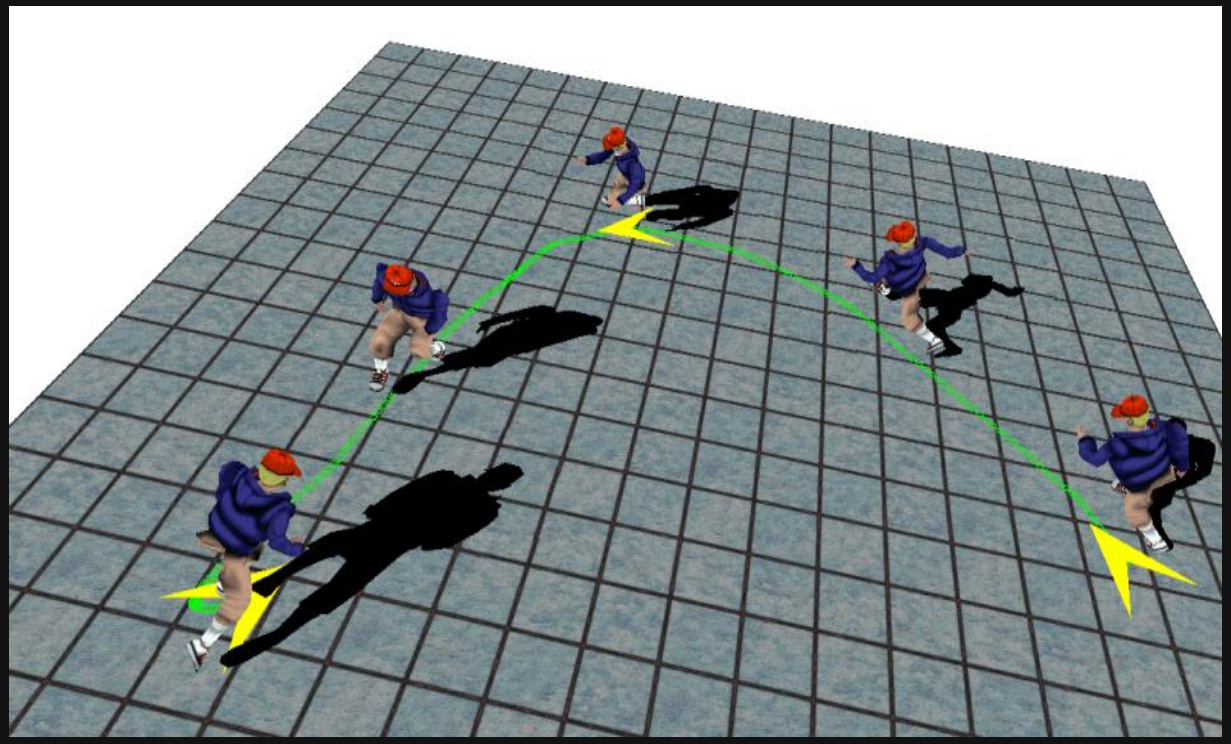
A State Transition Example



Motion Matching (2010s)

- Main idea: Searches a proper future motion segment from database

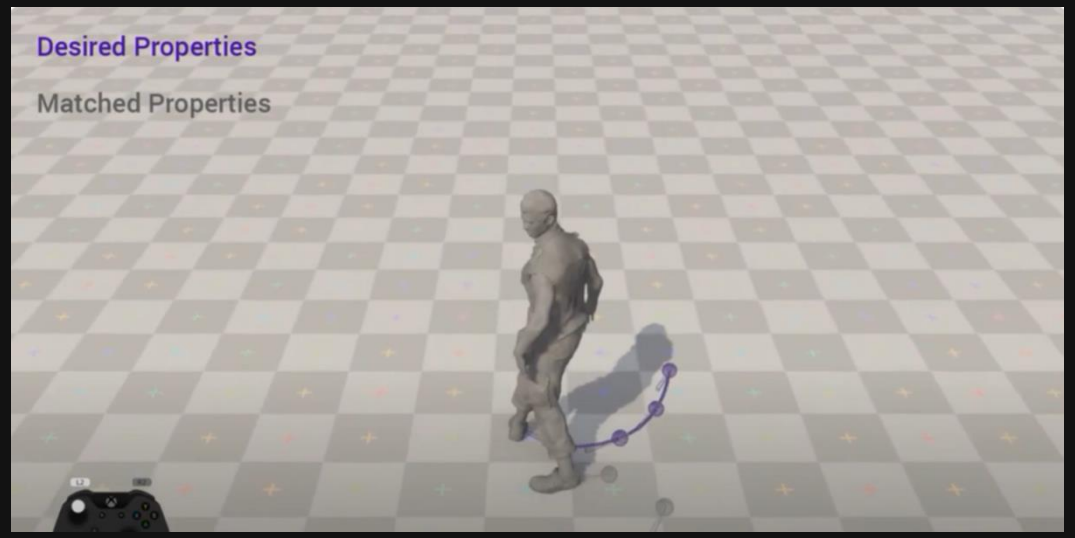




Motion Matching (2010s)

- Motion matching searches a proper future motion segment from database

Which matches current motion clip and trajectories



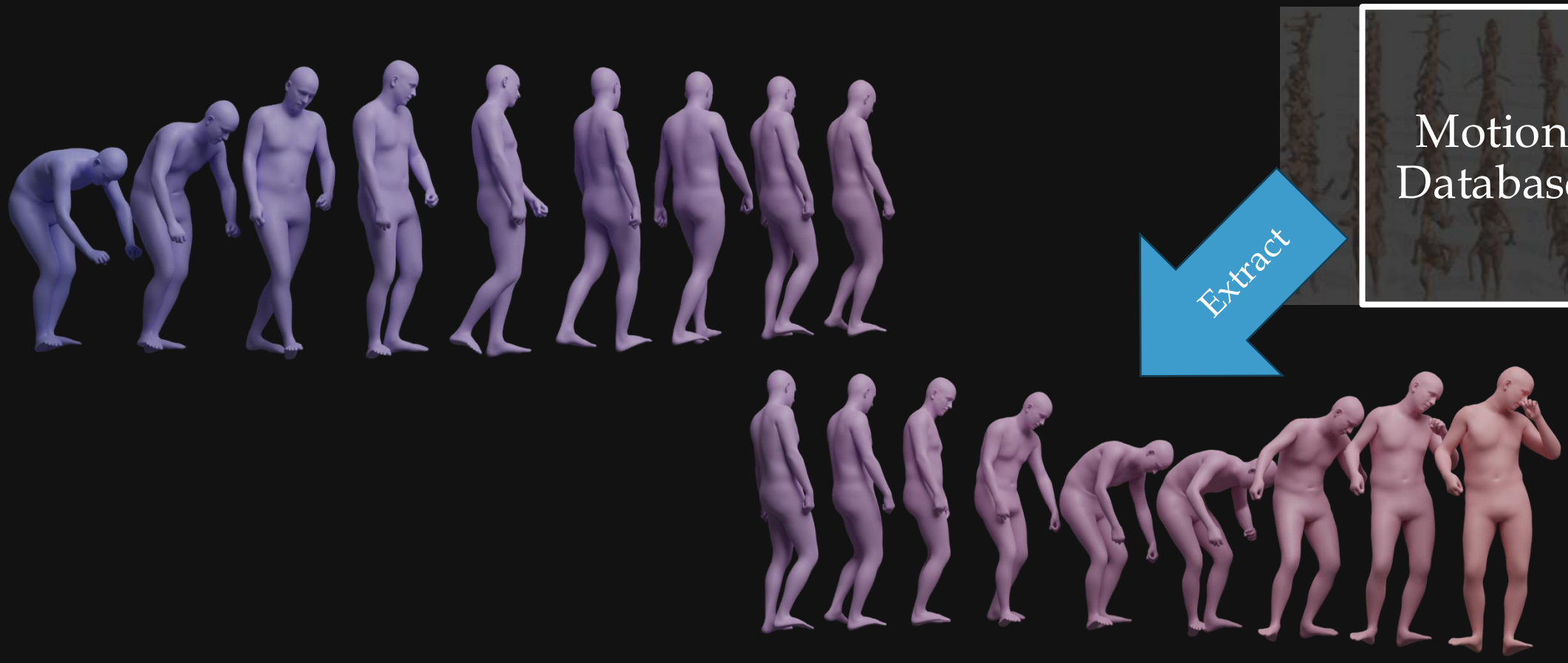
Which performs specific tasks



Learned Motion Matching. ACM Transactions on Graphics. 2020.

Motion Matching (2010s)

- Motion matching searches for next motion segment from motion database based on the features of previous segment



Motion Matching (2010s)

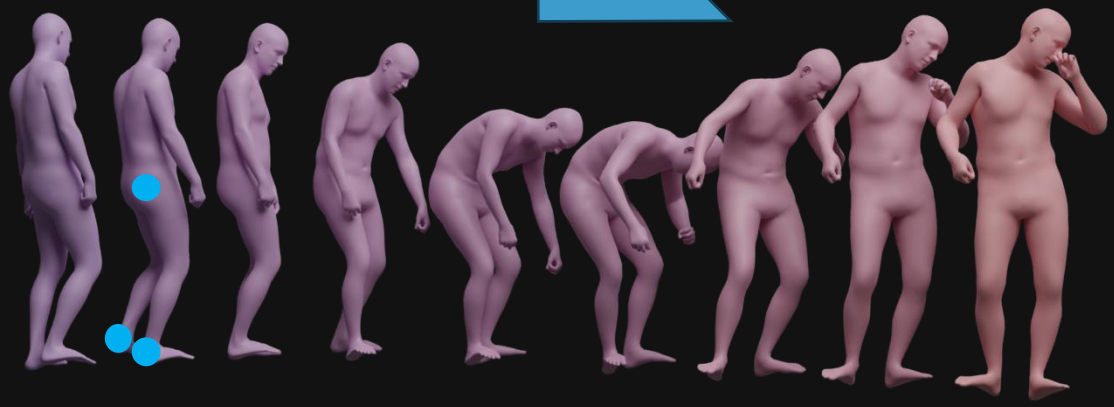
- Motion matching searches for next motion segment from motion database based on the features of previous segment



The feature contains:
root position
foot position



Extract



Motion Matching (2010s)

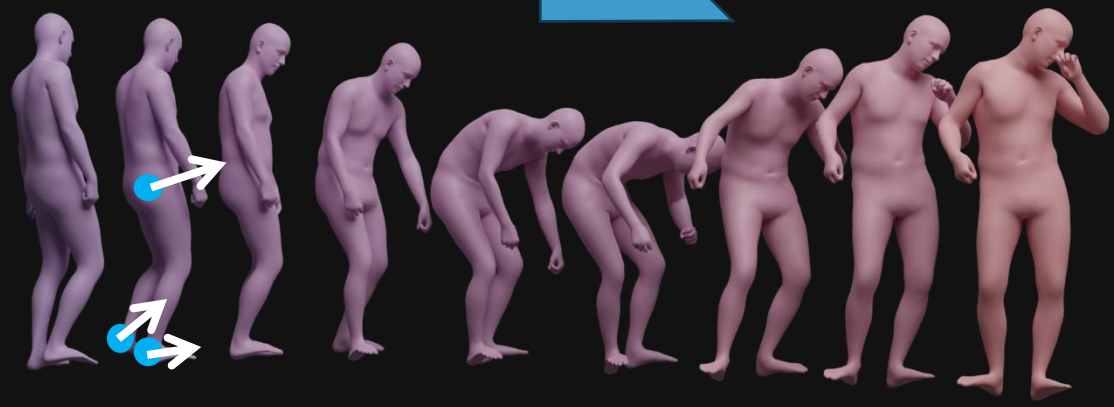
- Motion matching searches for next motion segment from motion database based on the features of previous segment



The feature contains:
root position
foot position
root velocity
foot velocity

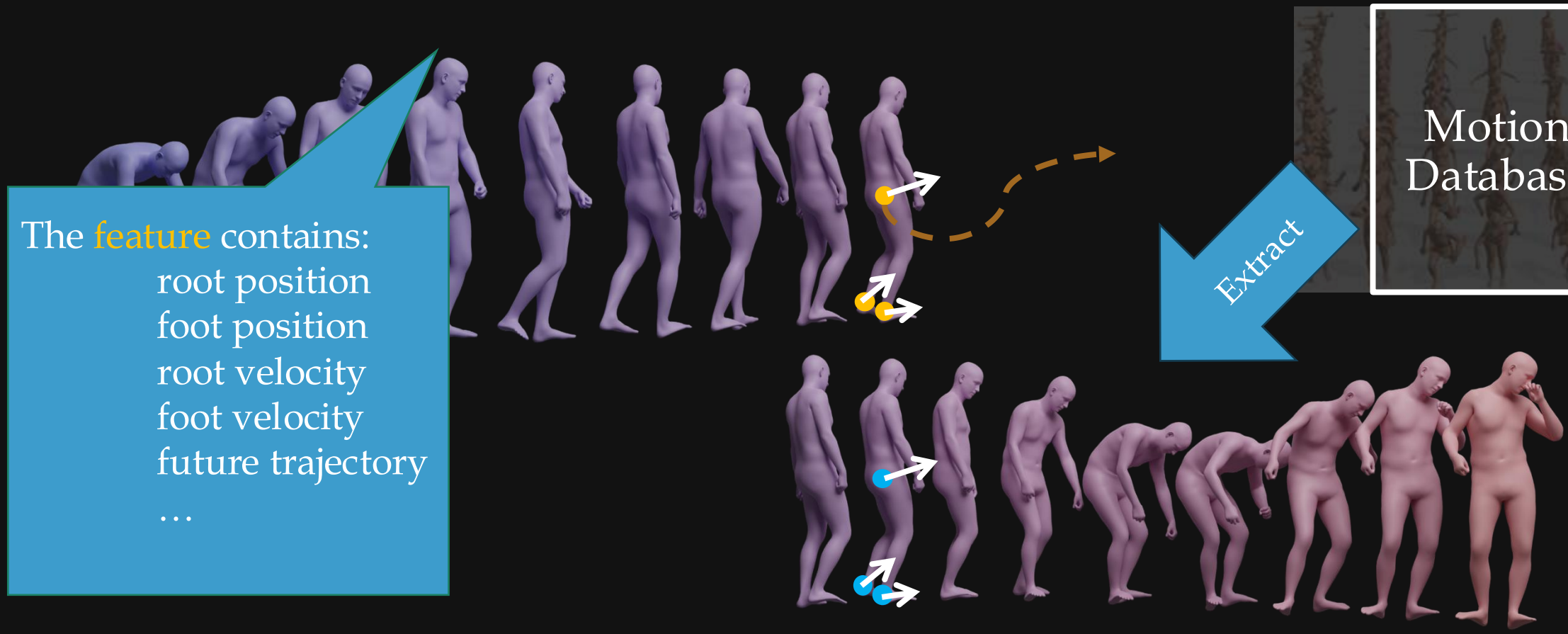


Extract



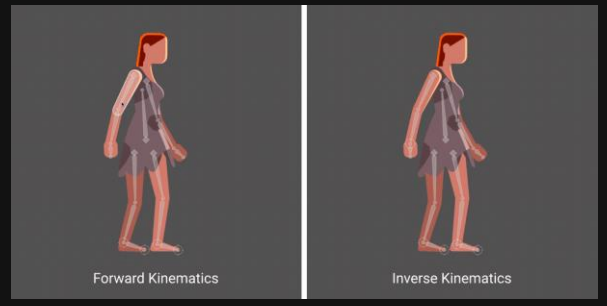
Motion Matching (2010s)

- Motion matching searches for next motion segment from motion database based on the features of previous segment

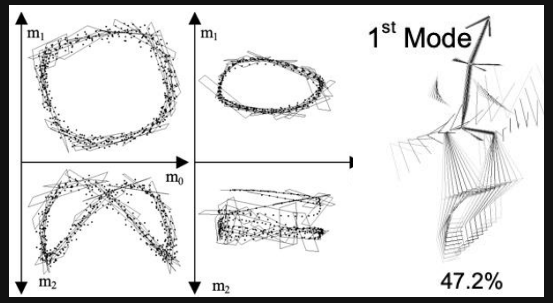


Generative AI Methods (2020s)

Retrieval-based



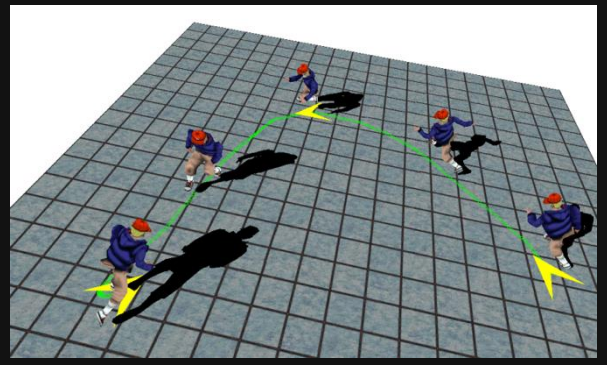
Wang and Chen, 1991
Inverse Kinematics



Arikan and Forsyth, 2002
Statistic Models

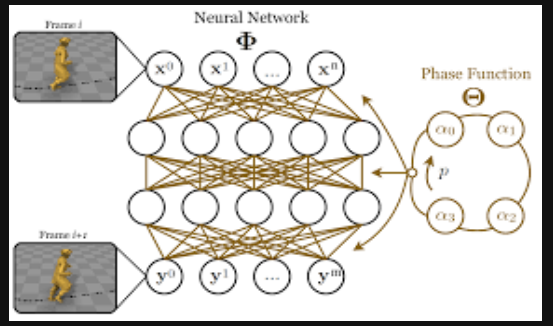


Simon et al, 2016
Motion Matching

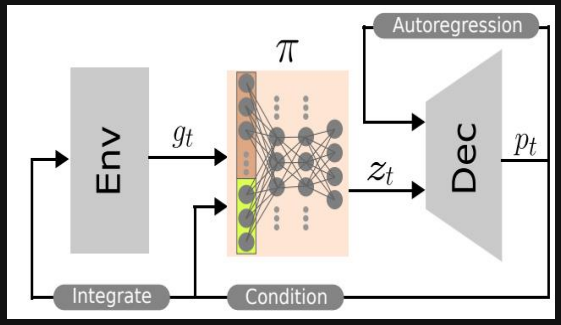


Kovar et al, 2016
Motion Matching

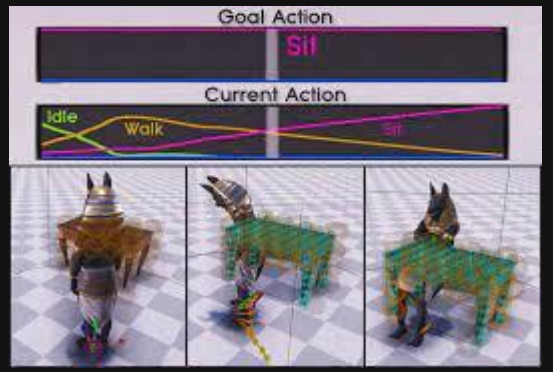
Learning-based



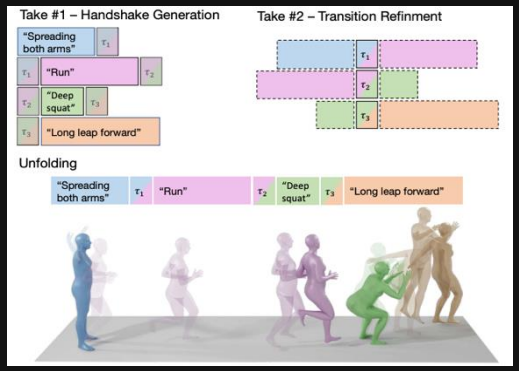
Sheng et al, 2021
PFNN



Ling et al, 2021
Motion VAE

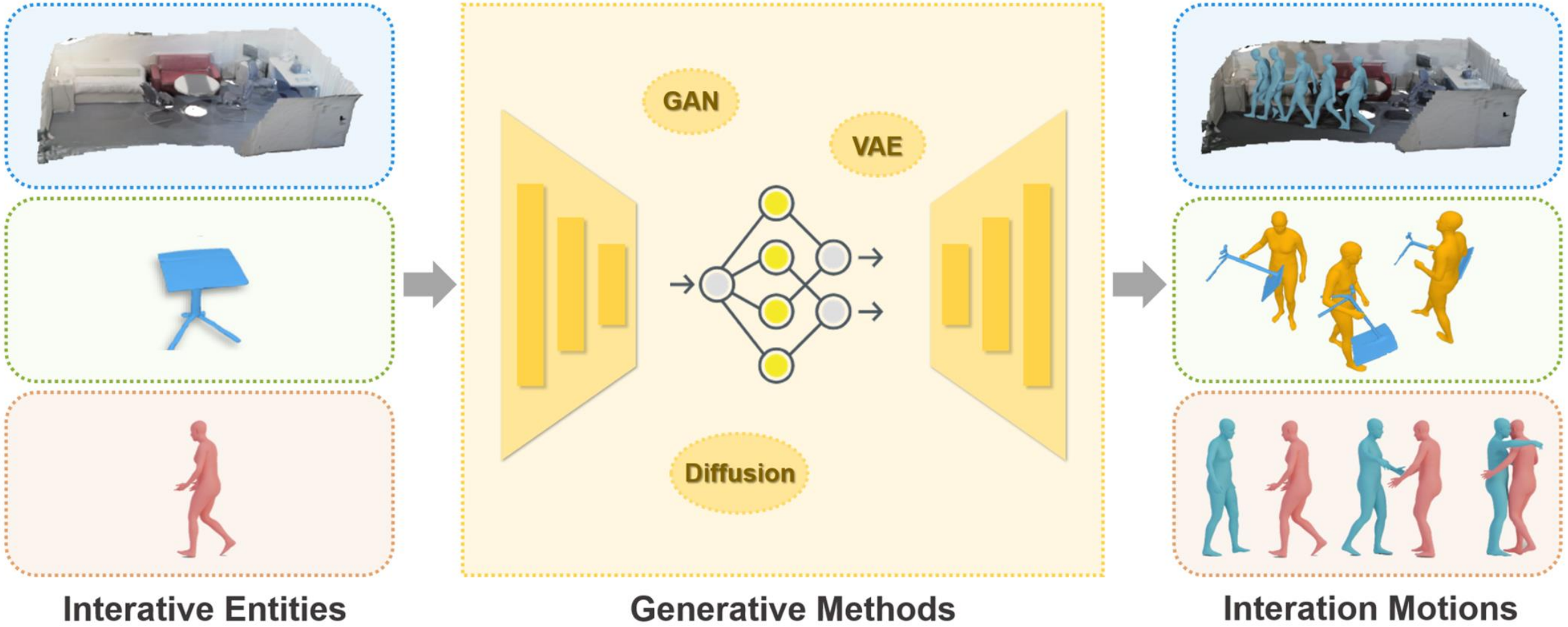


Starke et al, 2019
Neural State Machine

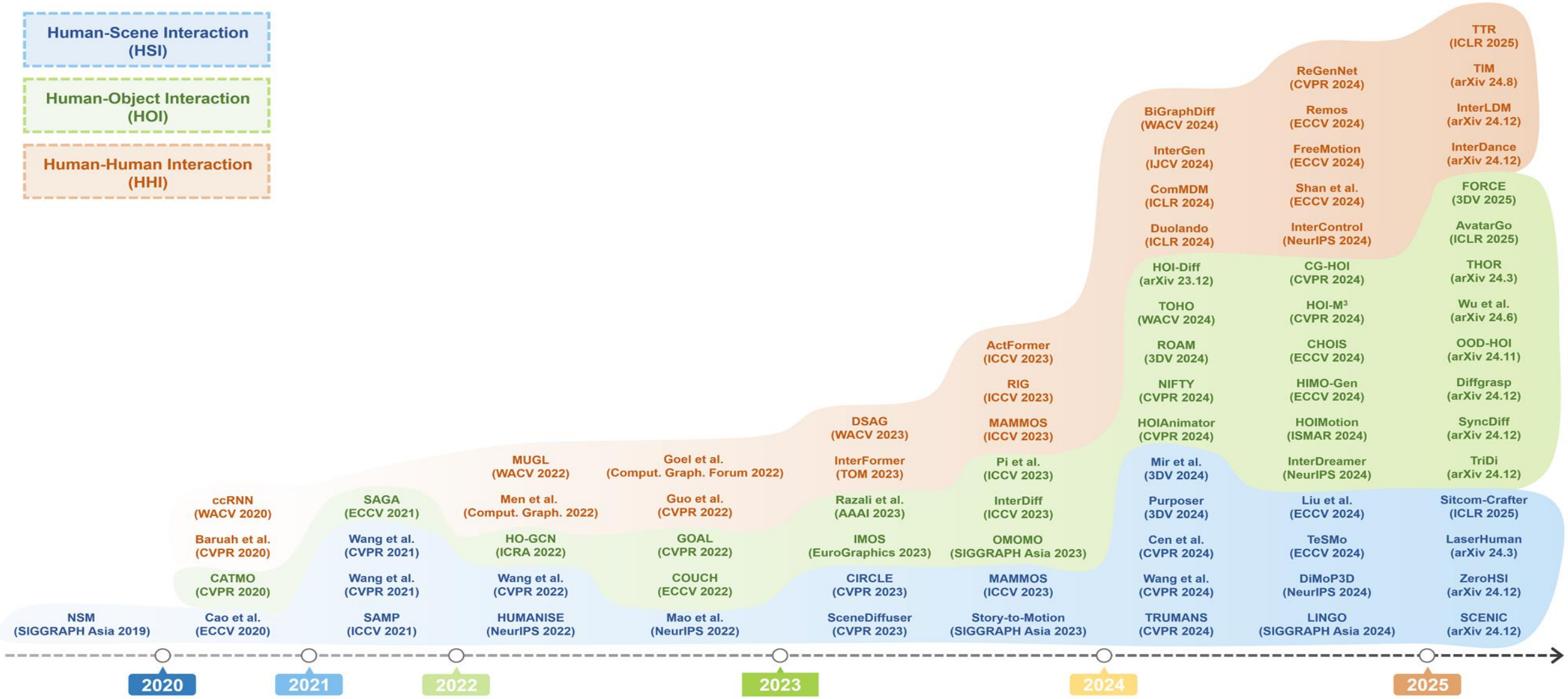


Shafir et al, 2023
Diffusion Model

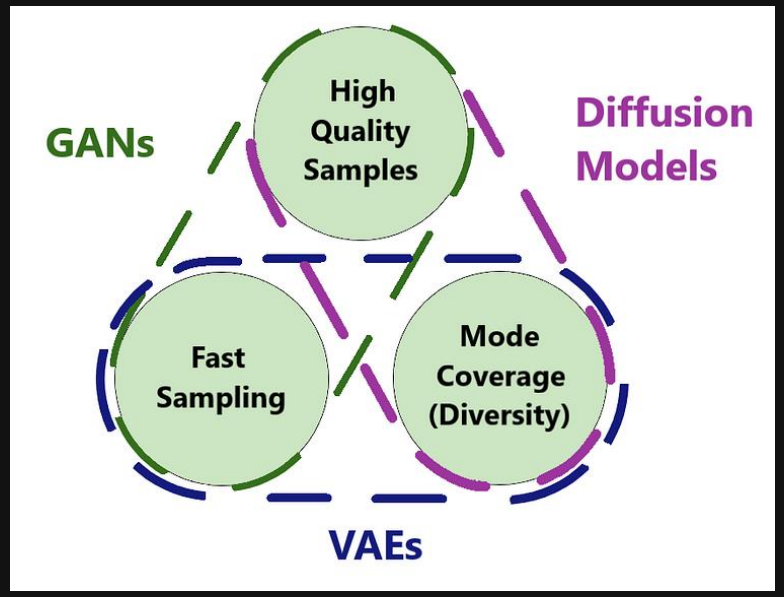
Generative AI Methods (2020s)



Generative AI Methods (2020s)

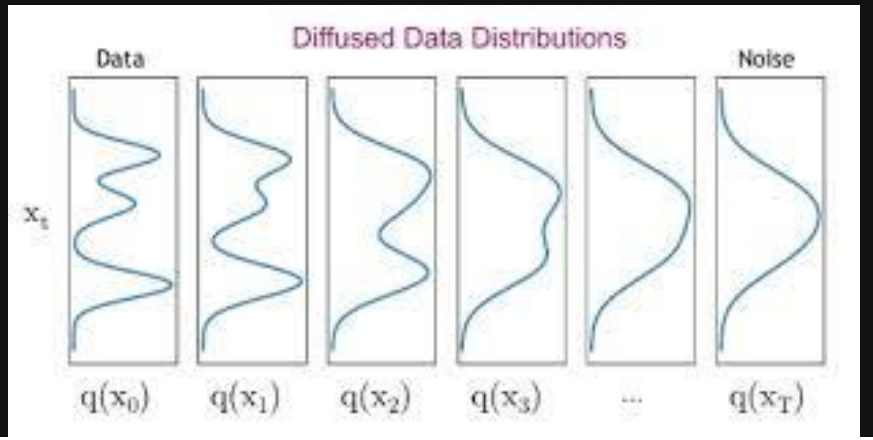


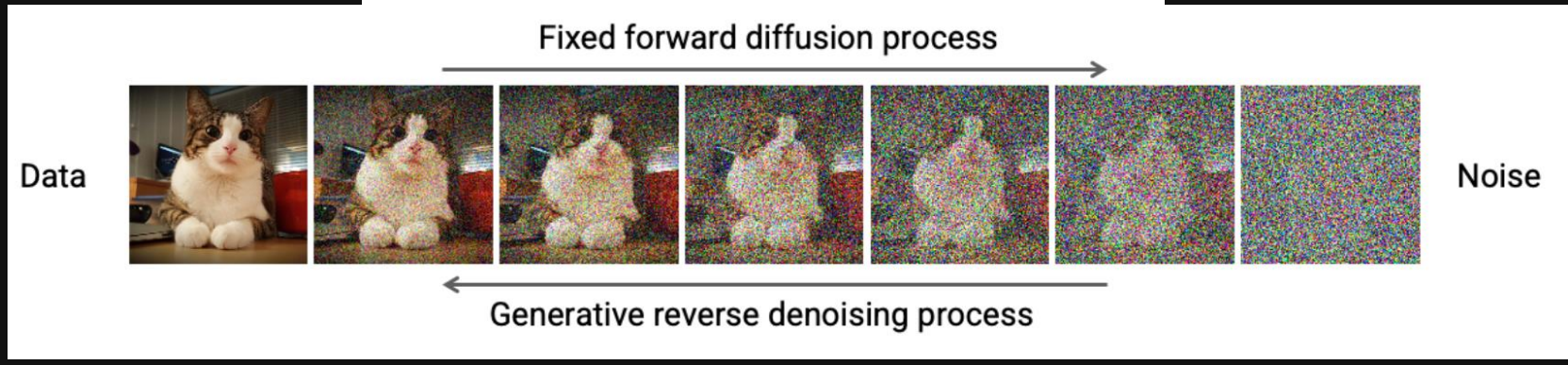
Generative AI Methods (2020s)



Three types of generative models

Diffusion model can fit data distribution better





Denoising Diffusion-based Generative Modeling: Foundations and Applications

Generative AI Methods (2020s)

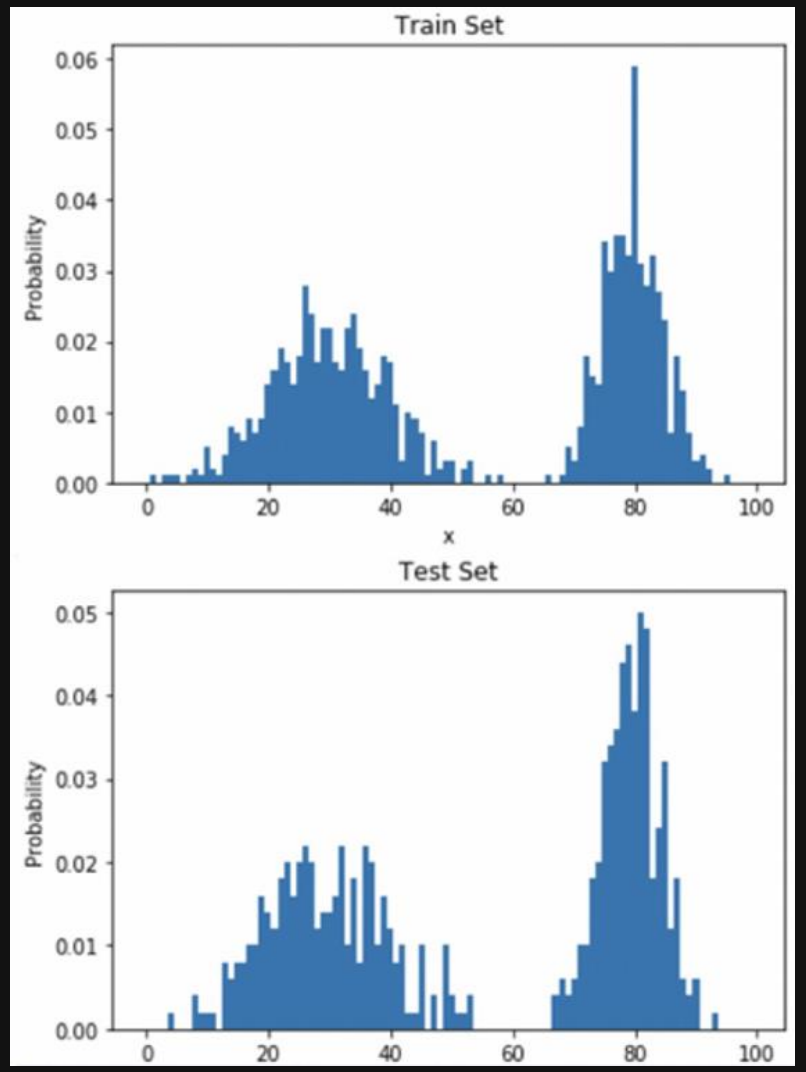
An example of denoising process using diffusion model



Recall: Data Synthesis

Let's say we have some training set of data following some distribution p(x)

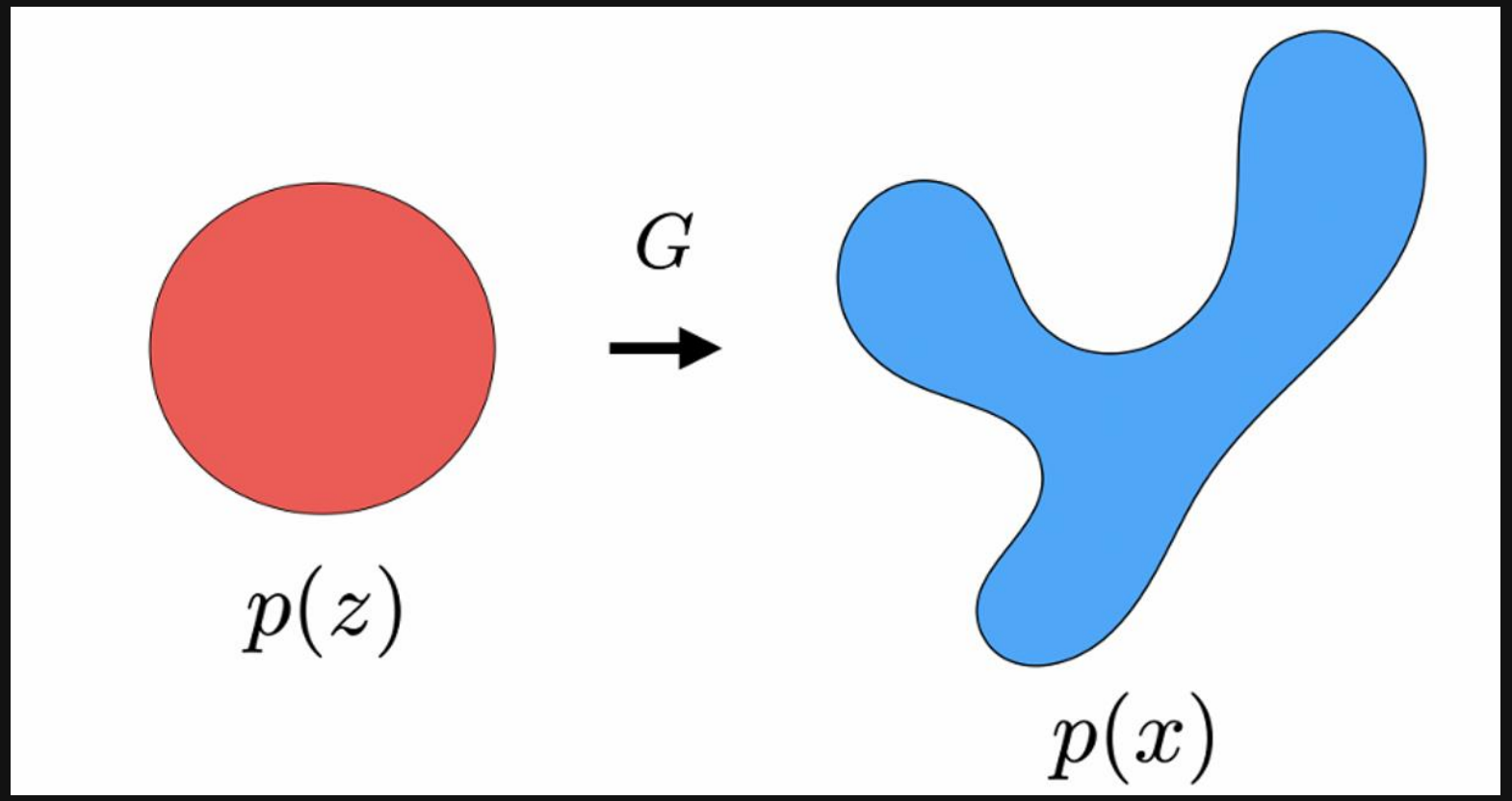
The goal of generative models is to learn this distribution to the best of their ability



We generate new data by sampling from this learned distribution

Sampling from Noise

Source distribution Target distribution

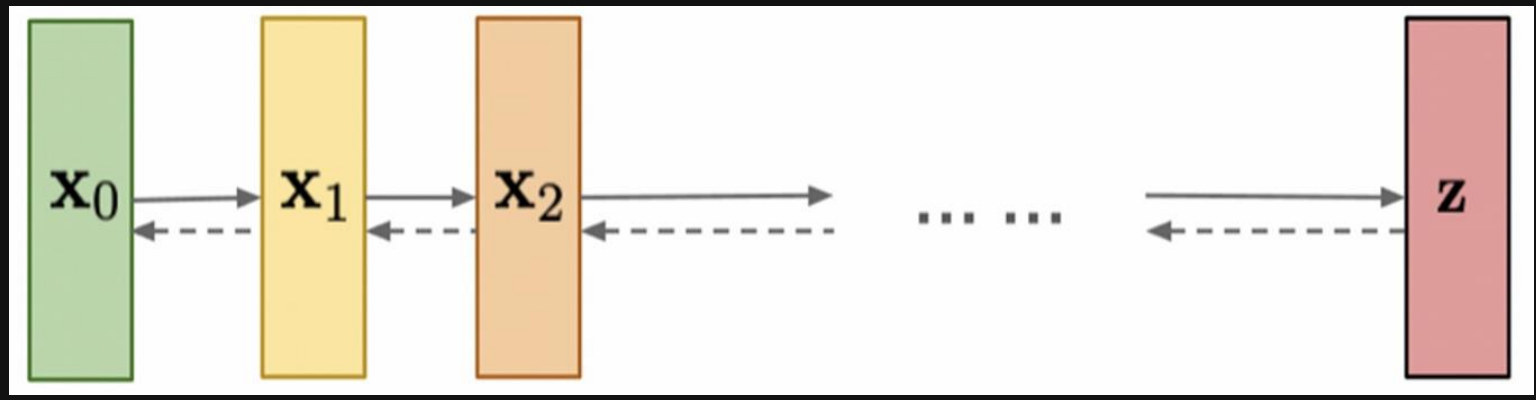


Normally, the source distribution is relatively simple, and the target distribution can be much more complex

Intuition of Diffusion Model

- Estimating and analyzing small step sizes is more tractable/easier than a single step from random noise to the target distribution
- Convert a well-known and simple base distribution (like Gaussian distribution) to the target distribution iteratively, with small steps, via a Markov chain

Diffusion models: gradually add Gaussian noise and then reverse



Forward Process

- Noise added can be parameterized by:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}) \quad \{\beta_t \in (0,1)\}_{t=1}^{T}$$

The parameter (mean and std) of Gaussian is varied according to the noise scheduler

- You can prove with some math that as T approaches infinity, you end up with an Isotropic Gaussian (i.e., pure random noise)
- Note: forward process is fixed

Reparameterization Trick

- Do we have to add noise iteratively to get to some timestep t? Nope!
- Forward process can be written in one step:

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$\alpha_t = 1 - \beta_t$$
$$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$$

- This will be useful during training!

Implementing Forward Process

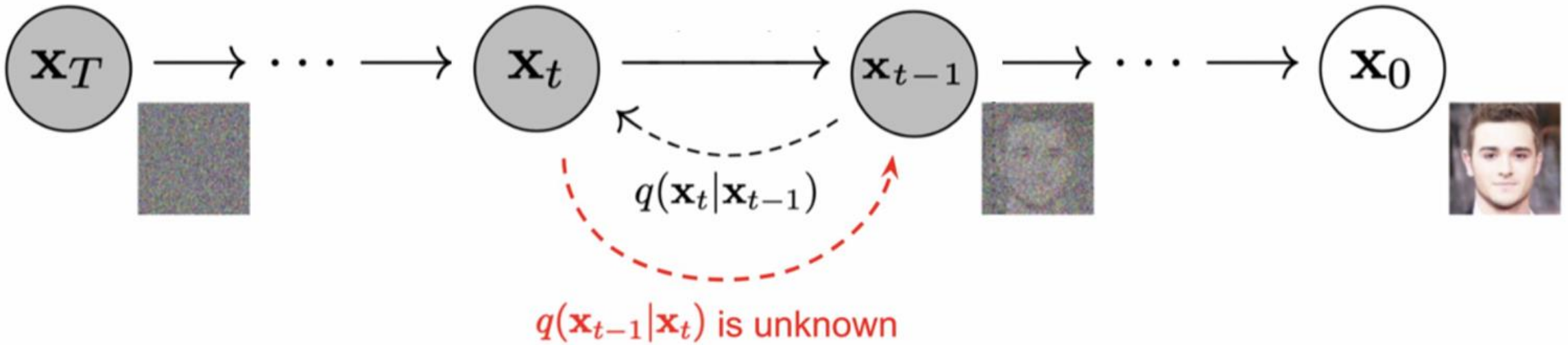
$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0,\ (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$\alpha_t = 1 - \beta_t$$
$$\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$$

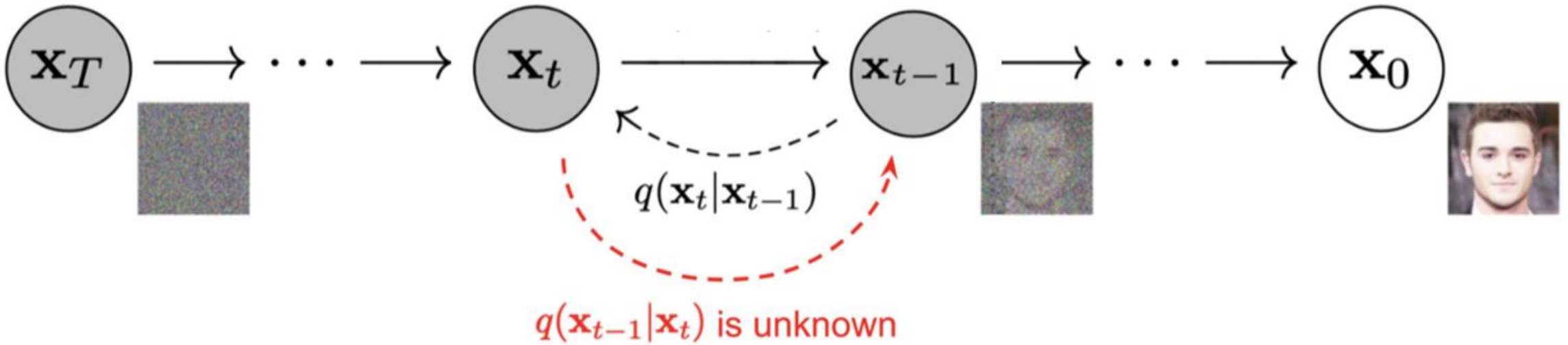
- Let's take image generation as an example

1. Sample an image from the training dataset
2. Sample noise from a standard normal distribution
3. Add the scaled image and noise together to get a 'noisy' image

Reverse Process

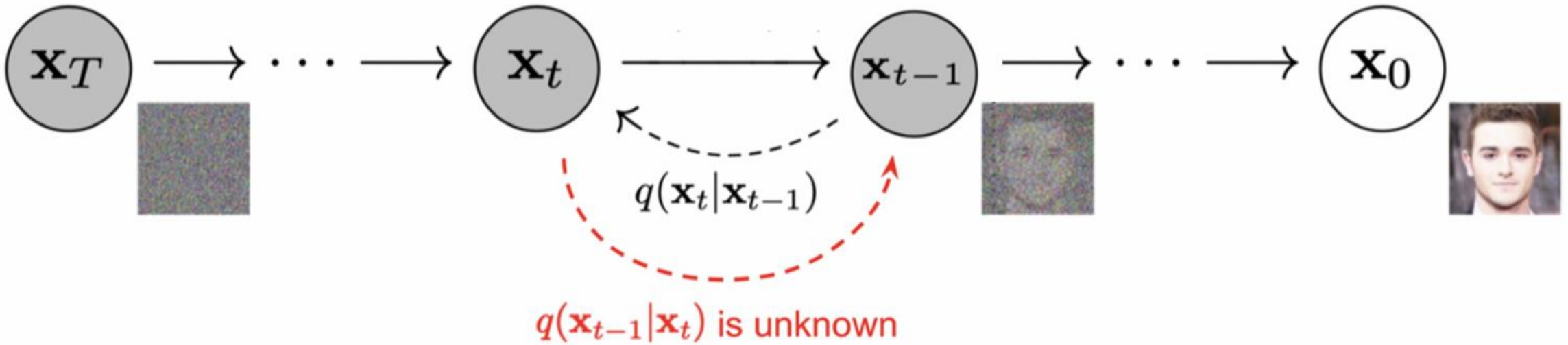


Reverse Process



- The goal of diffusion model is to learn the reverse denoising process to iteratively undo the forward process
- In this way, the reverse process as if it is generating new data from random noise

Reverse Process



- In practice, we can parameterize the learned reverse process as:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

- Therefore, we can use neural networks to estimate the mean and std

Learning Objective: Intuition

- We can get the loss function look like this:

$$L_{\text{VLB}} = L_T + L_{T-1} + \cdots + L_0$$

$$\text{where } L_T = D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))$$

$$L_t = D_{\text{KL}}(q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1})) \text{ for } 1 \leq t \leq T-1$$

$$L_0 = -\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)$$

Note that we have ignored a huge amount of math here

Learning Objective: Intuition

$$L_t = D_{\mathrm{KL}}(q(\mathbf{x}_t|\mathbf{x}_{t+1}, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_t|\mathbf{x}_{t+1})) \text{ for } 1 \leq t \leq T-1$$

- KL divergence: measure the "distance" between two distributions

- If high: very dissimilar distributions
- If low: very similar distributions

Our goal is to drive this very low

Reverse Process

- Recall that we use a network to parameterize the mean of the noisy data distribution:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$$

- Therefore, we minimize a mean-square error between the predicted mean

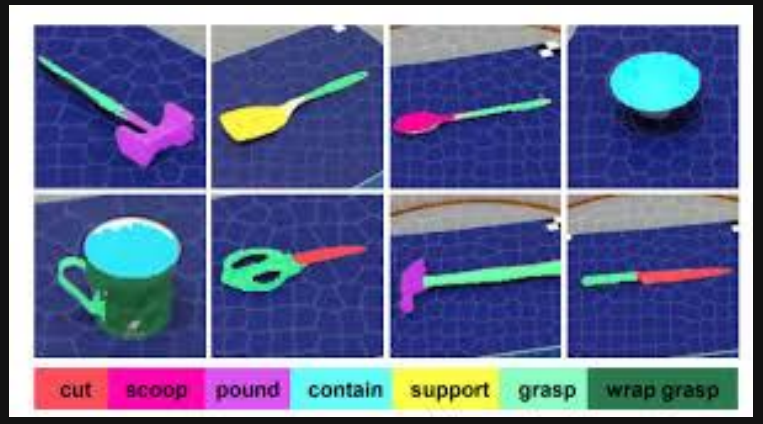
$$\mathrm{MSE}(\mu_\theta(x_t, t), x_{t-1})$$

Note that sometimes we can only fit the mean and provide a pre-defined std

Outline

- Section 1: 2D HOI
  - General formulation
  - HOI detection
- Section 2: 3D HOI
  - 3D representation
  - 3D reconstruction from 2D image
  - From perception to generation
- Section 3: From HOI to HSI
  - Scene-conditioned human motion generation
  - Human-centric scene synthesis

Recap: Object Affordance vs. Scene Affordance



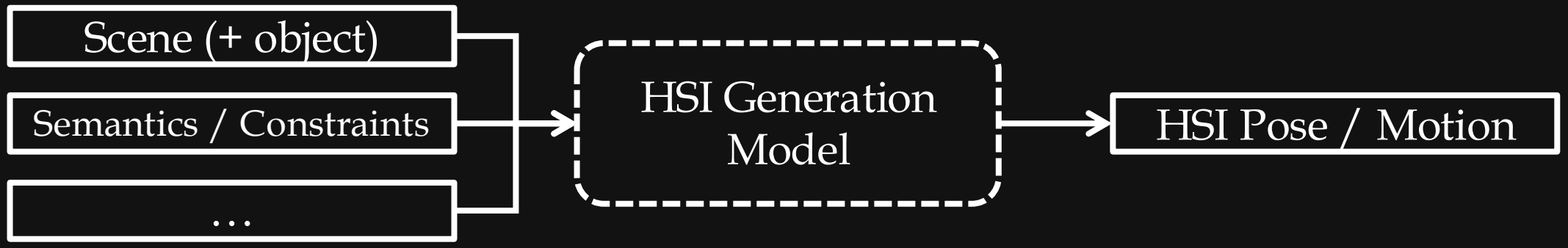


- Object Affordance
  - potential interactions of objects.
  - based on objects properties.
    - shape, size, articulation, etc.

Scene Affordance

- potential interactions within a broader context or environment.
- based on combination and arrangement of multiple objects.

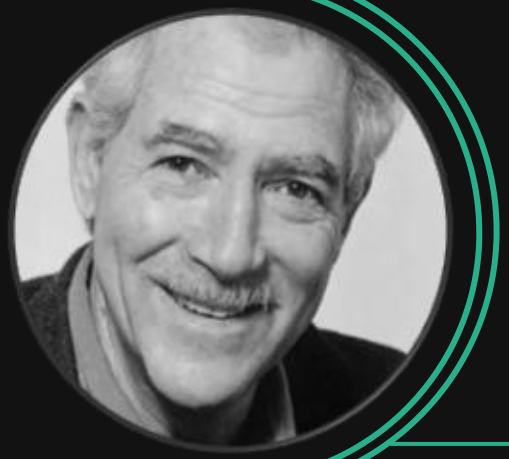
From HOI to Human-Scene Interaction (HSI)







Three stages of HSI

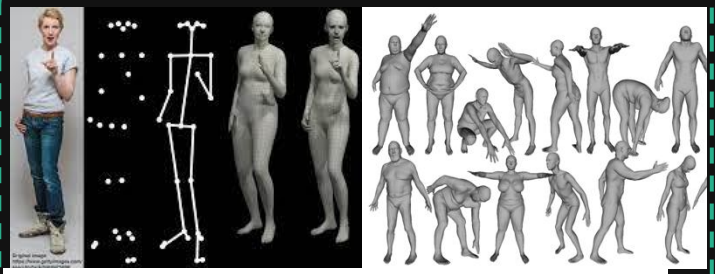


> The infant first experiences the emergent self (0-2 months), followed by the subjective self capable of intersubjectivity (2-15 months), and finally the verbal self engaged in symbolic interaction.

Stage	Infants develop understanding of themselves and the external world	Infants begin to actively explore control of bodily movements based on surrounding scenes	Children are able to understand and respond to high-level, symbolic semantic instructions
Research Problem	 Modeling and joint representation of human body and scenes	 Modeling action probability distributions that conform to scene constraints	 Text-guided motion generation

Recap: 3D representation

Human



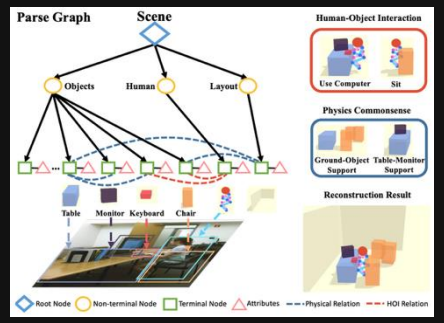
Matthew et al, 2015

Pose params θ
Joint rotations

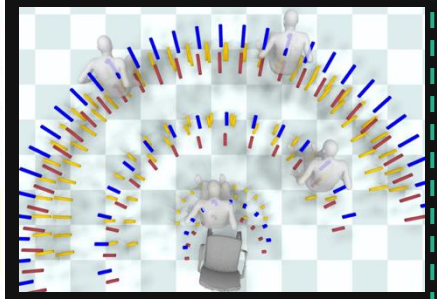
Shape params β
Body shape PCAs

Humam 3D model
Mapping θ and β to human surface mesh

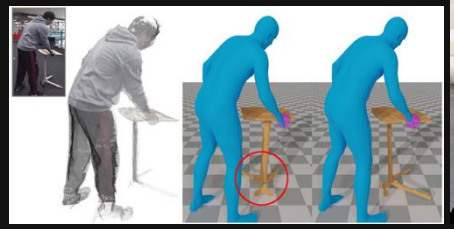
Interaction



Qi et al, 2016
Parse Graph



Kulkarni et al, 2023
Interaction Field

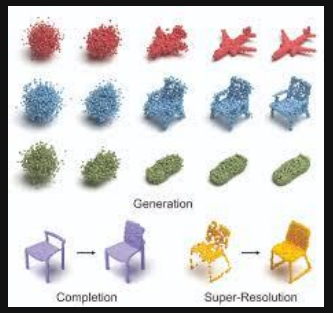


Bharat et al, 2018
Contact Points

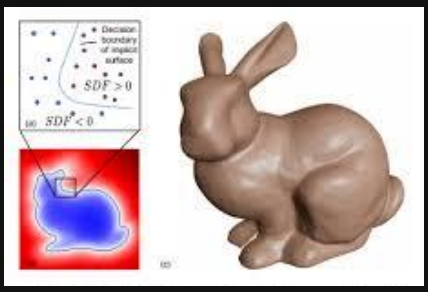


Wang et al, 2024
Affordance

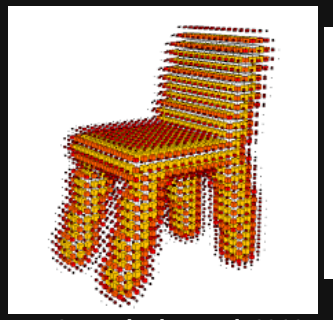
Scene



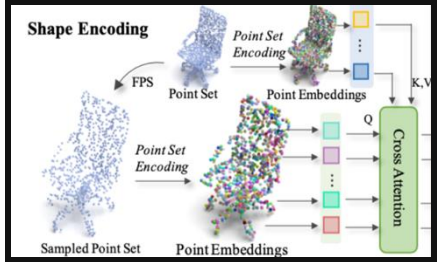
Qi et al, 2016
Point Cloud



Qi et al, 2016
SDF



Qmescheder et al, 2018
Occupancy



Zhang et al, 2023
3DShapeToVecSet

Human-Scene Interaction

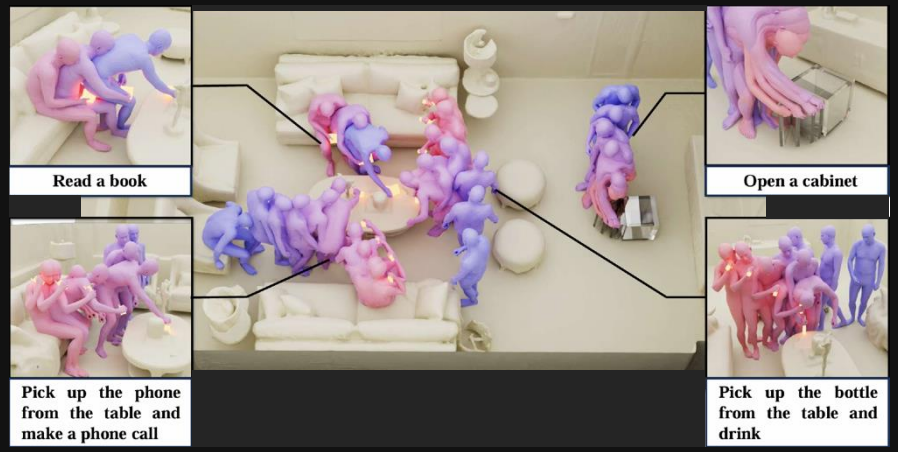
- The Human-Scene Interaction space is defined as

$$\Omega = (M, S, P, L)$$

Motion
Scene
Pose
Language

$M \in \mathbb{R}^{T \times J \times 3}$	Human motion representation, J represents num of joints, T represents frame num
$S \in \{0,1\}^{N \times N \times N}$	Scene voxel grid, N represents resolution of voxel
$P \in \mathbb{R}^{J \times 3}$	Key pose of human
$L \in \mathcal{L}$	Language space, contains the text description of task

Human-Scene Interaction Modeling



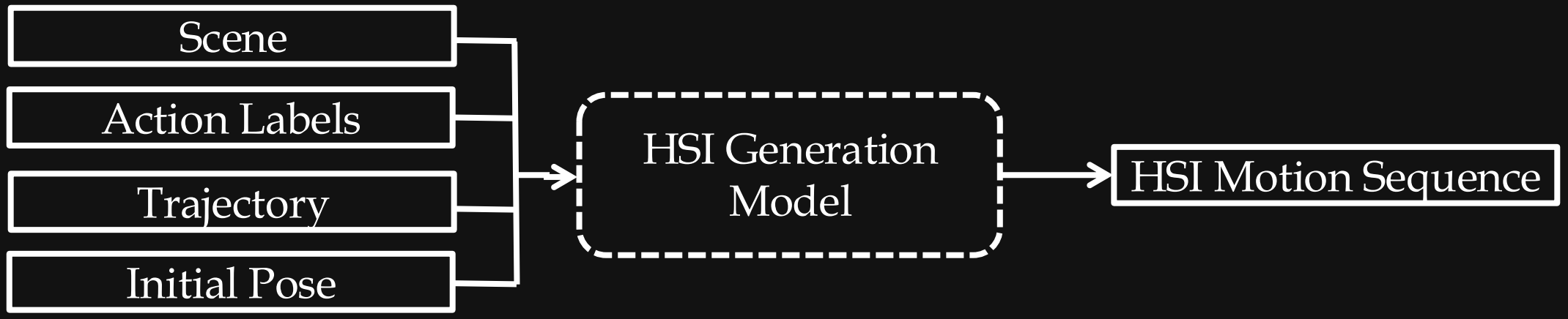


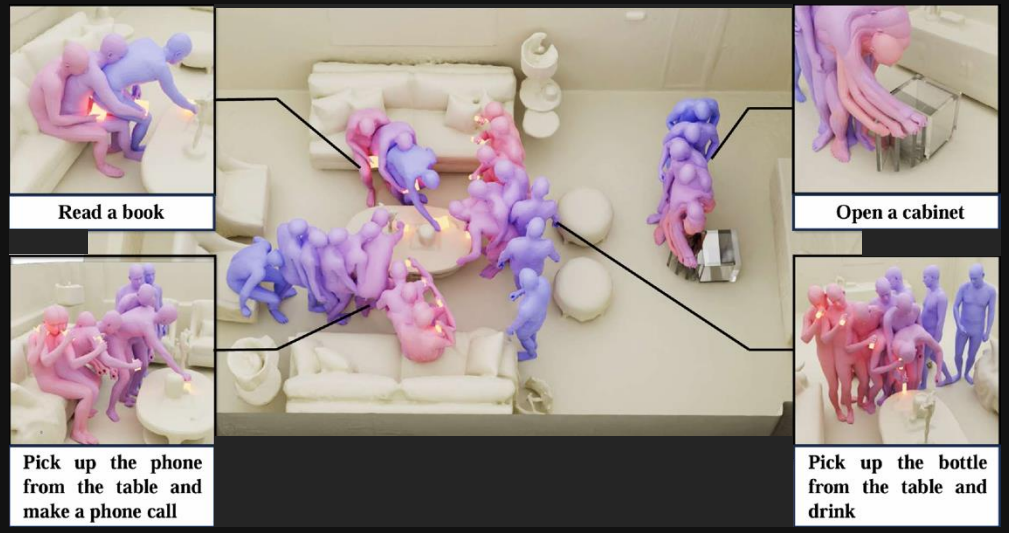






General Formulation: P(M | P, S)







MoCap Dataset

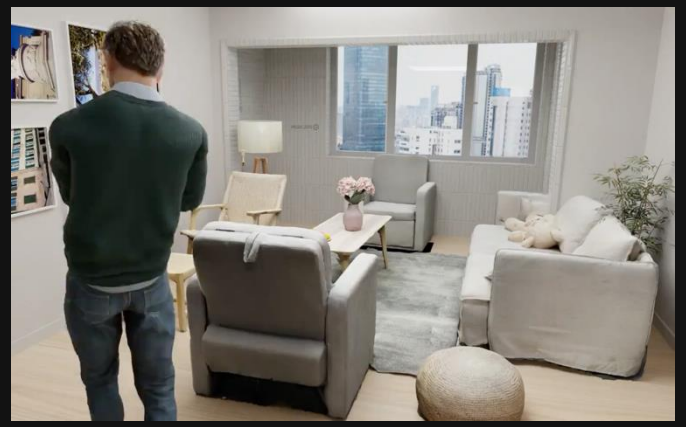
TRUMANS Dataset: modeling human-object interactions across entire scenes

TRUMANS Dataset

TRacking hUMan ActioNs in Scenes







Rich human-scene interaction dataset with diverse annotations (15+ hours) across 100 indoor scenes with extensible objects and backgrounds

MoCap Dataset

Easy to scale up by replacing objects and backgrounds

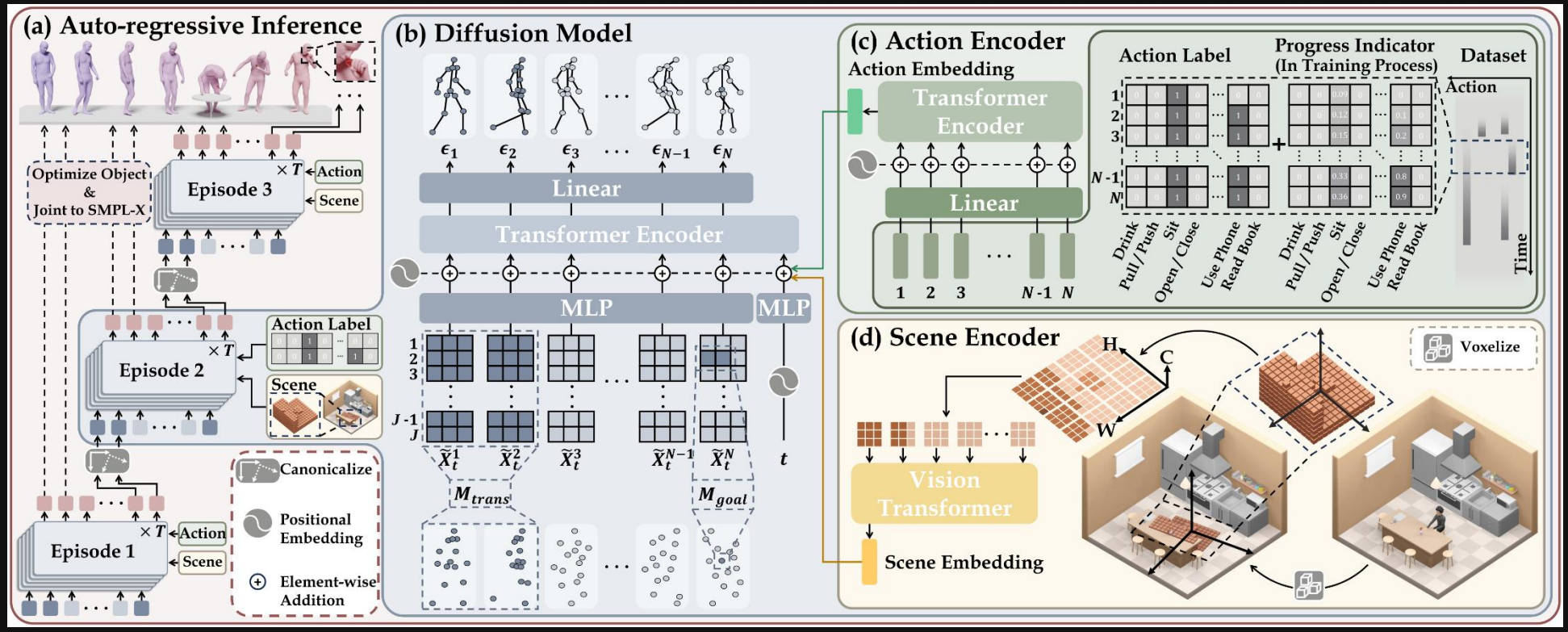
Objects Replacement



Background Replacement



Auto-regressive Diffusion

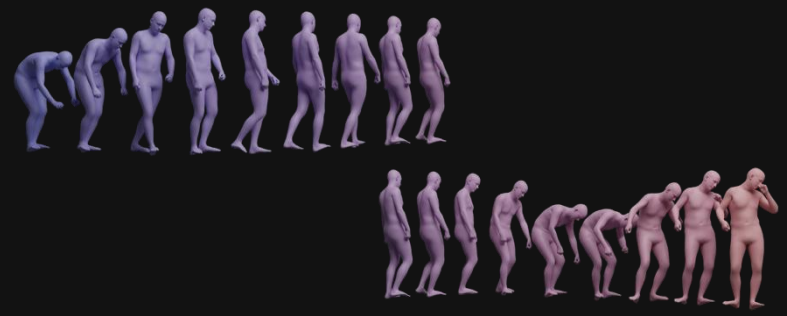


Using autoregressive conditional diffusion model as motion generation framework

Deep understanding and generation of complex human-scene interaction patterns

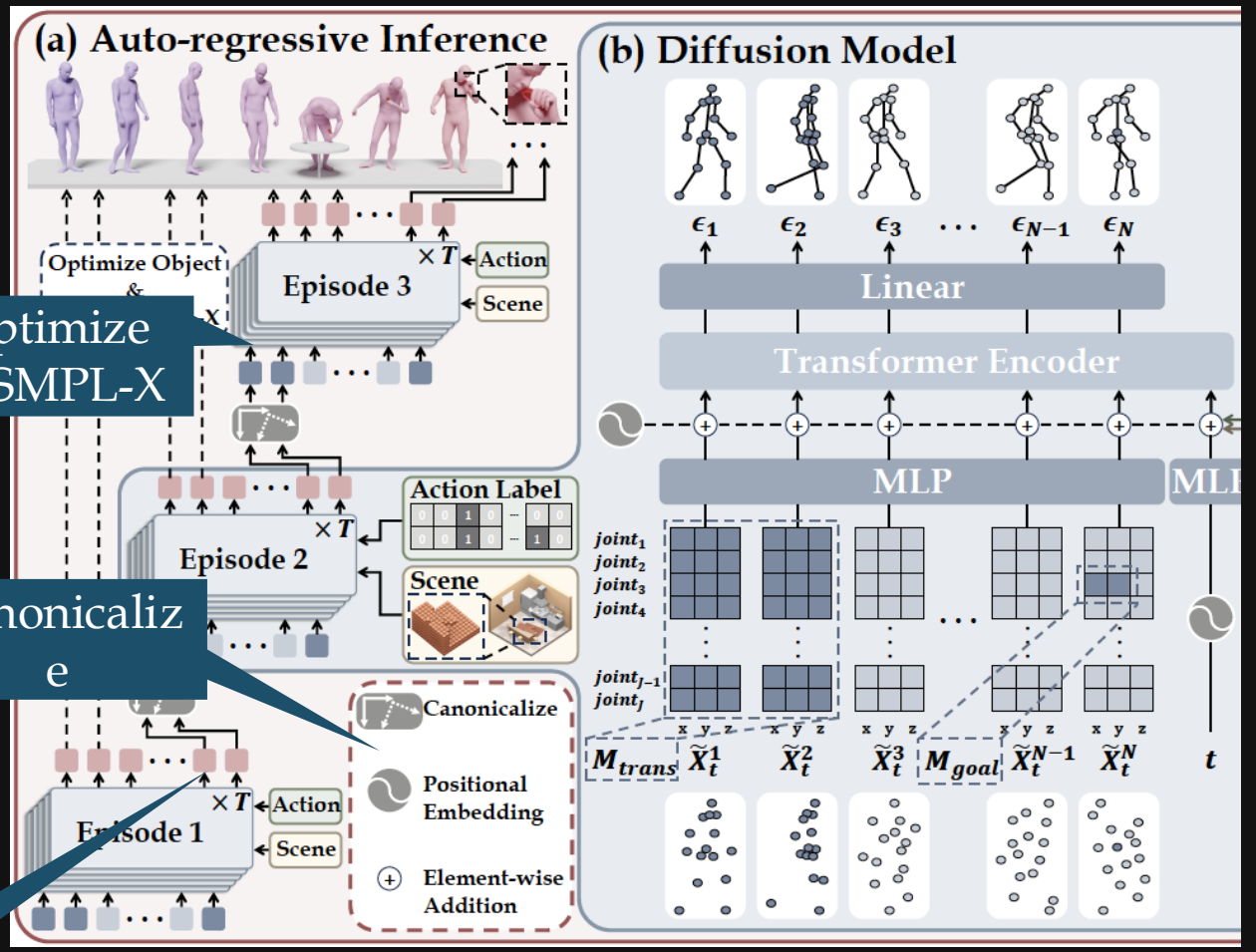
Auto-regressive Diffusion

Each motion segment connects to the end of the previous one, ensuring motion continuity





Generating motions from given body part trajectories

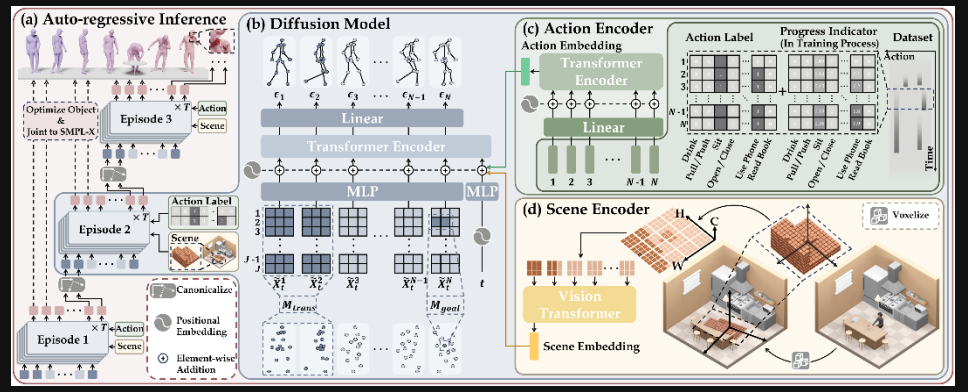


Results

TRUMANS can generate complex, diverse, and natural human-scene interaction sequences



+









Results



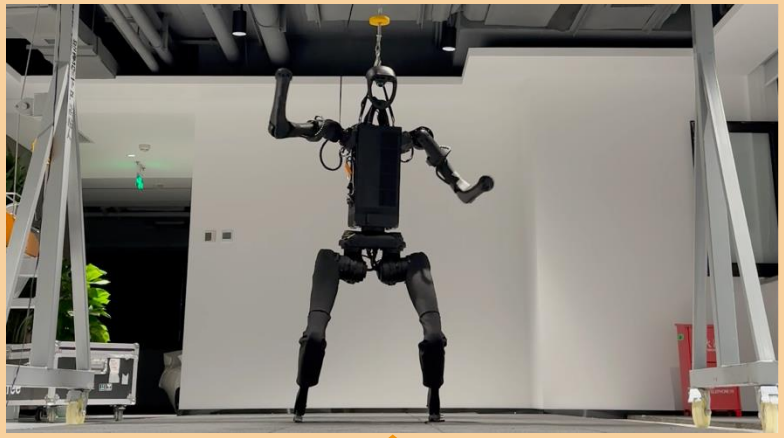
HSI and Humanoids

Planning and Editing: Generating Interactive Motion



Reality

Executing: Enabling Executable Physics

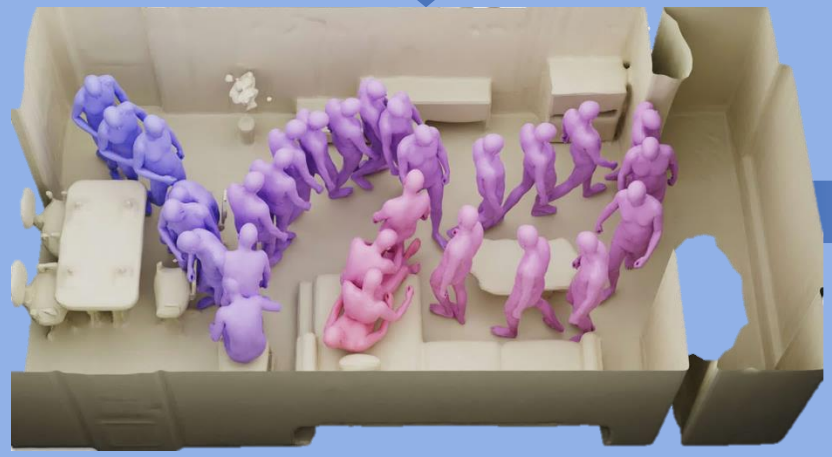


Human

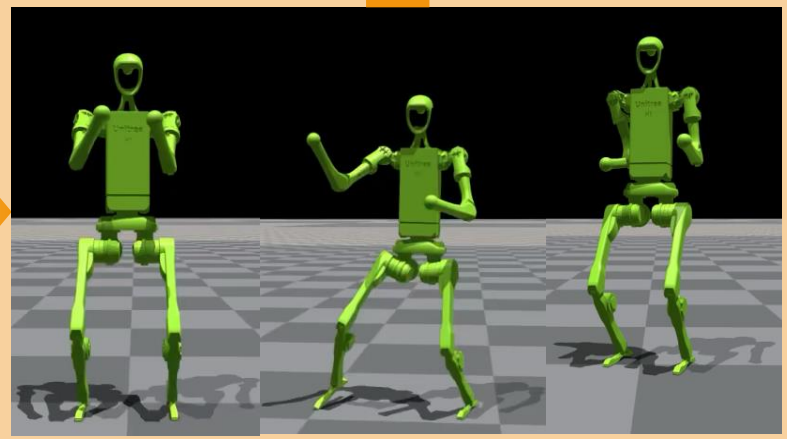
Robot

Motion Generation, Editing

Sim-to-real Transfer



Imitation Learning



Simulator

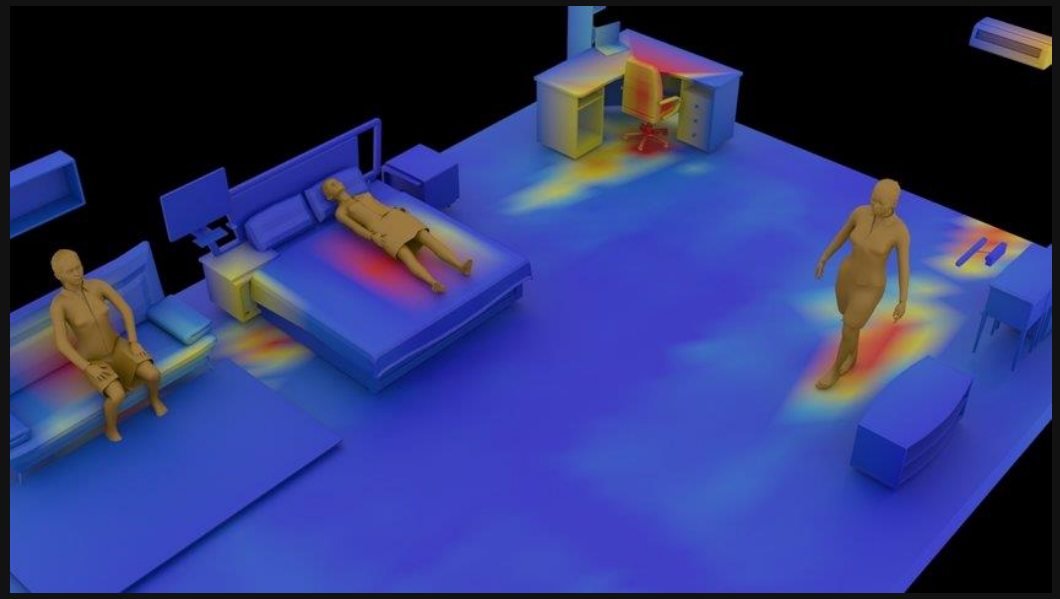
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Recap: Scene Affordance

- “Human-made scenes are functional spaces that serve human activities, whose objects exist primarily to assist human actions.”

— — J. J. Gibson





Recap: Scene Affordance

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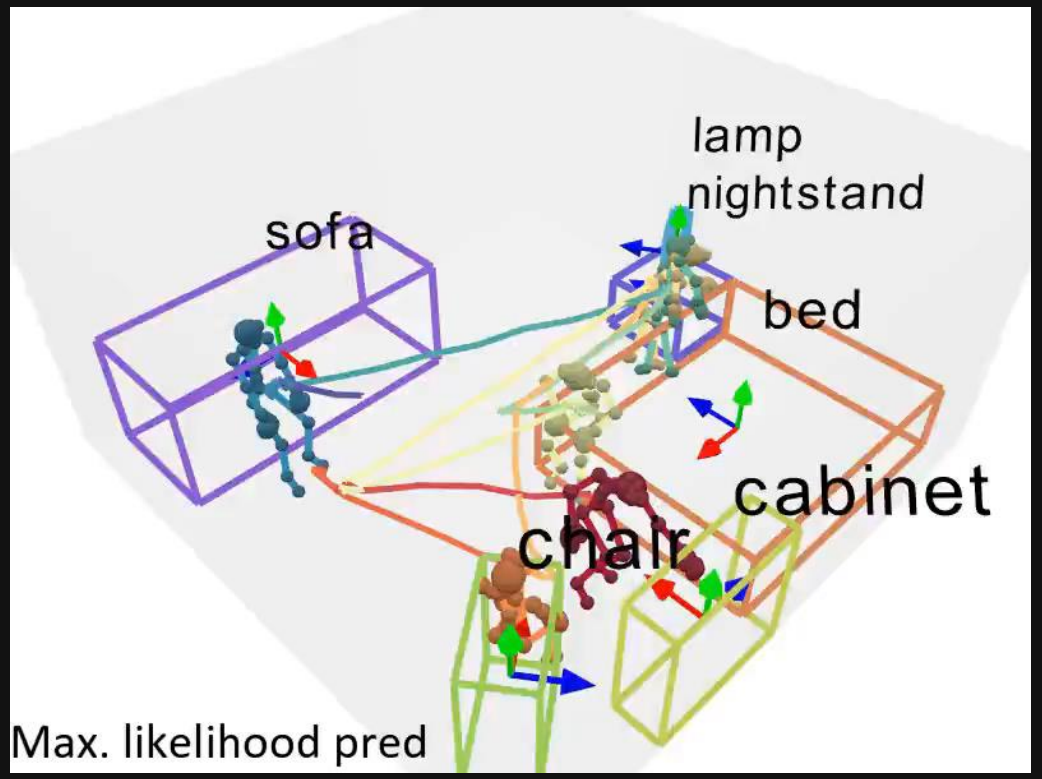
Human cues serve as valuable constraints for human-made scene synthesis

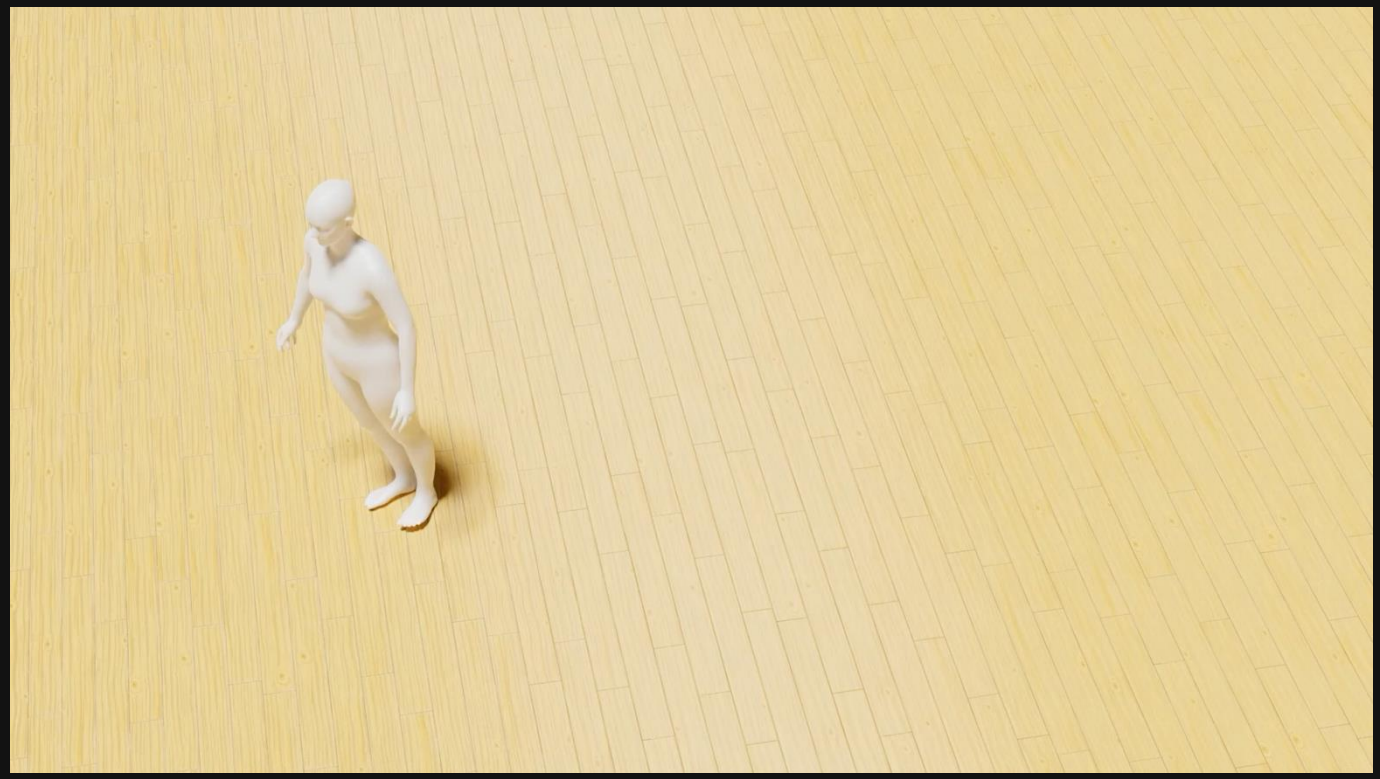


Make it Home


Make it Home:
Automatic Optimization of
Furniture Arrangement
Lap-Fai Yu* Sai-Kit Yeung*
Chi-Keung Tang** Demetri Terzopoulos*
Tony F. Chan** Stanley Osher*
*University of California , Los Angeles
**The Hong Kong University of Science and Technology

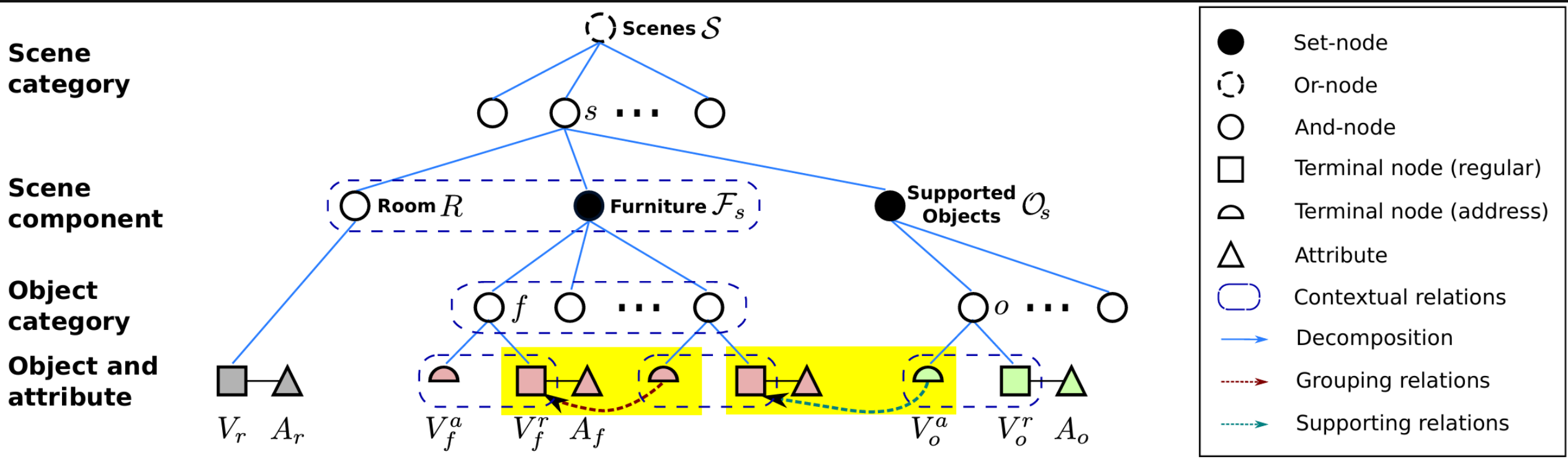
Human-centric Scene Synthesis



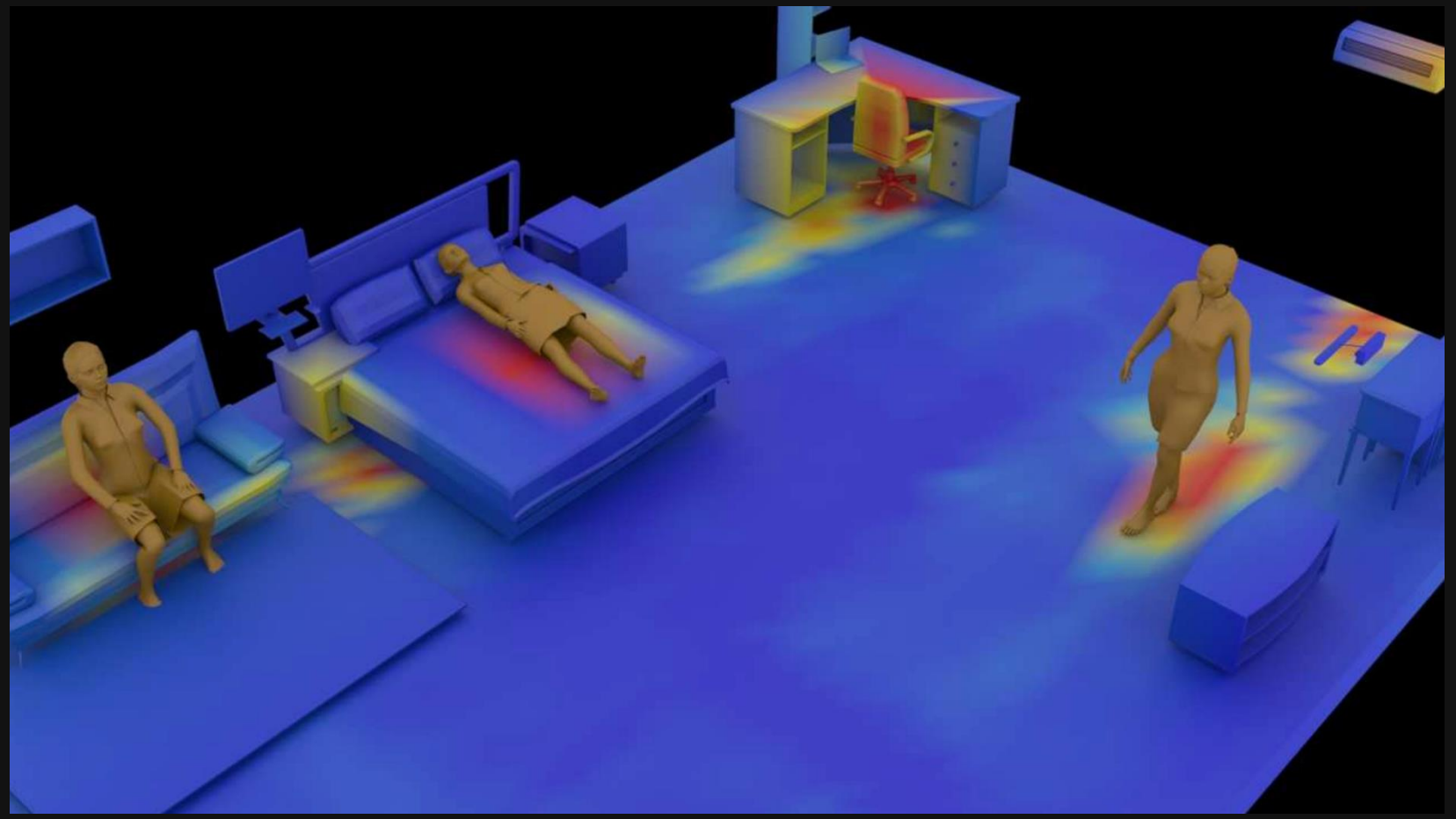


Case study: Human-centric Indoor Scene Synthesis

Represent indoor scenes using spatial And-Or Graph (S-AOG)

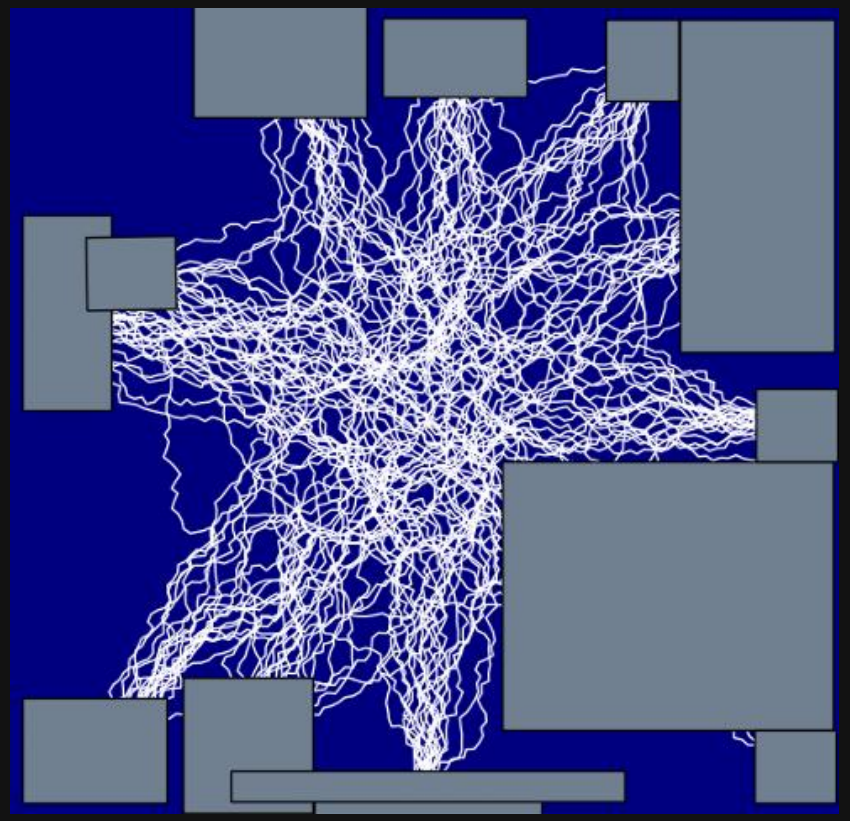


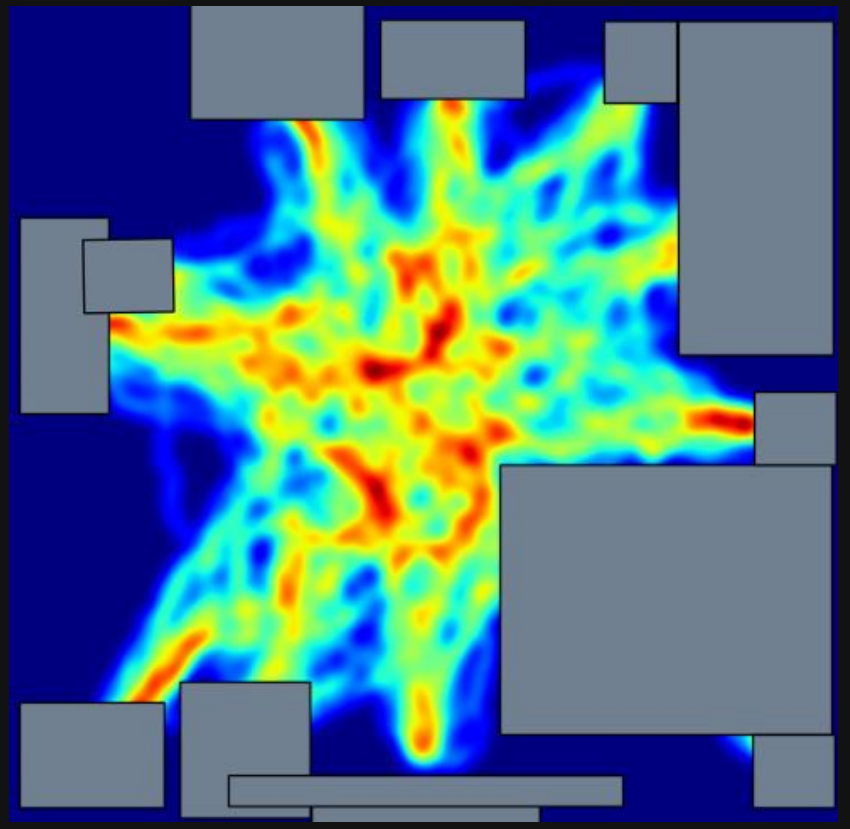
Task: Sample and synthesize human-centric 3D room layouts



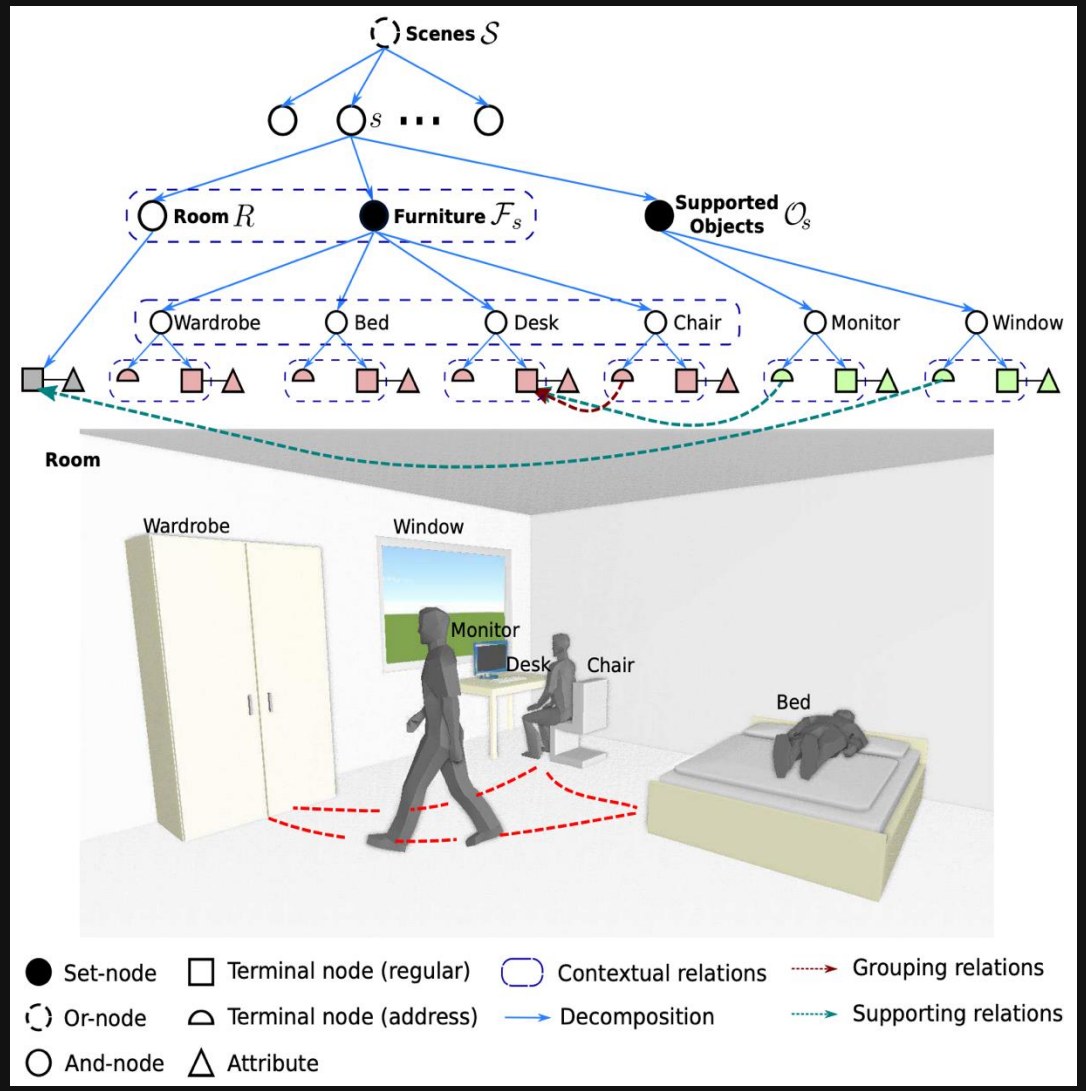
Case study: Human-centric Indoor Scene Synthesis

- Generating human activity probability map with trajectory planning





Approach

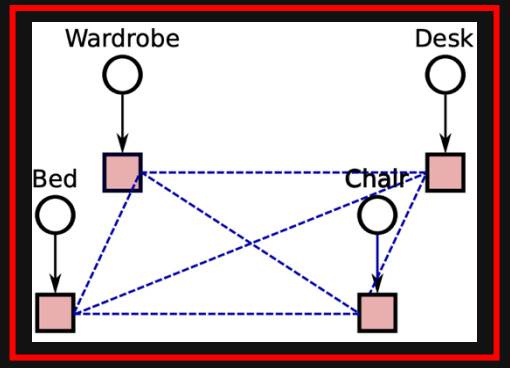


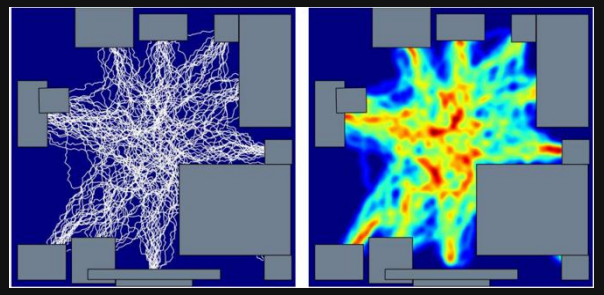
$$p(E_{pt}|\Theta) = \frac{1}{Z}\exp\{-\mathcal{E}(E_{pt}|\Theta)\}$$
$$= \prod_{c\in C_f}\phi_f(c)\prod_{c\in C_o}\phi_o(c)\prod_{c\in C_g}\phi_g(c)\prod_{c\in C_r}\phi_r(c)$$

Define the contribution of possible human activity towards functional group

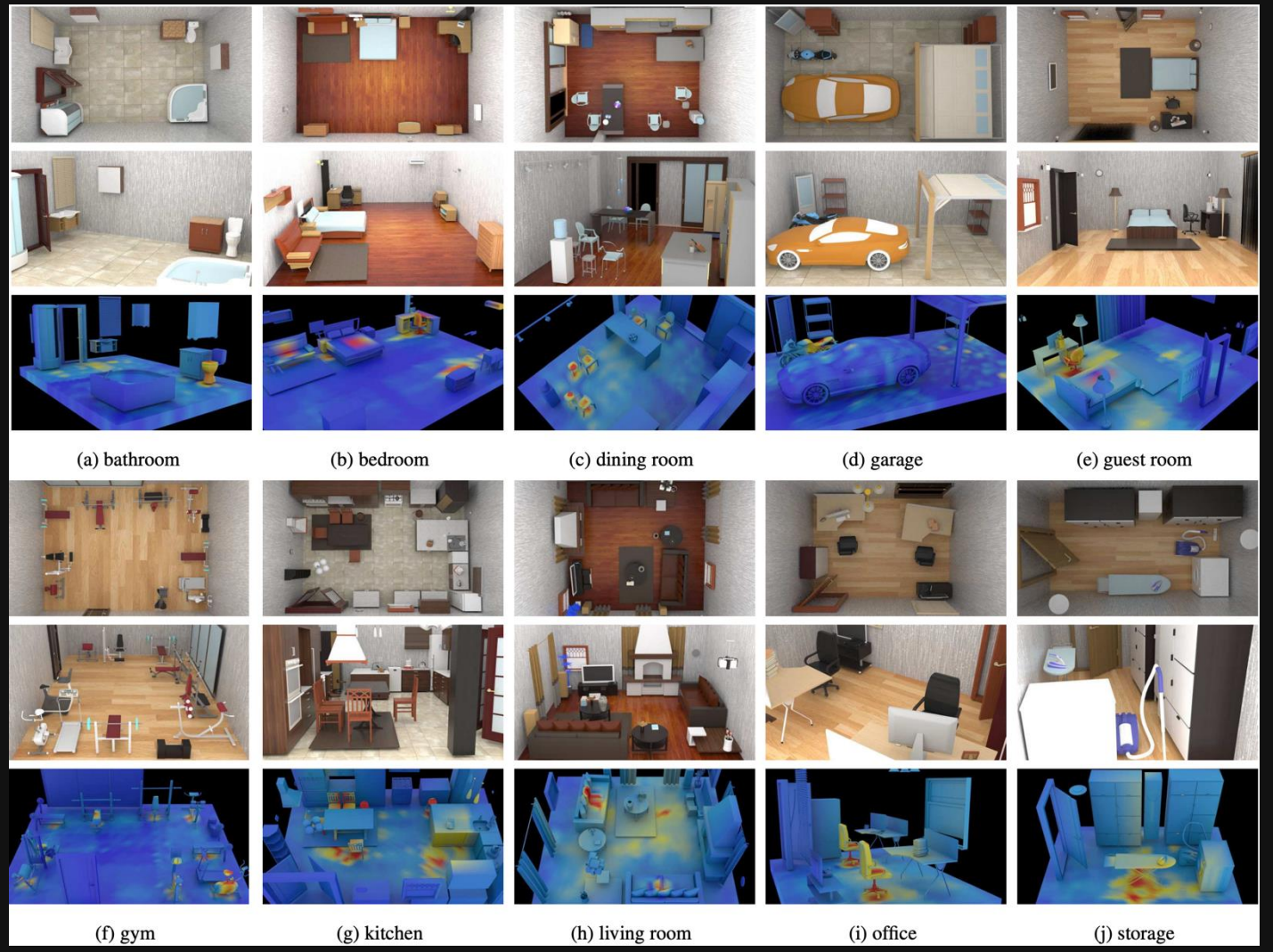
Results

Contextual relations forms a Markov Random Field (MRF)

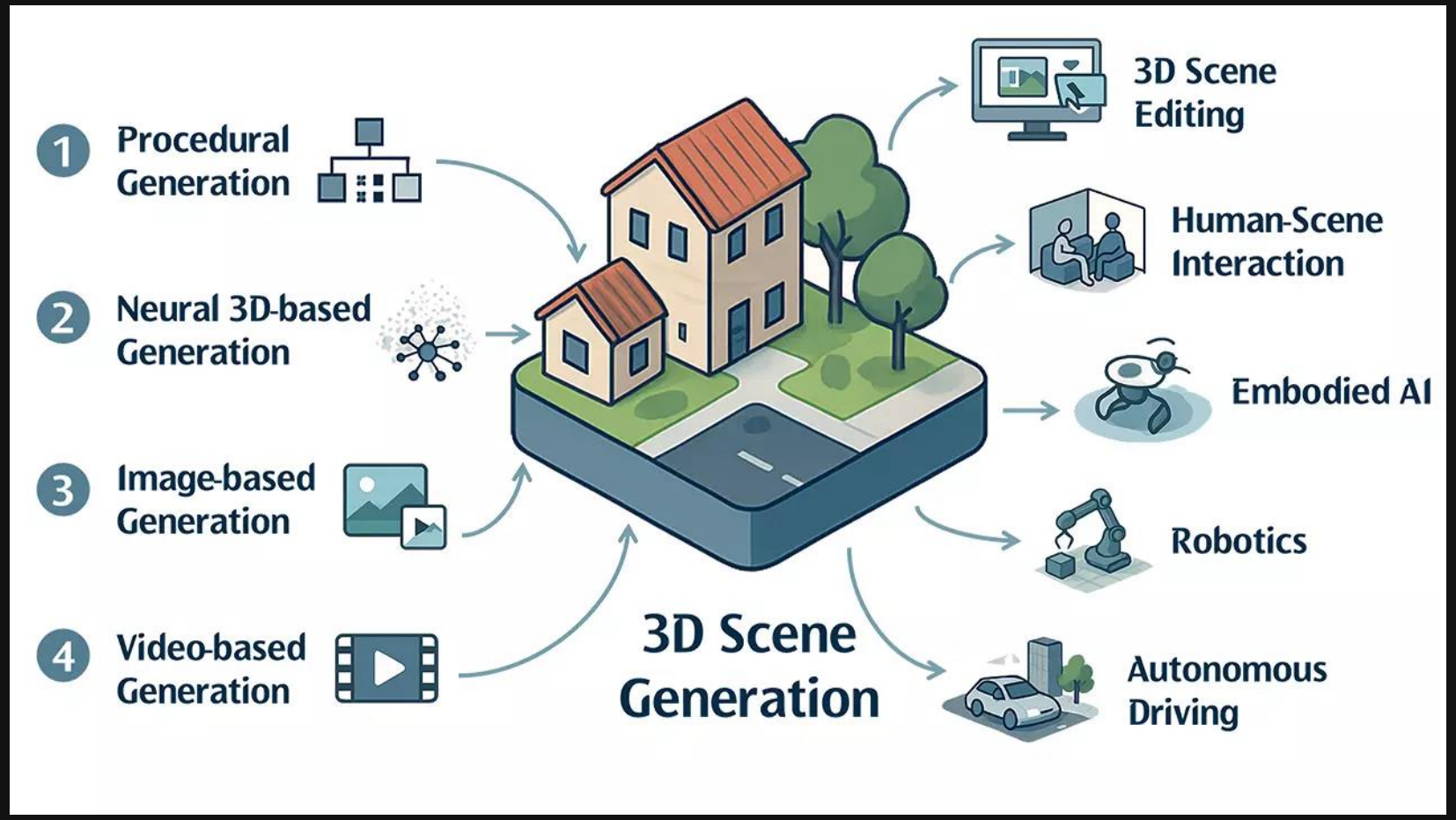




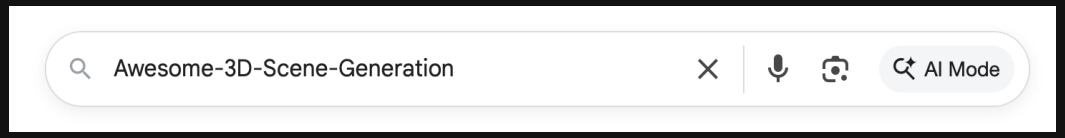
Sampled activity trajectory over the cliques in the MRF



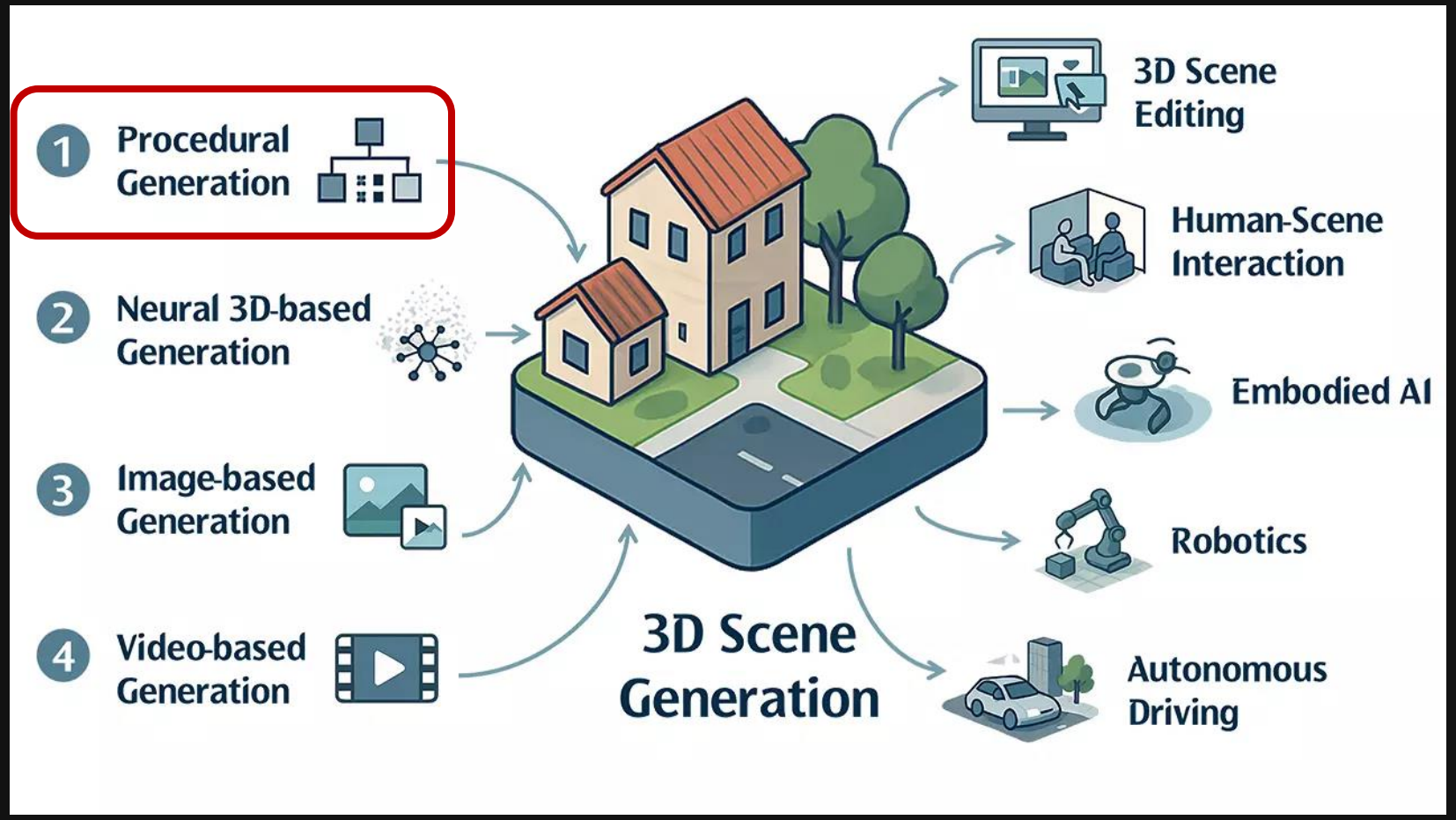
3D Scene Generation: Methods and Usages



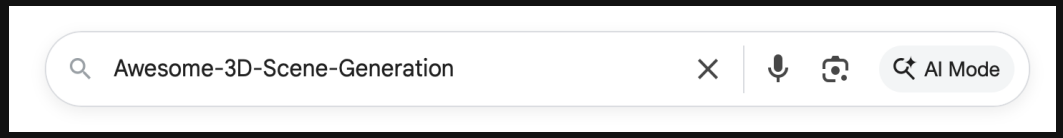
For a complete list of papers, please search on Google:



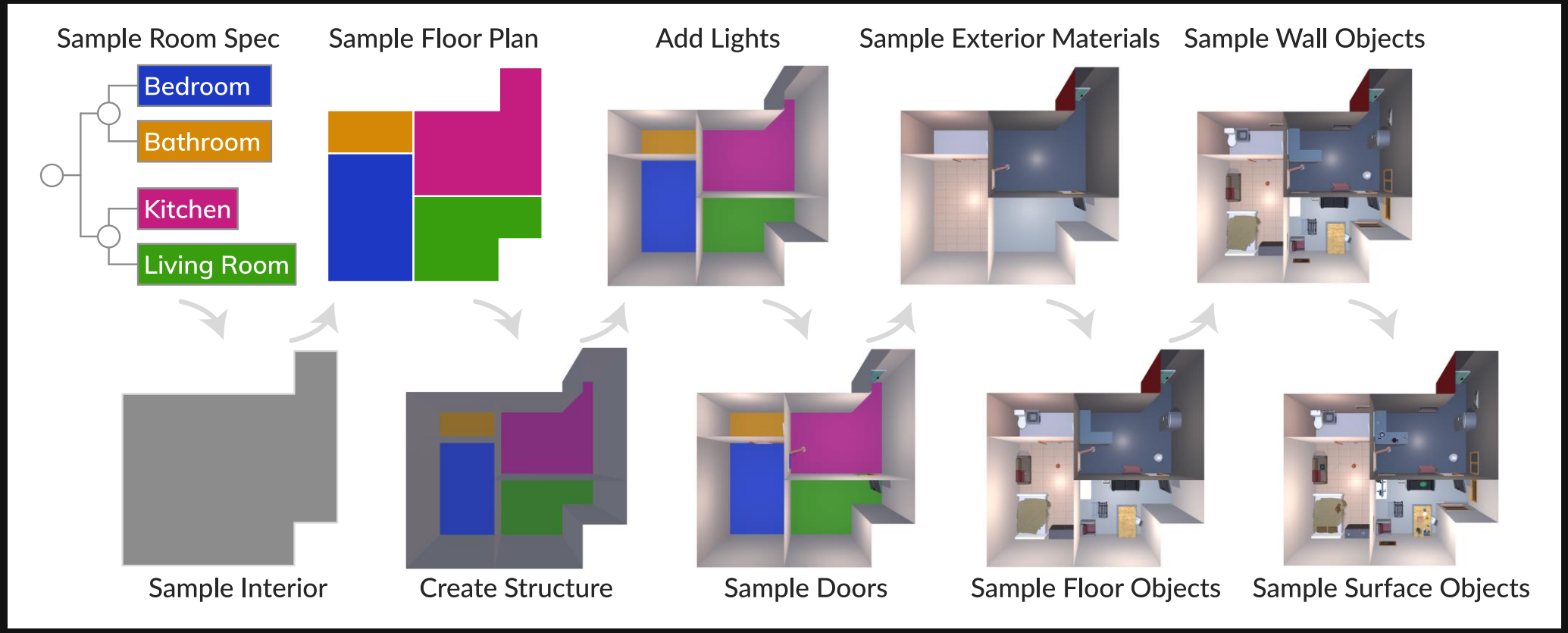
3D Scene Generation: Methods and Usages



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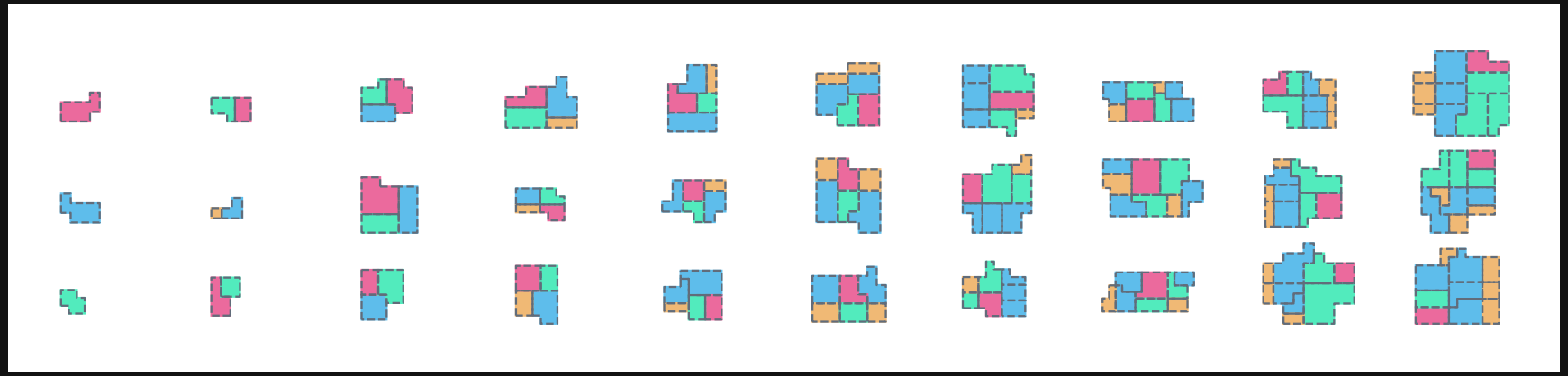
Procedural Generation: Method



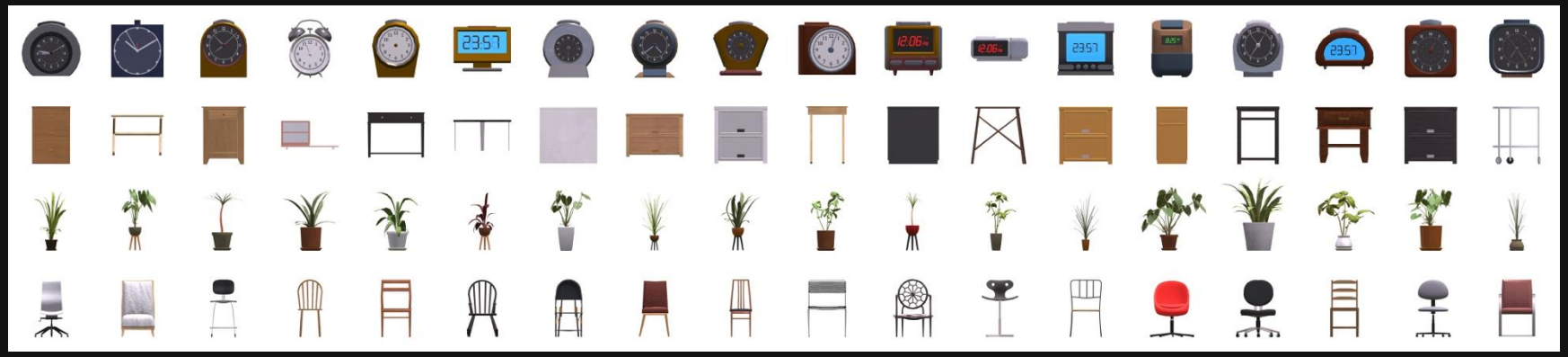
Generate the 3D scene step by step using an asset library and handcrafted rules.

Deitke, Matt, et al. "ProcTHOR: Large-Scale Embodied AI Using Procedural Generation." NeurIPS, 2022.

Procedural Generation: Examples



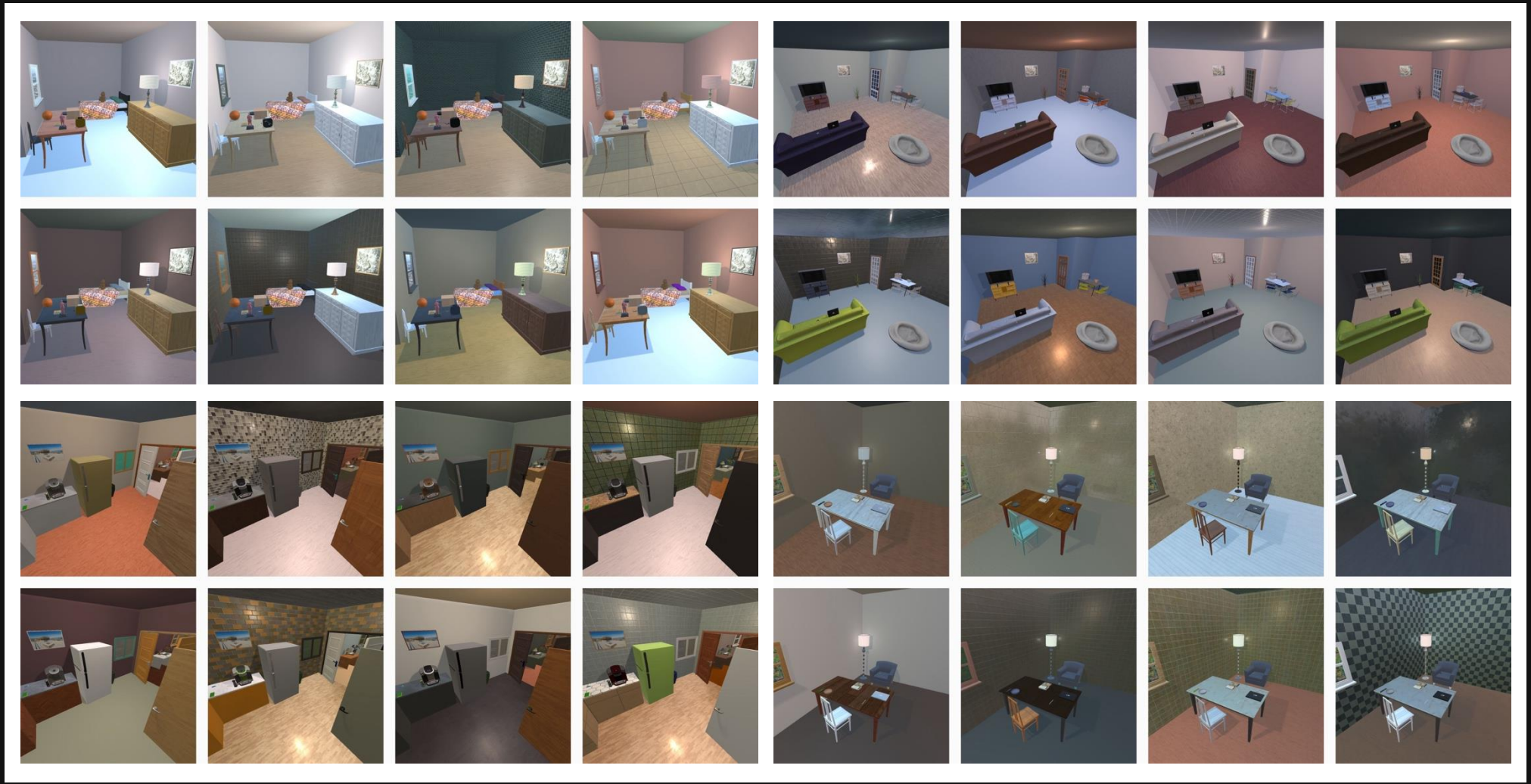
Example of generated floorplans



Example of asset library

Deitke, Matt, et al. “ProcTHOR: Large-Scale Embodied AI Using Procedural Generation.” NeurIPS, 2022.

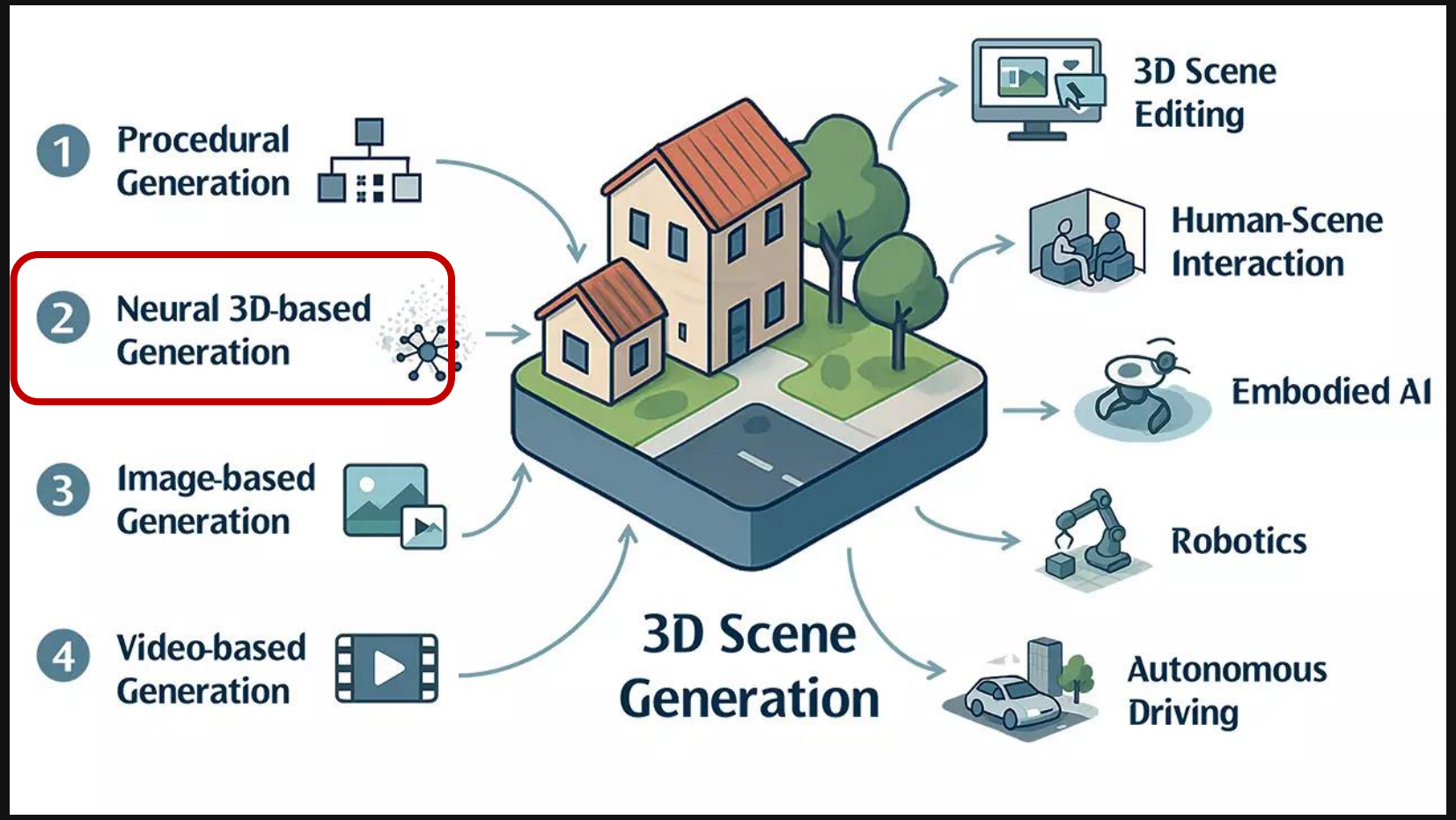
Procedural Generation: Examples



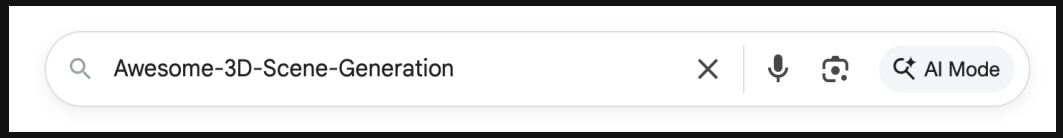
Example of generated floorplans with ProcTHOR

Deitke, Matt, et al. “ProcTHOR: Large-Scale Embodied AI Using Procedural Generation.” NeurIPS, 2022.

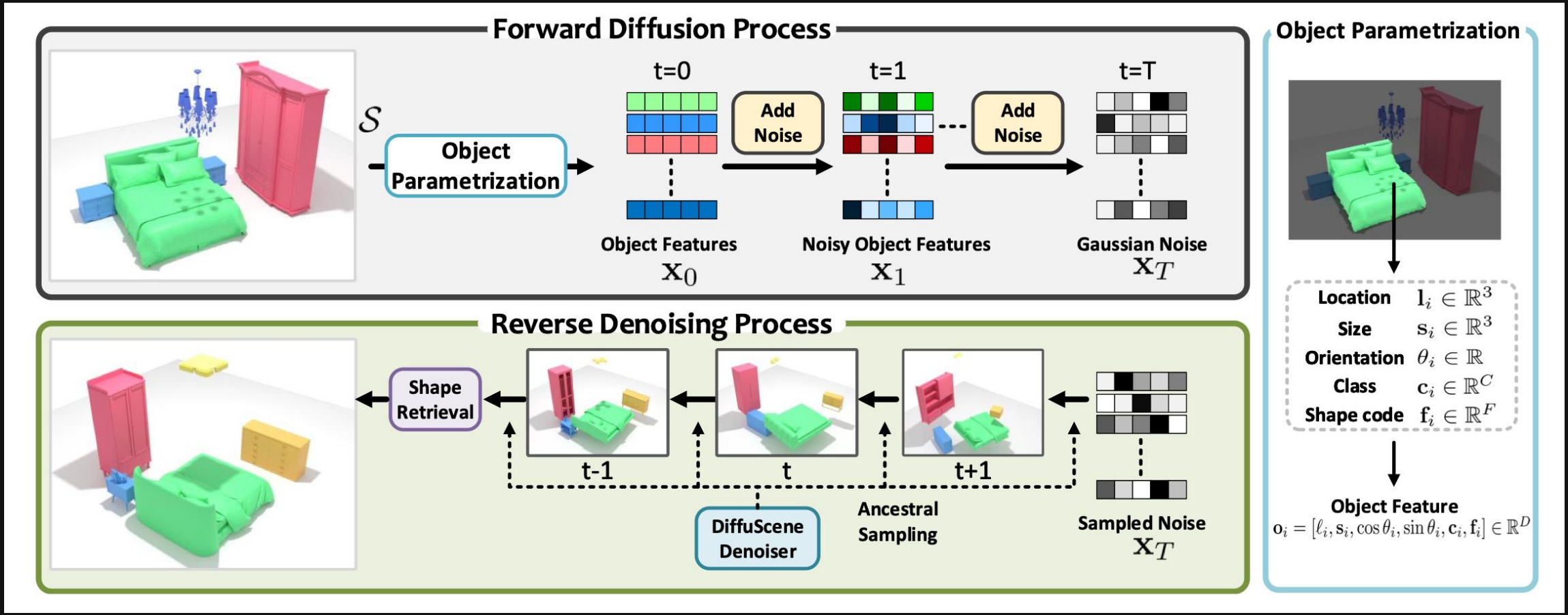
3D Scene Generation: Methods and Usages



For a complete list of papers, please search on Google:



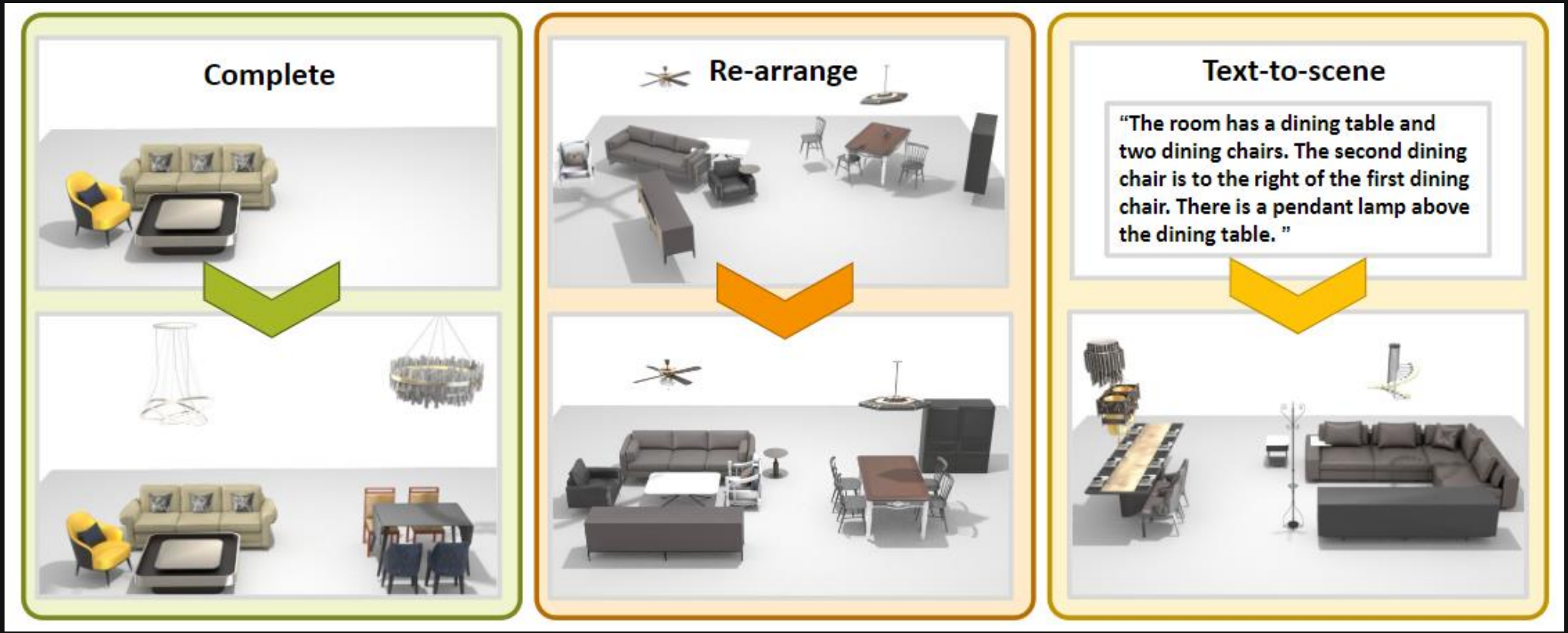
Neural 3D-based Generation: Method



Generate the 3D scene with learned distributions

Tang, Jiapeng, et al. "Diffuscene: Denoising Diffusion Models for Generative Indoor Scene Synthesis." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 20507–18.

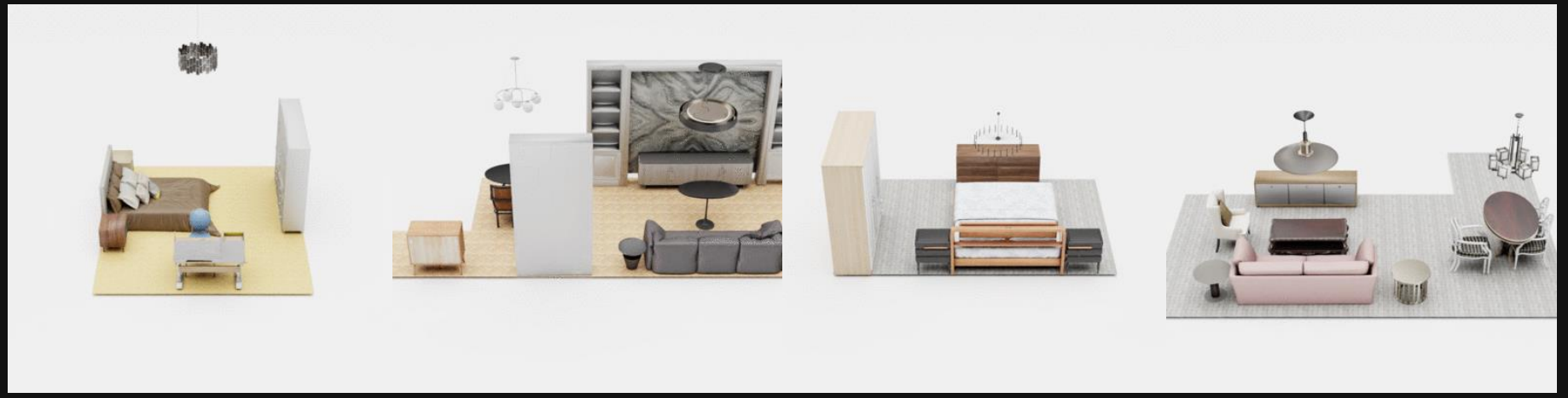
Neural 3D-based Generation: Examples



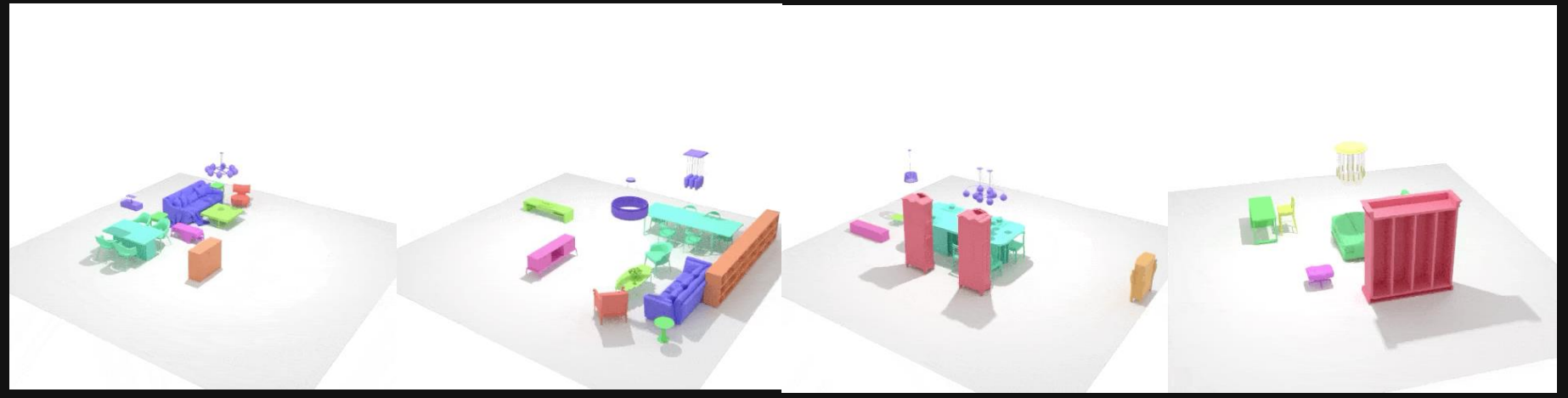
Support multiple types of input

Tang, Jiapeng, et al. "Diffuscene: Denoising Diffusion Models for Generative Indoor Scene Synthesis." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 20507–18.

Neural 3D-based Generation: Examples



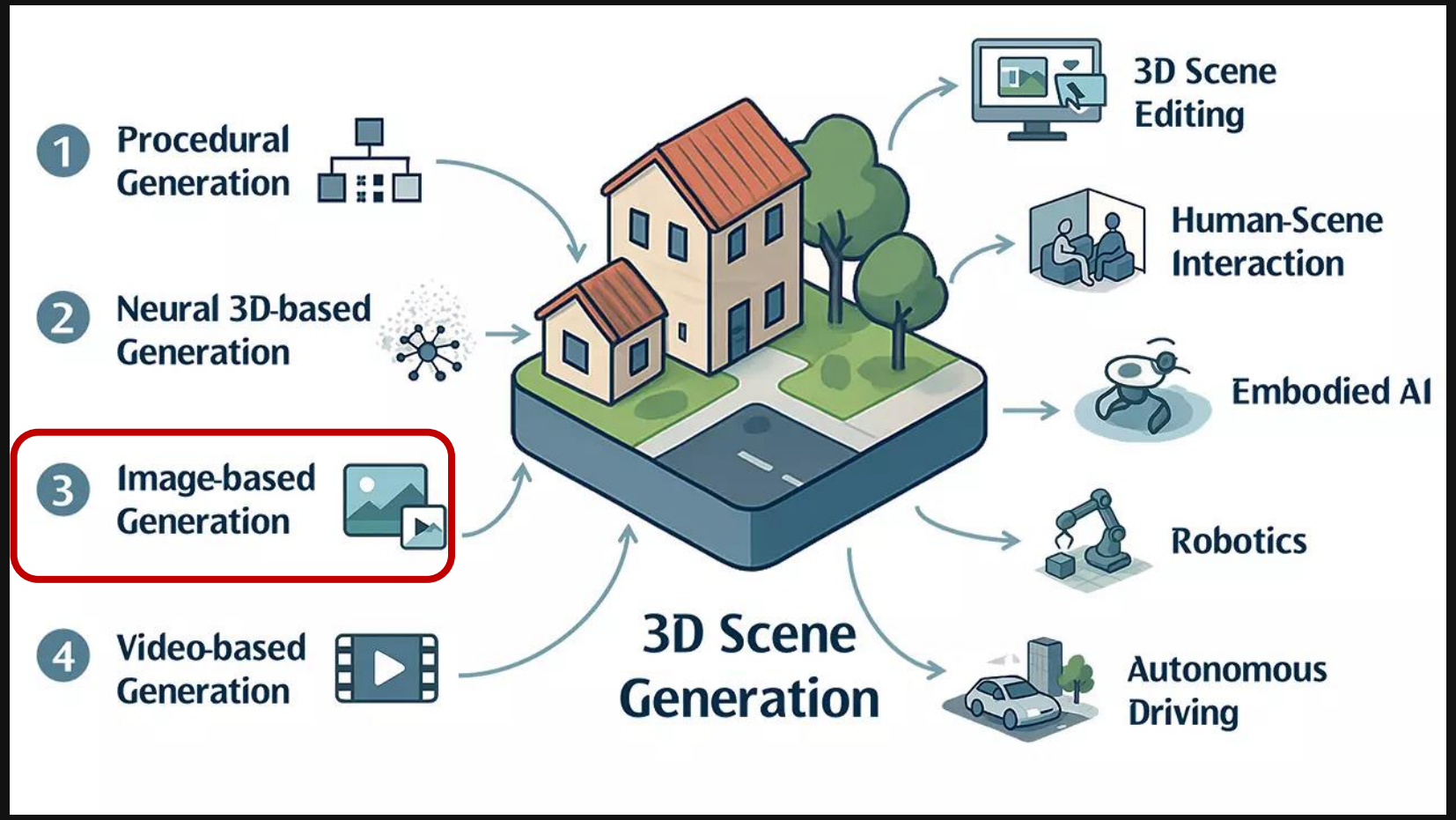
Example scenes generated by ATISS[1]



Example scenes generated by DiffuScene[2]

[1] Paschalidou, Despoina, et al. "Atiss: Autoregressive Transformers for Indoor Scene Synthesis." Advances in Neural Information Processing Systems, vol. 34, 2021, pp. 12013–26.
[2] Tang, Jiapeng, et al. "Diffuscene: Denoising Diffusion Models for Generative Indoor Scene Synthesis." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 20507–18.

3D Scene Generation: Methods and Usages



For a complete list of papers, please search on Google:

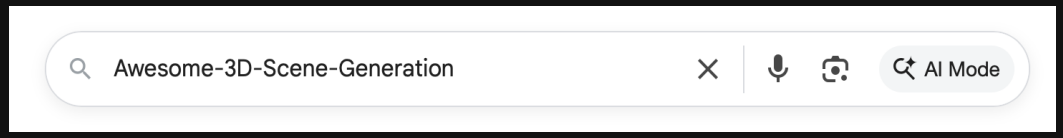
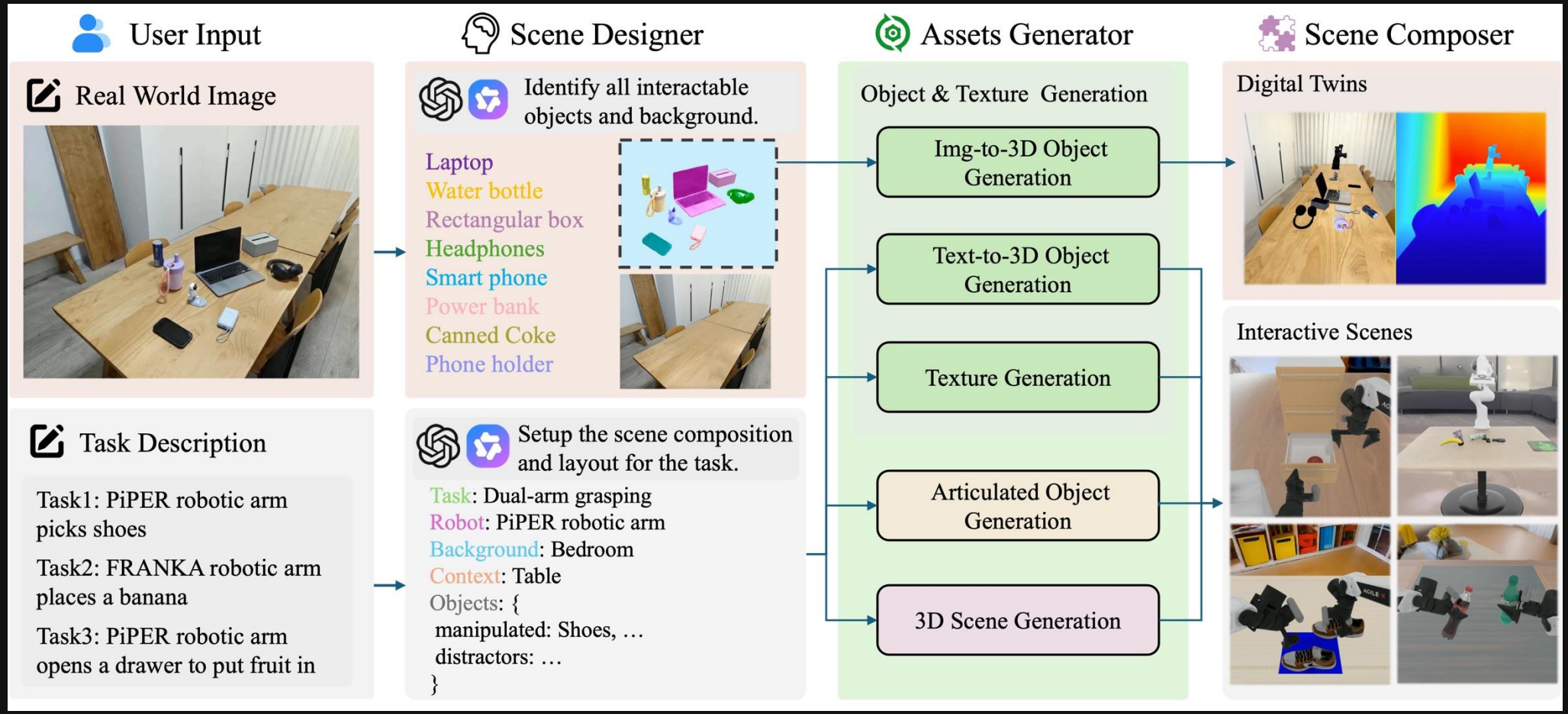


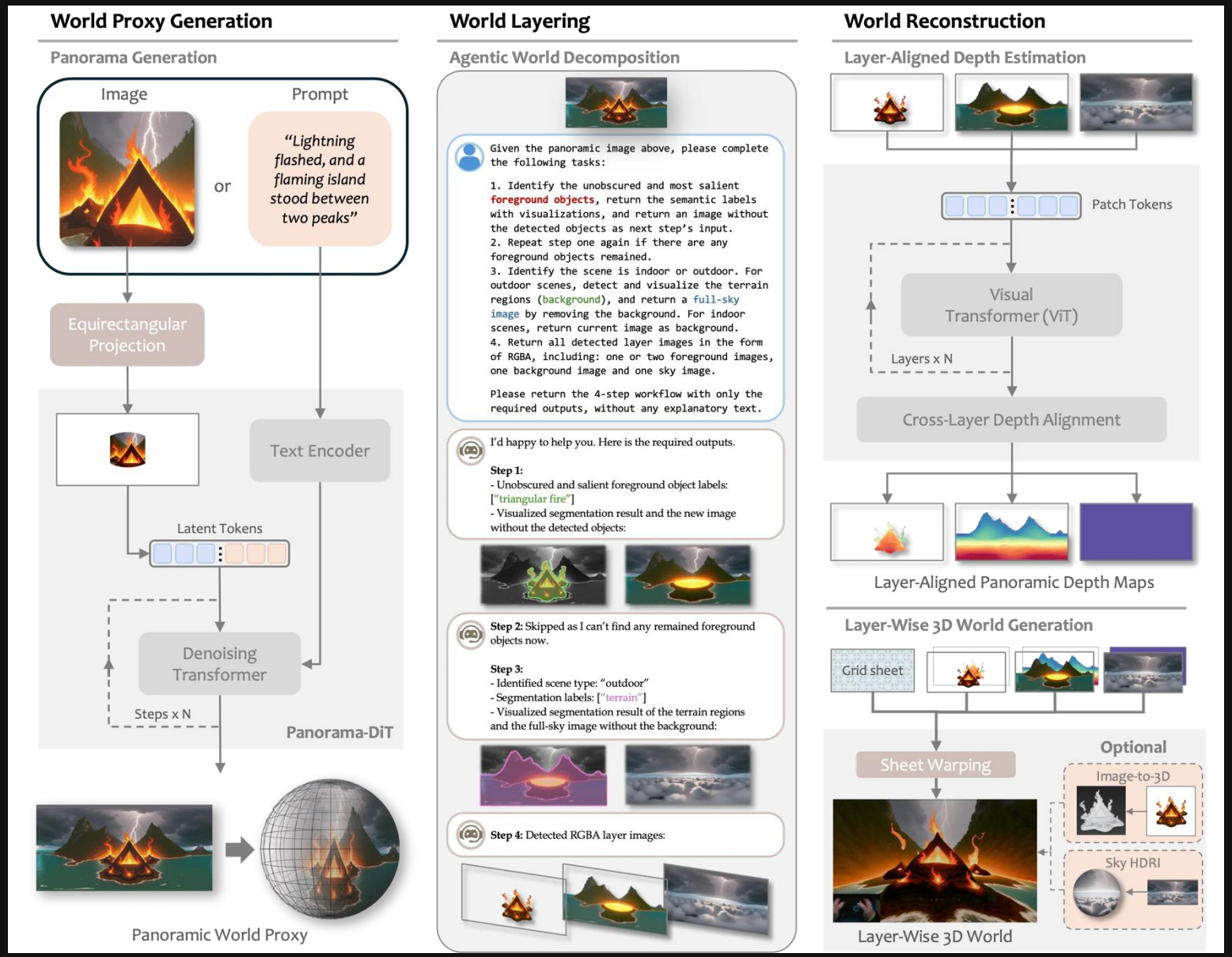
Image-based Generation: Method



Generate the 3D scene with image and text information

Wang, Xinjie, et al. "EmbodiedGen: Towards a Generative 3D World Engine for Embodied Intelligence." arXiv Preprint arXiv:2506.10600, 2025.

Image-based Generation: Method



Support generation with image + text, multiple images, or decomposing the image.

Team, HunyuanWorld, et al. "HunyuanWorld 1.0: Generating Immersive, Explorable, and Interactive 3D Worlds from Words or Pixels." arXiv Preprint arXiv:2507.21809, 2025.

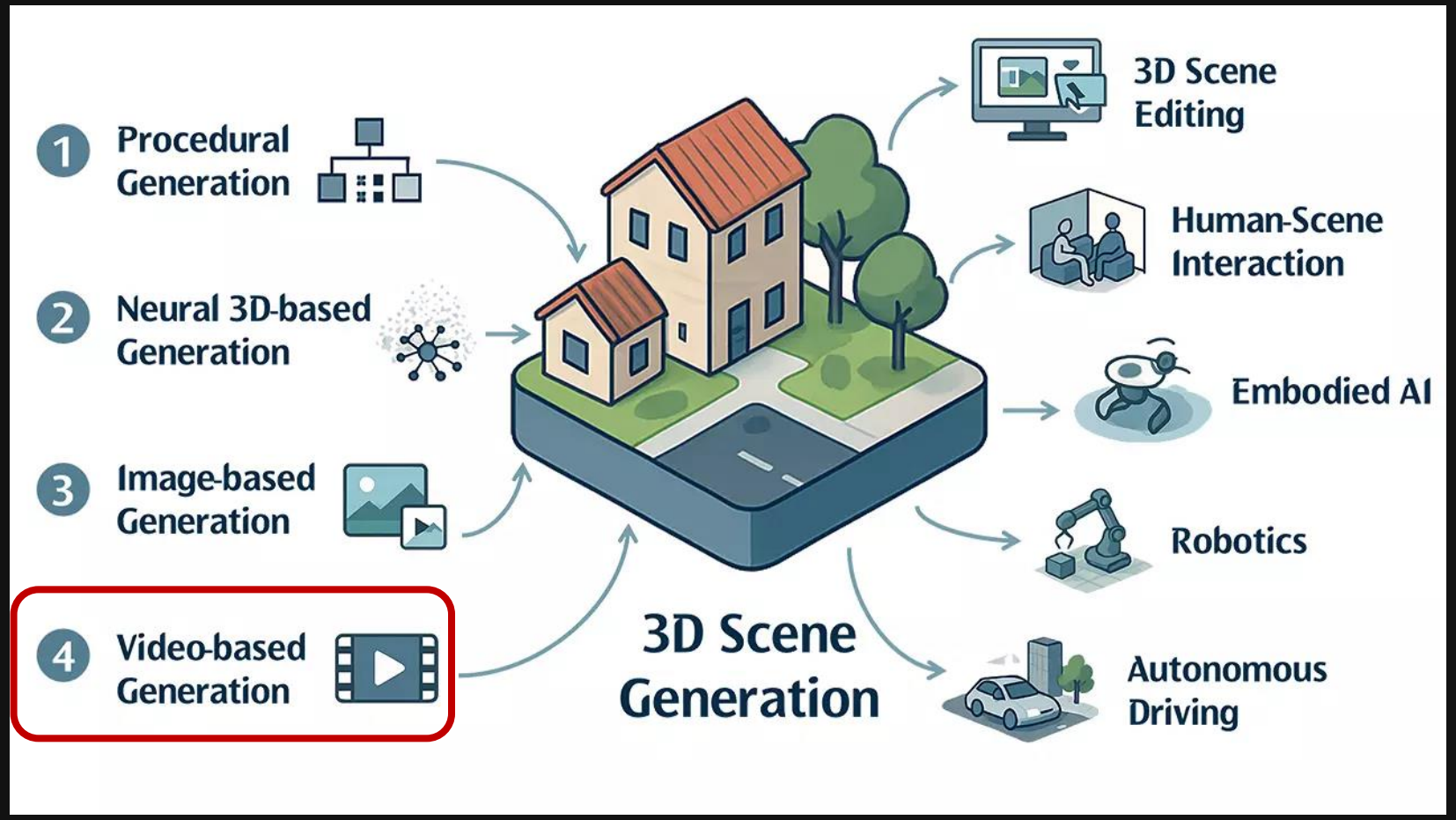
Image-based Generation: Examples



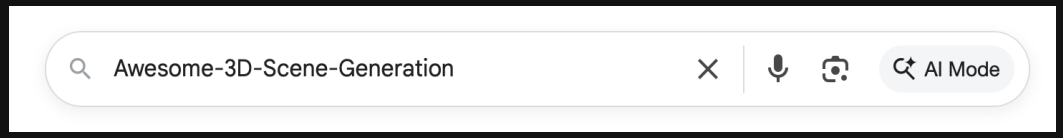
Generated scene with HunyuanWorld.

Team, HunyuanWorld, et al. “HunyuanWorld 1.0: Generating Immersive, Explorable, and Interactive 3D Worlds from Words or Pixels.” arXiv Preprint arXiv:2507.21809, 2025.

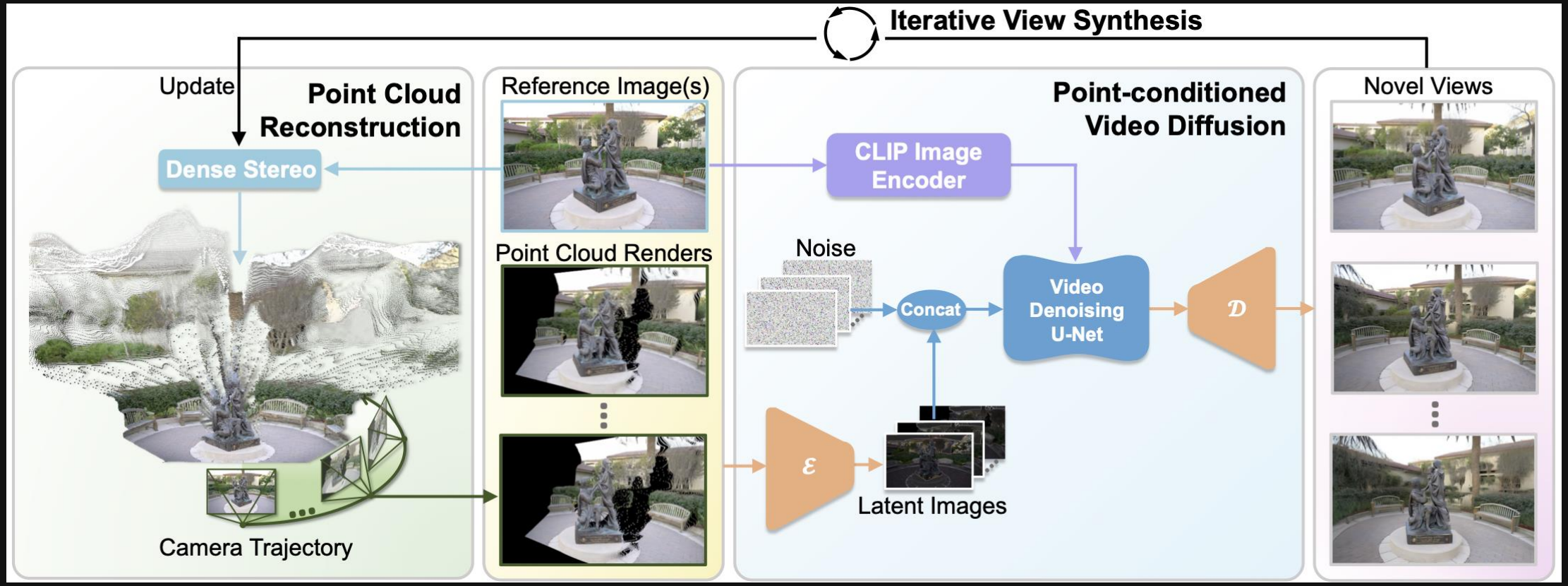
3D Scene Generation: Methods and Usages



For a complete list of papers, please search on Google:



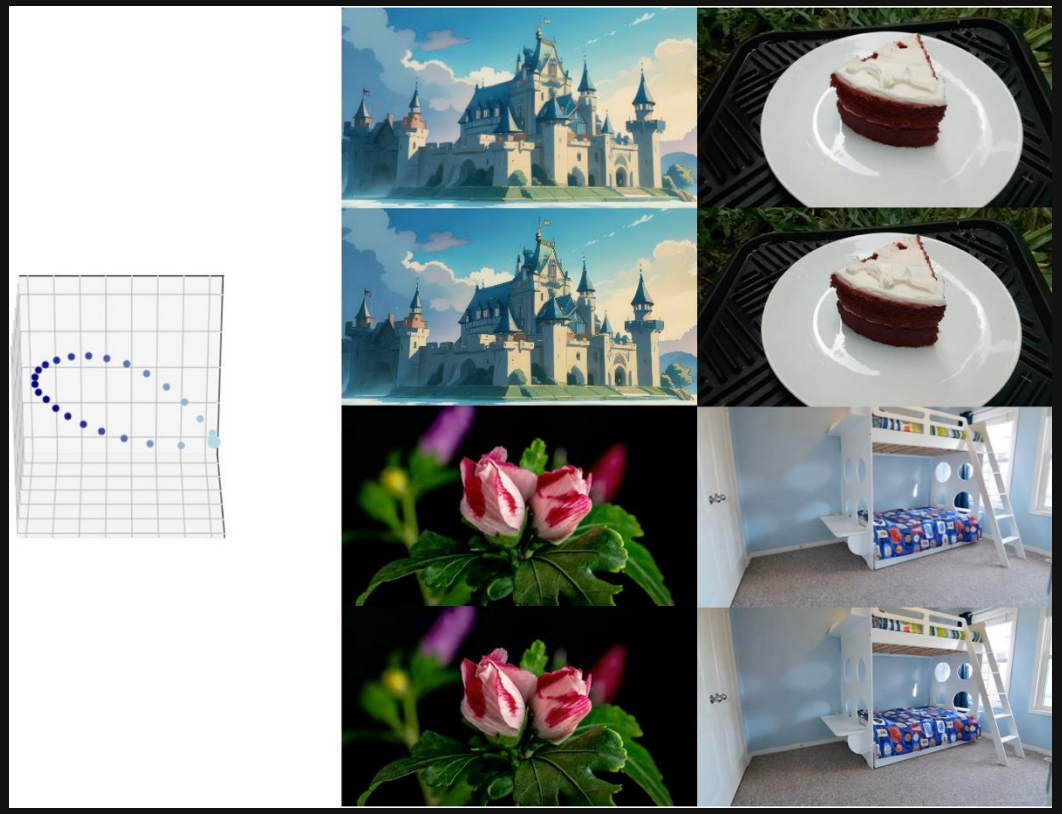
Video-based Generation: Method



Generate the 3D scene with video as final output.

Yu, Wangbo, et al. "Viewcrafter: Taming Video Diffusion Models for High-Fidelity Novel View Synthesis." arXiv Preprint arXiv:2409.02048, 2024.

Video-based Generation: Examples





Generate video using given camera trajectory (using one or more images)

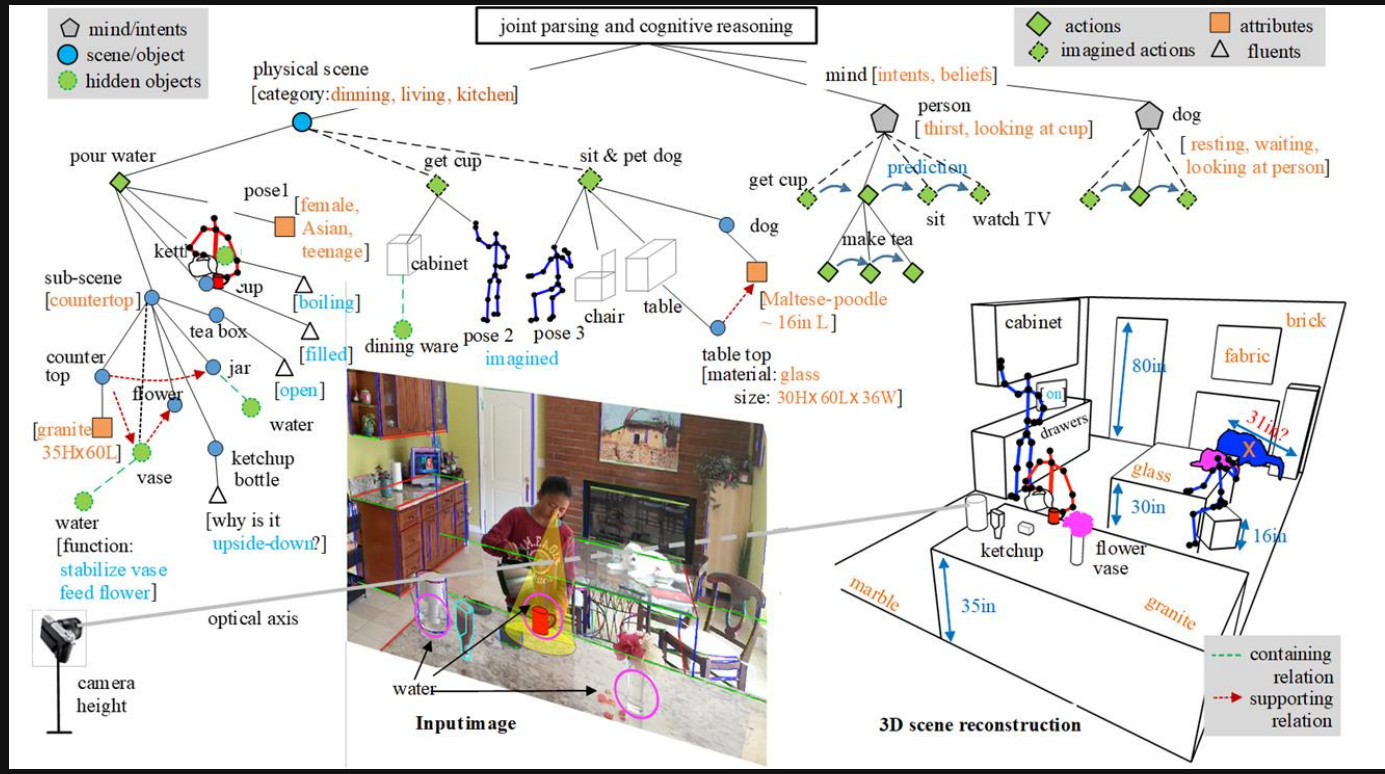
Yu, Wangbo, et al. "Viewcrafter: Taming Video Diffusion Models for High-Fidelity Novel View Synthesis." arXiv Preprint arXiv:2409.02048, 2024.

Outline

- Section 1: 2D HOI
  - General formulation
  - HOI detection
- Section 2: 3D HOI
  - 3D representation
  - 3D reconstruction from 2D image
  - From perception to generation
- Section 3: From HOI to HSI
  - Scene-conditioned human motion generation
  - Human-centric scene synthesis

Back to where we began with ...





1. What do we need to parse this image?
2. What do we need to understand the joke?
3. What do we need to understand the purpose of the image?

Thanks for your listening and sleeping